

Chapter 1

JES and JER from W - analysis overview

The use of hadronically decaying W boson for jet calibration was already proposed long before Run 1 [1]. Although this technique has never been included in the nominal JES calibration, it was used as a cross-check in the 7 TeV JES measurement, as described in Section 15 in [2]. In addition, several ATLAS [3–5], CMS [6] or CDF (at Tevatron) [7] top-quark mass measurements have exploited the precisely known W -boson mass to constrain the JES uncertainties. While the corrections derived in the top-quark mass measurements are not universal, they demonstrate the sensitivity of the reconstructed W -boson mass to the jet calibration.

Single-lepton $t\bar{t}$ events, shown in Figure 1.1, are used for the presented measurements. In these events, each top quark¹ decays into a b -quark and a W boson. One of the W bosons decays leptonically into a charged lepton and a neutrino, while the other W boson decays hadronically into two light-quark² jets. The two jets are used to reconstruct the invariant mass of the W boson, which is very sensitive to the scale and resolution of the jets. The single-lepton $t\bar{t}$ production has a large cross section of 830 pb, with a relative uncertainty of 4.6%, at $\sqrt{s} = 13$ TeV at the LHC [8]. It features a distinctive experimental signature consisting, at LO, of two b -jets, one isolated charged lepton that is used for triggering, missing transverse momentum arising from the undetected neutrino, and two light-flavoured jets originating from the hadronic decay of the W -boson. Additional jets can arise from final-state radiation or initial-state

¹Throughout this analysis, the terms top quark, b -jet, lepton and neutrino are used generically to denote both particles and their antiparticles: top and anti-top quarks, b - and $\bar{b}$ -initiated jets, leptons and antileptons, and neutrinos and antineutrinos, respectively.

²In this analysis, the term light-quark jet is used to denote jets initiated by u , d , s , or c quarks (and their corresponding antiquarks). While charm quarks are generally not classified as light quarks, they are included in the light-quark jet definition adopted here, which simply excludes b -quarks.

radiation. This signature allows a clean event selection, with only a small contribution from background processes (described in Section sec:background).

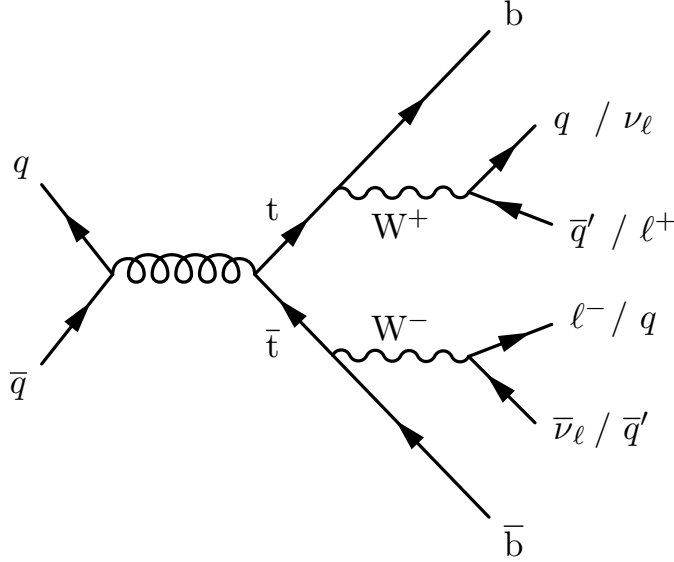


FIGURE 1.1: Feynman diagram of $t\bar{t}$ production in proton-proton collisions, with top-quark decay in the single-lepton decay channel.

Unlike in the standard ATLAS *in situ* calibration techniques described in Chapter ??, there is no well-measured reference object to balance the jet system. Instead, a forward-folding technique (Chapter 2) is used to produce the reconstructed W -boson mass distributions under different JES or JER assumptions, referred to as the W -mass templates (Chapter 5).

The object definitions specific to the presented measurement are introduced in Chapter 3. These definitions establish the objects used throughout the analysis and form the basis for the subsequent event selection introduced in Chapter 4. In the selected events, the two jets originating from the hadronic W -boson decay are identified, and the invariant mass of the W boson is reconstructed from these jets.

The mean of the W -boson mass distribution is used to determine the JES correction from fit to data, while the standard deviation is used to determine the JER correction (Chapter 6).

To allow for a future combination with other ATLAS *in situ* calibration methods, the JES and JER correction factors are measured separately.

For the JES measurement, the best available *in situ* JER calibration is applied to simulated samples, while no *in situ* JES calibration is applied to data, as the JES

correction is the quantity being measured. The *in situ* JER calibration consists of smearing the p_T of jets in simulation to reproduce the typically worse JER in data, as described in Section ???. However, the corresponding smearing has not yet been derived for the current ATLAS reconstruction software release and it is therefore not used. The difference in JER between data and simulation is currently accounted for in the considered systematic uncertainties. Conversely, for the JER measurement, the best available *in situ* JES calibration is applied to data, while no *in situ* JER calibration is applied to simulation.

Once the *in situ* JER calibration has been derived for the current ATLAS reconstruction software release, two versions of the simulated samples will be used. One will include the *in situ* JER calibration and will be used for the JES measurement. The other will not include the *in situ* JER calibration and will be used for the JER measurement. Thus, the JES measurement will compare data without an *in situ* calibration to simulation with the *in situ* JER calibration applied, whereas the JER measurement will use data with the *in situ* JES calibration applied and simulation without the *in situ* JER calibration.

To account for the correlations between JES and JER, the JER systematic uncertainties (see Chapter ??) are considered in the JES measurement and vice versa.

Chapter 2

Forward-folding method

Forward-folding method [9–11] is used to simulate the changes in the reconstructed jet p_T due to different assumptions of JES or JER. The forward-folding formula (Equation 2.1) is applied individually on the reconstructed p_T^{reco} of jets, taking into account their true momenta p_T^{truth} and taking an assumption on JES or JER. The invariant mass of the W boson, which is very sensitive to JES and JER, is then reconstructed from the folded jets, producing the W -boson mass templates corresponding to the different JES and JER assumptions (Section 5).

2.1 Reco-to-truth matching

In order to perform the forward-folding, the *reco* jets (defined in Section ??) have to be paired with corresponding *truth* jets. This pairing is based on angular distance ΔR between the *reco* jet and the *truth* jet. The closest *truth* jet, i.e. the *truth* jet with lowest ΔR within $\Delta R = 0.2$, is matched to the *reco* jet. If there is no *truth* jet within $\Delta R = 0.2$, the *reco* jet is left unmatched. In case the same *truth* jet is matched to various *reco* jets in the event, these jets (both *reco* jet and *truth* jet) are considered unmatched as well.

When multiple *truth* jets are located close to the *truth* jet matched to a *reco* jet, the physical origin of p_T^{reco} cannot be uniquely assigned to a single *truth* jet¹. To avoid this ambiguity and ensure reliable one-to-one jet matching, the *truth* jets considered in the matching procedure are required to be well isolated and thus no other *truth* jet is allowed within $\Delta R = 0.5$ of the *truth* jet that is being matched to the *reco* jet.

In addition, no prompt *truth* electron or muon are allowed within $\Delta R = 0.5$ of the *reco* jet to avoid any prompt lepton being included in the jet reconstruction.

¹In particular, p_T^{reco} may correspond to the p_T of a single *truth* jet, to the combined p_T of multiple *truth* jets, or to only a fraction of the p_T associated with one or more *truth* jets.

The forward-folding is not applied on jets that do not satisfy the matching and isolation criteria described above, and they are treated as a background for JES or JER, as they do not provide any separation between the different assumptions for JES and JER.

2.2 Forward-folding formula

The reconstructed transverse momenta p_T^{reco} of the detector level jets are transformed with the forward-folding formula, assuming various corrections to JES and JER used in simulation, s and r , respectively:

$$p_T^{\text{folded}} = s \cdot p_T^{\text{reco}} + (p_T^{\text{reco}} - \mathcal{R} \cdot p_T^{\text{truth}}) \cdot (r - s), \quad (2.1)$$

where p_T^{truth} is the momentum of *truth* jet matched to the *reco* jet (the matching procedure is described in Section 2.1), $\mathcal{R}$ is the mode of the jet response distribution:

$$\mathcal{R} = \operatorname{argmax} \left\{ \frac{p_T^{\text{reco}}}{p_T^{\text{truth}}} \right\}.$$

The mode of the response distribution is used instead of the mean because it is more stable against changes in the tail of the response distribution, which can cause the mean to shift while leaving the peak (mode) of the distribution unchanged.

Equation 2.1 can be rewritten as:

$$p_T^{\text{folded}} = s \cdot \mathcal{R} \cdot p_T^{\text{truth}} + r \cdot (p_T^{\text{reco}} - \mathcal{R} \cdot p_T^{\text{truth}}), \quad (2.2)$$

where the term $\mathcal{R} \cdot p_T^{\text{truth}}$ is the reconstructed p_T corresponding to the peak (mode) of the response distribution, while $p_T^{\text{reco}} - \mathcal{R} \cdot p_T^{\text{truth}}$ is the event-by-event fluctuation around this central value $\mathcal{R} \cdot p_T^{\text{truth}}$. Shifting $\mathcal{R} \cdot p_T^{\text{truth}}$ by s corresponds to shifting the mode of the response distribution which defines the JES, while modifying the fluctuation around the central value $p_T^{\text{reco}} - \mathcal{R} \cdot p_T^{\text{truth}}$ by r corresponds to modifying the width of the response distribution which defines the JER.

The jet four-momentum is scaled by $p_T^{\text{folded}}/p_T^{\text{reco}}$, yielding the *folded* jet.

The corrections s and r range from 0.95 to 1.05 and 0.60 to 1.40, respectively. When smearing jets with different JES (s) values, the JER (r) values are set to unity and vice-versa to generate pure JES and pure JER variations. The values of $\mathcal{R}$ are estimated by fitting a Gaussian function around the peak of the jet response (defined by

Equation ?? in Chapter ??) distribution in the nominal MC simulation, separately for p_T bins 0–20 GeV, 20–35 GeV, 35–50 GeV, 50–70 GeV, 70–100 GeV, 100–150 GeV, 150–200 GeV and > 200 GeV, η bins $|\eta| < 0.8$, $|\eta| < 1.3$ and $|\eta| < 2.5$ and for b -jets, c -jets, gluon jets or light (u, d, s)-jets (the truth flavour labels are available for jets in simulation as explained in Section ??).

The Run 2 response distributions for $|\eta| < 0.8$ bin and the 50–70 GeV p_T bin are shown in Figure 2.1. The response distribution for the nominal $t\bar{t}$ sample is shown with the red Gaussian function around the peak and compared to several other $t\bar{t}$ samples, separately for the different types of jets. The samples ending with "FS" are the full-simulation samples and samples ending with "AF" are fast-simulation samples, hence the different responses. Run 2 response distributions for all the considered p_T and $|\eta|$ bins are shown in Appendix A. Only Run 2 distributions are shown as the Run 3 distributions are very similar.

The $\mathcal{R}$ values used in the forward-folding formula (Equation 2.1) are listed in Tables 2.1 and 2.1 for Run 2 and Run 3, respectively. The b -jet response in the 0–20 GeV p_T bin is not relevant for this analysis as we require at least 30 GeV for the b -jets. No uncertainties on the obtained response values are considered in the forward-folding formula, however, several systematic uncertainties that impact jet response are considered for the JES and JER measurement (see Chapter ??). Several trends can be observed. The response for lower p_T jets differs from unity, while it approaches unity with increasing p_T . The b -jet response is lower than one due to b -jets decaying also semi-leptonically into a charged lepton and a neutrino, with the neutrino being undetected. The gluon-jet response shows the best compatibility with unity across the whole p_T range due to the *MC JES* calibration (which aims to correct the response to unity, see Section ??) being derived using dijet events, which are dominated by gluon-initiated jets.

TABLE 2.1: The mean values of Run 2 simulated response $\mathcal{R}$ from nominal MC for different p_T bins and different jet flavours.

jet p_T [GeV]	b -jets	c -jets	gluon jets	light jets
0 - 20	0.97	1.21	1.20	1.24
20 - 35	1.08	1.03	1.01	1.07
35 - 50	0.97	1.00	0.98	1.03
50 - 70	0.98	1.01	0.99	1.03
70 - 100	0.98	1.01	0.99	1.02
100 - 150	1.00	1.01	0.99	1.01
150 - 200	1.00	1.00	0.99	1.01
> 200	0.99	1.00	0.99	1.00

TABLE 2.2: The mean values of Run 3 simulated response $\mathcal{R}$ from nominal MC for different p_T bins and different jet flavours.

jet p_T [GeV]	b -jets	c -jets	gluon jets	light jets
0 - 20	1.11	1.18	1.18	1.23
20 - 35	1.08	1.04	1.02	1.07
35 - 50	0.97	1.01	0.99	1.04
50 - 70	0.98	1.01	0.99	1.03
70 - 100	0.98	1.01	0.99	1.02
100 - 150	1.00	1.01	0.99	1.01
150 - 200	0.99	1.00	0.99	1.00
> 200	0.99	1.00	0.99	1.00

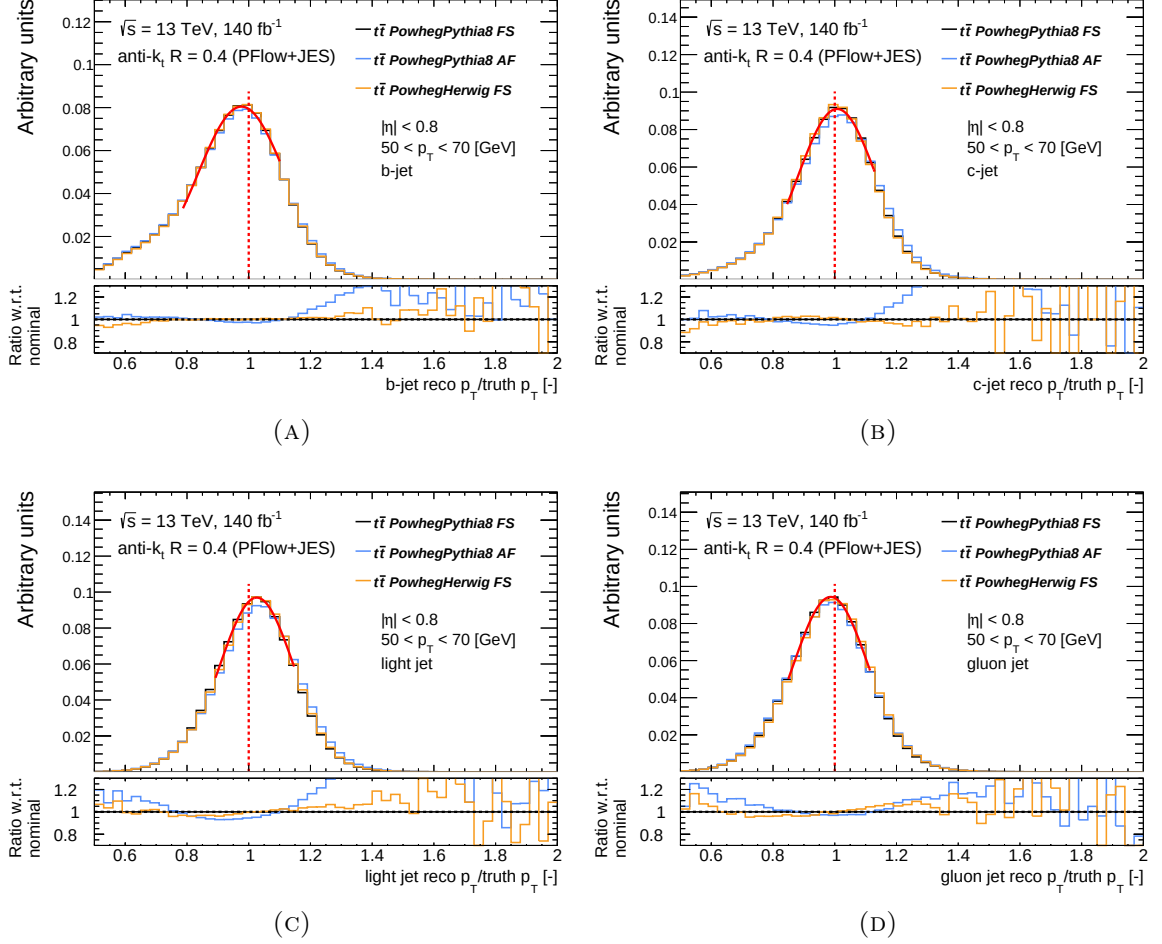


FIGURE 2.1: The Run 2 simulated jet response distribution for jets with $|\eta| < 0.8$ and p_T from 50 – 70 GeV bin. The distributions are shown for the nominal $t\bar{t}$ sample produced with Powheg+Pythia (black), the fast-simulation $t\bar{t}$ sample produced with Powheg+Pythia (yellow), the fast-simulation $t\bar{t}$ sample produced with Powheg+Herwig (orange), the fast-simulation $t\bar{t}$ sample produced with Powheg+Pythia using the recoilToTop recoil model in Pythia (light blue), and the fast-simulation $t\bar{t}$ sample produced with Powheg+Pythia with the Pythia underlying-event variation Var1Down (dark blue). The response distributions are shown separately for (A) b -jets, (B) c -jets, (C) light jets, and (D) gluon jets. The red line represents the Gaussian fit to the response distribution of the nominal $t\bar{t}$ sample. The bottom panel shows the ratios of jet response distributions in the alternative samples over the jet response in the nominal $t\bar{t}$ Powheg+Pythia sample.

Chapter 3

Analysis-specific object definitions

The electrons in the presented measurement are defined using the `tight` identification working point (defined in Section ??) and `Tight_VarRad` isolation working point (defined in Section ??) and maximum transverse impact parameter significance w.r.t. the beam line `maxD0significance` (defined in Section ??) less or equal to five. The tracks associated to an electron must satisfy a requirement on the longitudinal impact parameter (defined in Section ??) w.r.t. the primary vertex such that $|z_0 \sin \theta| < 0.5$ mm, where θ is the polar angle of the track (defined in Section ??). These electrons are required to have $p_T > 27$ GeV and $|\eta| < 2.47$ and not come from the crack region between the barrel and end-cap calorimeters ($1.37 < |\eta| < 1.52$).

The muons are identified using the `Medium` identification working point (defined in Section ??) and `Tight_VarRad` isolation working point (defined in Section ??) and maximum transverse impact parameter significance w.r.t the beam line `maxD0significance` less or equal to three. The longitudinal impact parameter w.r.t the primary vertex is required to satisfy $|z_0 \sin \theta| < 0.5$ mm. The muons are required to have $p_T > 27$ GeV, $|\eta| < 2.5$.

The selected electron and muon working points ensure that the contribution from fake leptons is sufficiently suppressed, while retaining high efficiency for prompt leptons. The analysis would not benefit significantly from the additional statistics provided by looser working points, whereas the increased contribution from fake-leptons would reduce the selection purity. The electron and muon p_T requirement is imposed in order to have fully efficient single-electron and single-muon triggers (see Section ??).

The `Tight` E_T^{miss} working point is used.

The presented analysis uses two event selections, the *low- p_T* selection is applied to the MC simulated events, and the *high- p_T* selection, applied to data. The two event selections use their corresponding object definitions: the *low- p_T* and the *high- p_T* objects.

The primary difference between the two object definitions is the minimum p_T requirement for jets. In the *high- p_T* objects, jets are required to satisfy $p_T > 20$ GeV, while in the *low- p_T* objects, jets satisfy $p_T > 15$ GeV. This lower threshold is chosen because the forward-folding procedure (described in Section 2), applied only on the simulated events, modifies the jet p_T . Jets with $15 \text{ GeV} < p_T < 20 \text{ GeV}$ are kept, as they may be shifted above the 20 GeV threshold after the forward-folding procedure. The differences in the remaining reconstructed objects between the *high- p_T* and *low- p_T* definitions arise only from the overlap removal procedure (described in Section ??). The overlap removal between the *low- p_T* jets and electrons and muons, as well as between the *high- p_T* jets and electrons and muons is performed separately, resulting in two sets of jets, electrons and muons, corresponding to the *low- p_T* and *high- p_T* definitions, which are used in this analysis.

The detector-level jets, referred to as the *reco* jets (defined in Section ??), correspond to the *PFlow* jets reconstructed by clustering particle flow objects (see Section ??) with the anti- k_T algorithm using a $R=0.4$ radius parameter (see Section ??), implemented in the FASTJET software package (see Section ??). To suppress jets originating from pile-up, jets with $p_T < 60$ GeV are required to satisfy a neural-network-based jet vertex tagger (NNJVT) discriminant (see Section ??). The default NNJVT working point *FixedEffPt* is used. In simulated samples, jets with $p_T > 15$ GeV and $|\eta| < 2.5$ are preselected. In subsequent analysis stages, following the forward-folding procedure where applicable, jets are required to satisfy $p_T > 20$ GeV, while *b*-jets must satisfy $p_T > 30$ GeV. The 20 GeV threshold is chosen because it is the minimum p_T for which jet calibration is available in ATLAS. The lower preselection threshold of 15 GeV ensures the inclusion of jets that cross the $p_T > 20$ GeV requirement only after the forward-folding. The 30 GeV requirement for *b*-jets prevents them from falling, after forward-folding, below 20 GeV threshold required for *b*-tagging. The *b*-jets with $p_T < 20$ GeV cannot be *b*-tagged and would therefore be excluded from the analysis. Because *b*-jets have typically higher p_T , this requirement does not result in a significant loss of events. The two jets that are used for the reconstruction of the invariant mass of W boson are required to have $|\eta| < 0.8$ since only central jets are used for absolute *in situ* jet calibration in ATLAS, while the relative η -intercalibration corrects the forward jets ($|\eta| < 4.5$) to the scale of the calibrated central jets, as explained in Section ??.

The *b*-jets are identified by the GN2v01 flavour tagging algorithm (described in Section ??) with a working point corresponding to an efficiency of 85% in a $t\bar{t}$ reference sample.

3.0.1 Particle-level object definitions

Particle-level leptons, referred to as the *truth* leptons, are selected as leptons originating from the decay of a W or Z boson. The four-momentum of an electron or muon is summed with the four-momenta of all radiated photons within a cone of size $\Delta R = 0.1$ around its direction, excluding photons from hadron decays.

The particle-level jets, referred to as the *truth* jets, are reconstructed with the anti- k_T algorithm with a radius parameter of $R = 0.4$, using all stable particles (with $\tau \geq 30$ ps) except for the electrons, muons, and photons used in the definition of leptons and neutrinos originating from the W or Z bosons. The *truth* jets are required to have $p_T > 10$ GeV and $|\eta| < 2.5$.

Chapter 4

Event selection and hadronic W-boson reconstruction

Motivated by the signature of $t\bar{t}$ decays in the single-lepton channel, selected events are required to contain at least four jets, of which at least two must be b -jets, exactly one electron or muon, and one neutrino, which contributes to E_T^{miss} .

Two event selections are used to define the preselection for this analysis. The *high- p_T* event selection is applied to the recorded data, while the *low- p_T* event selection is applied to the MC simulated events. Each selection uses the corresponding object definitions. Specifically, the *high- p_T* event selection requires events to contain at least four *high- p_T* jets with $p_T > 20$ GeV, two of those must be b -tagged, and one *high- p_T* lepton, whereas the *low- p_T* event selection requires at least four *low- p_T* jets with $p_T > 15$ GeV, two of them must be b -tagged, and one *low- p_T* lepton (see Chapter 3 for the *high- p_T* and *low- p_T* object definition). No E_T^{miss} requirement is imposed at this stage.

Following the preselection described above, an additional set of event selection criteria is applied to both the data and to the simulated events. For the simulation, these criteria are applied after the forward-folding procedure, where applicable. At this point, all events are required to contain at least four jets with $p_T > 20$ GeV, of which at least two must be b -tagged and satisfy $p_T > 30$ GeV and E_T^{miss} is required to exceed 30 GeV. Finally, there have to be two jets identified as coming from the hadronic decay of the W boson with a matching quality of $\chi^2 < 1$. The identification of the jets from hadronic W -boson decay is described in the following Section 4.1. The 30 GeV E_T^{miss} requirement is imposed in order to suppress the fake-lepton contribution, as the real prompt leptons from W -boson decay are accompanied by a neutrino which translates to large E_T^{miss} .

The selection criteria described above define the *Inclusive* event selection. It is used to produce the control plots comparing recorded data with the prediction and to

determine the jet response in MC simulation (defined in Section 2 by Equation 2.2).

Figures 4.1 and 4.2 show the control plots for basic kinematic distributions in the *Inclusive* region for Run 2 data without the *in situ* correction applied. The remaining Run 2 control plots showing the data without the *in situ* correction applied, as well as the Run 3 versions of these plots are provided in Appendices B and C, respectively.

The uncertainty band represents all the considered uncertainties listed in Section ??.

By imposing the $\chi^2 < 1$ requirement (see Equation 4.1 in the following section) to select events where two light jets can be reliably identified as originating from a hadronic W -boson decay, background processes are further suppressed. Consequently, only 5% of the selected simulated events in the *Inclusive* region originate from background processes, with the dominant contribution coming from single-top quark production (specifically tW associated production; see Section ?? for a detailed description of the considered backgrounds). Figure 4.3 illustrates the event yields for the signal $t\bar{t}$ process and the considered backgrounds both before and after the application of this cut, showing how the $\chi^2 < 1$ requirement reduces the background contamination. A reasonable agreement between the predicted distributions and observed data is seen with some known features such as mismodelling in the lepton p_T spectrum that is likely related to missing higher order calculations in the MC generators [12].

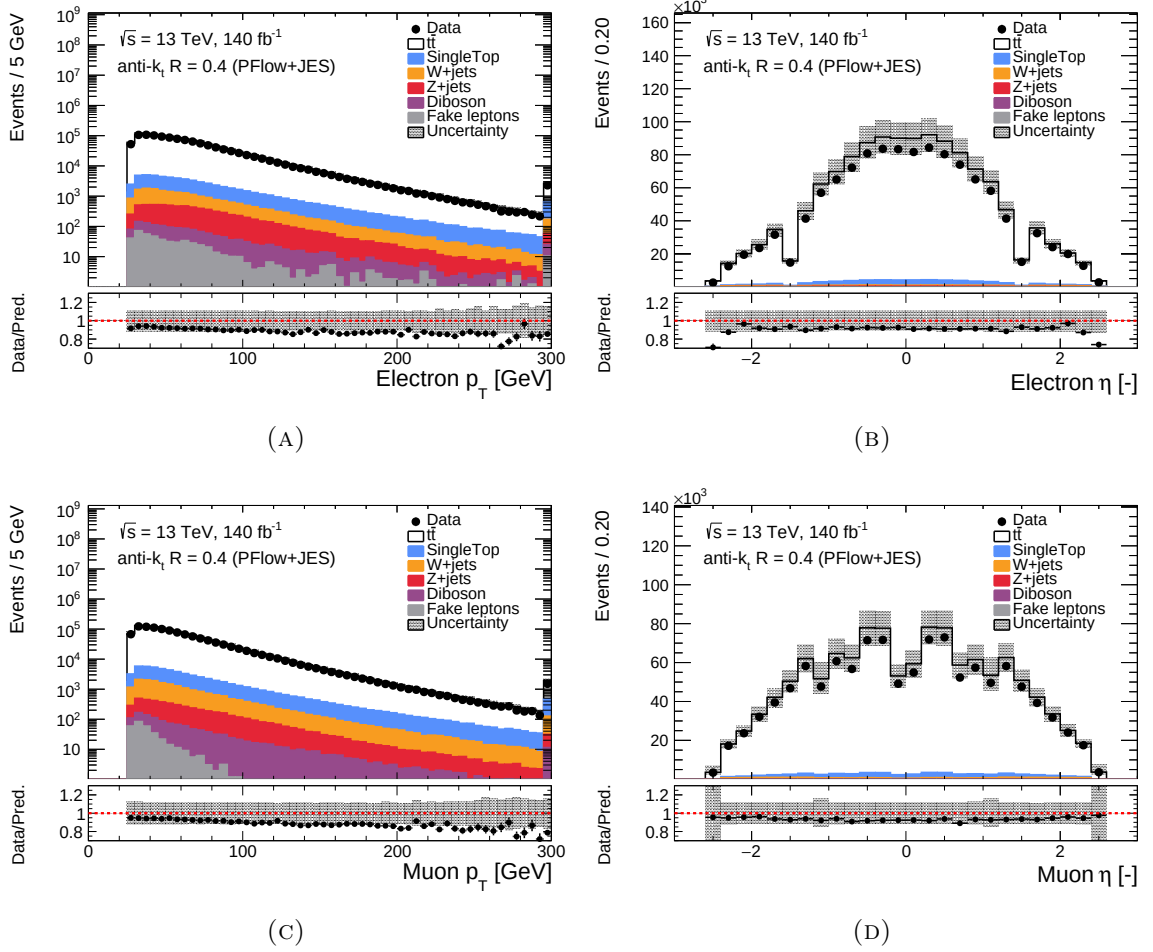


FIGURE 4.1: The Run 2 distributions of the reconstructed leptons in data (without JES *in situ* correction) and MC in the *Inclusive* region with no p_T or η cut on the jets from W boson. (A) Electron p_T in logarithmic scale, (B) electron η , (C) muon p_T in logarithmic scale and (D) muon η . The uncertainty represents all considered systematic uncertainties as described in Section ???. The bottom panels show the ratio of the data to the MC prediction.

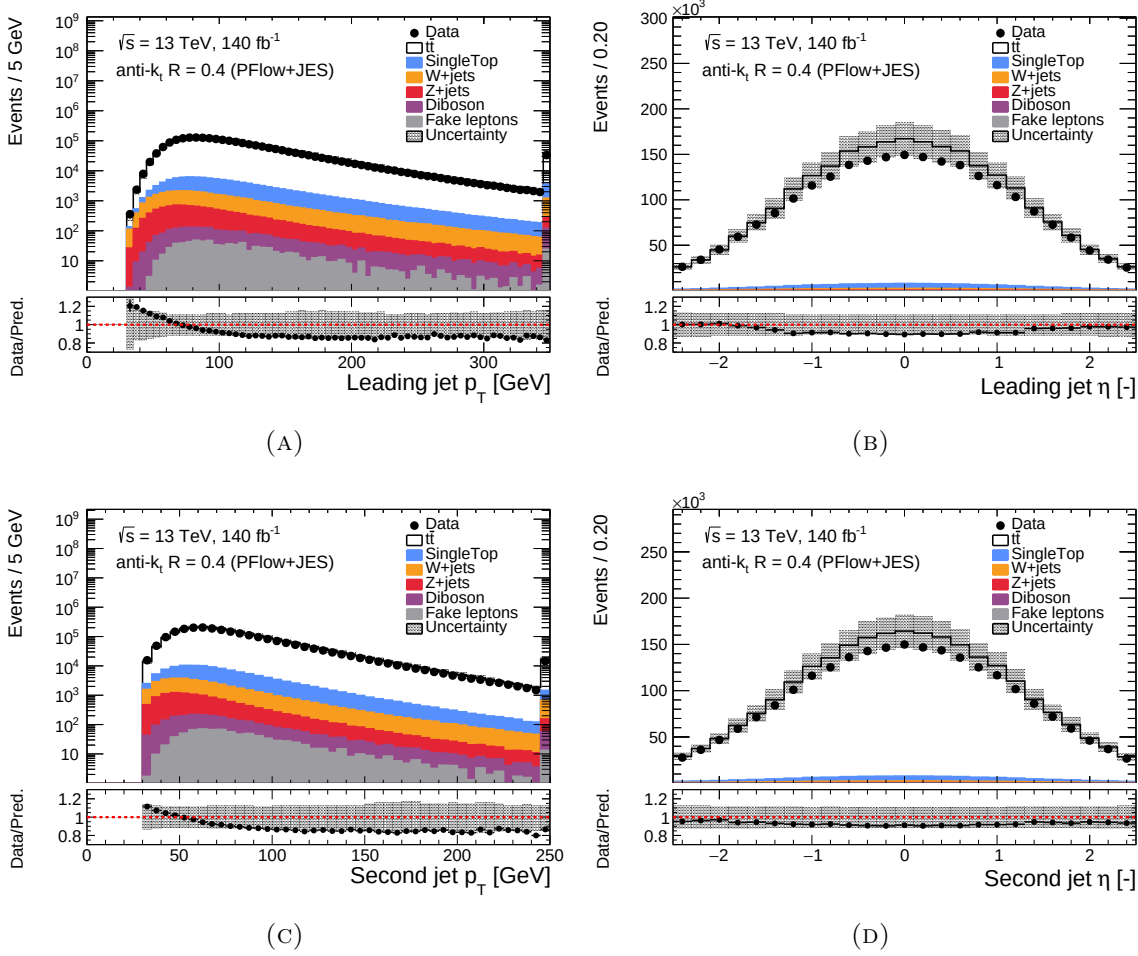
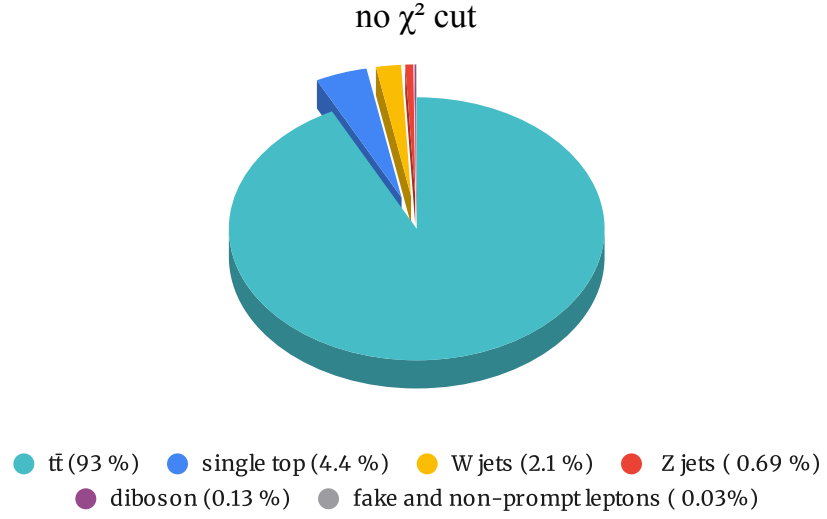
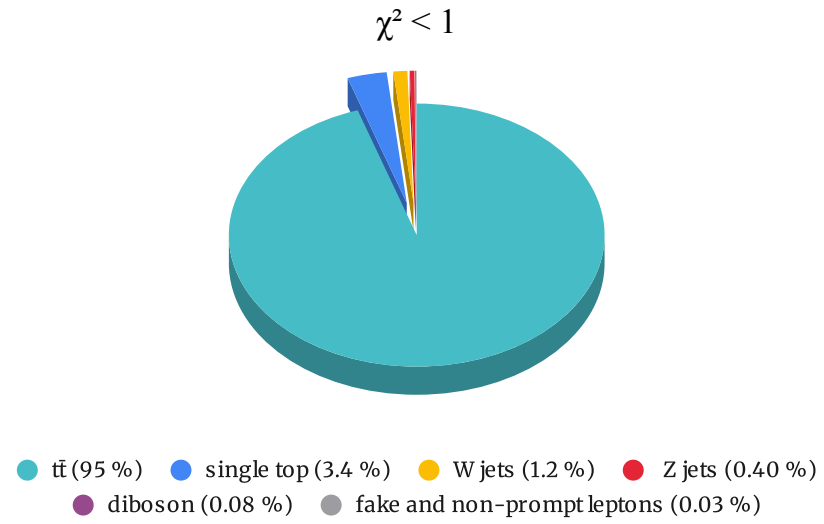


FIGURE 4.2: The Run 2 distributions of the reconstructed jets in data (without JES *in situ* correction) and MC in the *Inclusive* region with no p_T or η cut on the jets from W boson. (A) p_T of the leading jet in logarithmic scale, (B) η of the leading jet, (C) p_T of the second leading jet in logarithmic scale and (D) η of the second leading jet. The uncertainty represents all considered systematic uncertainties as described in Section ???. The bottom panels show the ratio of the data to the MC prediction.



(A)



(B)

FIGURE 4.3

4.1 Hadronic W -boson kinematics reconstruction

At LO, the topology of the single-lepton $t\bar{t}$ decay consists of four jets, two of which are b -jets. Due to QCD gluon radiation and gluon splitting, additional jets may be present in the production or decay at higher orders, which are predominantly gluon or light-flavour jets. To reconstruct the invariant mass of the hadronically decaying W boson, the two jets from its decay need to be identified, that is, the reconstructed jets need to be matched to the W -boson decay products.

There is no unique algorithm for this matching and several algorithms have been developed, such as those introduced in [13] or [14]. In this analysis, a simple χ^2 algorithm is used for the matching. The algorithm takes advantage of the known masses of the W boson and the top quark. It considers possible permutations of up to six highest p_T reconstructed jets and selects the one that gives the best agreement with the expected W -boson and top-quark masses. The level of agreement is quantified by the χ^2 value, defined as

$$\chi^2 = \frac{(m_t^{\text{reco}} - m_t^{\text{fixed}})^2}{\Sigma_t^2} + \frac{(m_W^{\text{reco}} - m_W^{\text{fixed}})^2}{\Sigma_W^2}, \quad (4.1)$$

where m_t^{reco} is the invariant mass of the reconstructed three-jet system from the top-quark decay, m_t^{fixed} is the fixed top-quark mass set to 172.5 GeV and the constant in the denominator Σ_t accounts for the decay width of the top quark, as well as for its approximate experimental resolution and its value is set to 35 GeV. Similarly, m_W^{reco} is the invariant mass of the reconstructed two-jet system from the W -boson decay, m_W^{fixed} is the fixed W -boson mass set to 80.38 GeV, and Σ_W accounts for the W -boson decay width and its approximate experimental resolution and its value is set to 25 GeV. It was tested that changing the Σ_t and Σ_W values to 15 GeV and 10 GeV, respectively and changing the χ^2 requirement to preserve the selection efficiency, does not significantly impact the reconstruction performance. If there are more than six jets in an event, at least two b -tagged jets (ordered in p_T) are always considered and then up to four other jets (ordered in p_T) are considered. Only the permutations where the b -tagged jets are in the position of a true b -jet (i.e. not in the position of the light jets from the W decay) are considered. Permutations that lead to invariant changes, e.g. if only the two light jets from the W are swapped, are skipped to save computing time. The permutation with the lowest χ^2 value is considered to be the correct one. In order to improve the matching efficiency, only events satisfying $\chi^2 < 1$ for the best permutation

are considered further in the analysis. This requirement removes approximately half of the events satisfying all the other selection criteria.

Figure 4.4 shows the comparison of W -boson mass distribution between data and prediction in Run 2 and Run 3 in the *Inclusive* region, for data with and without the *in situ* correction applied.

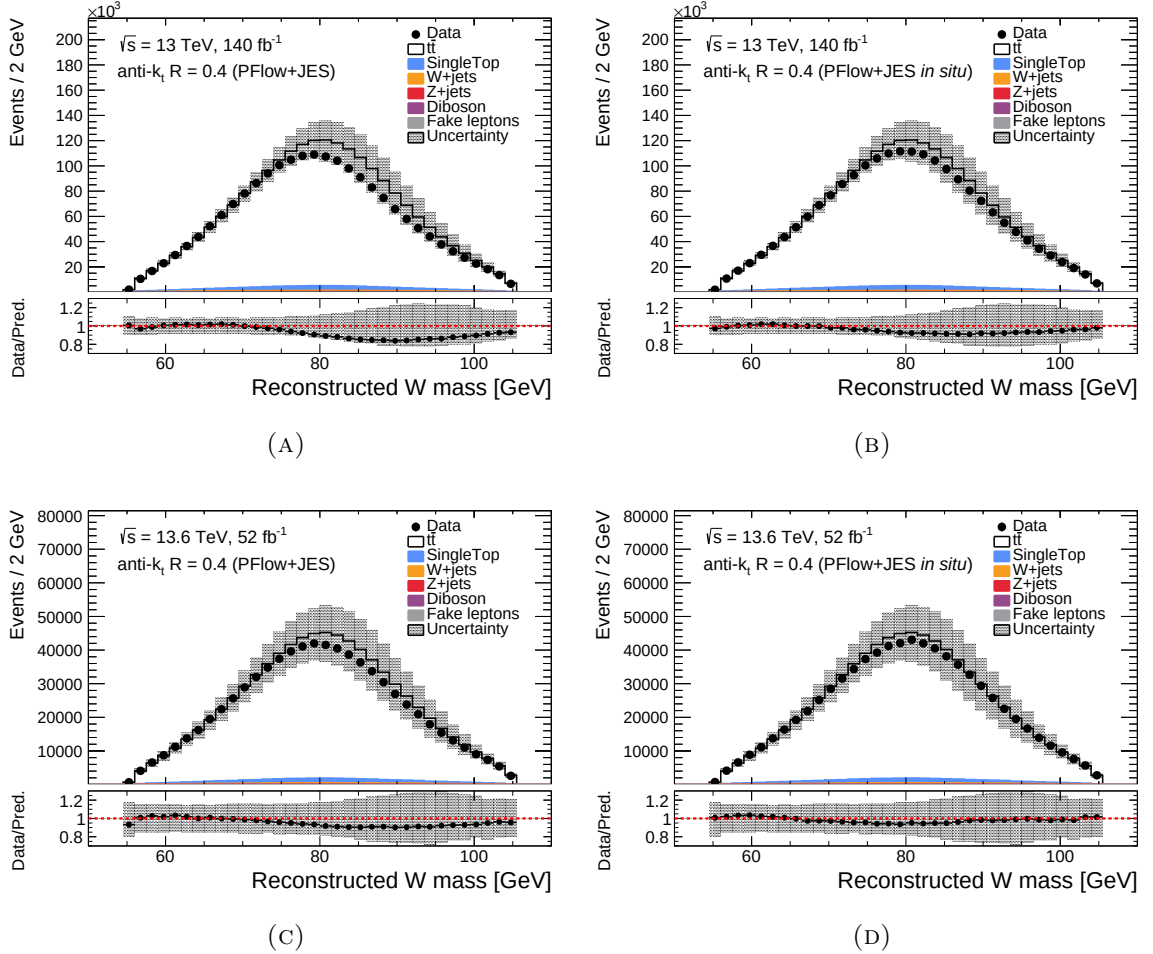


FIGURE 4.4: The Run 2 (top) and Run 3 (bottom) distributions of reconstructed W -boson mass in data and MC in the *Inclusive* region with no p_T or η cut on the jets from W boson, without (left) and with (right) the JES *in situ* calibration applied to data. The uncertainty represents all considered systematic uncertainties as described in Section ???. The bottom panels show the ratio of the data to the MC prediction.

The W -boson mass peak does not match well between the prediction and data without the *in situ* corrections, suggesting that JES needs to be corrected to match data well. However, the W -boson mass peak distribution is reasonably well covered

by the provided uncertainty band due to a large uncertainty originating from the $t\bar{t}$ modelling (mainly PYTHIA versus HERWIG comparison).

4.2 Analysis regions

The JES and JER *in situ* corrections are provided as a function of jet p_T , and only jets with $|\eta| < 0.8$ are used for *in situ* jet calibration in ATLAS. Building on the *Inclusive* selection, the analysis is further divided into analysis regions defined by the p_T and $|\eta|$ of the two jets from the hadronic W -boson decay. Six p_T bins are considered in this measurement: 20 – 35 GeV, 35 – 50 GeV, 50 – 70 GeV, 70 – 100 GeV, 100 – 150 GeV, and 150 – 200 GeV. The discrete correction factors obtained for each p_T bin will be used to derive a continuous correction function via interpolation. The region-specific jet p_T and $|\eta|$ criteria are applied on top of all the selection criteria described up to this point. Table 4.1 summarises the different selection criteria applied in the analysis.

TABLE 4.1: Summary of event selection criteria applied in different stages of the analysis for data and MC simulation. Preselection criteria are applied before forward-folding, while the Inclusive and Analysis region selection criteria are applied to simulated events after forward-folding.

Analysis stage	Data	MC simulation
Preselection	≥ 4 jets ($p_T > 20$ GeV)	≥ 4 jets ($p_T > 15$ GeV)
	≥ 2 b -tagged jets	≥ 2 b -tagged jets
	$= 1$ lepton ($p_T > 27$ GeV)	$= 1$ lepton ($p_T > 27$ GeV)
	no E_T^{miss} requirement	no E_T^{miss} requirement
Inclusive region	≥ 4 jets ($p_T > 20$ GeV)	
	≥ 2 b -tagged jets ($p_T > 30$ GeV)	
	$= 1$ lepton ($p_T > 27$ GeV)	
	$E_T^{\text{miss}} > 30$ GeV	
	2 jets from hadronic W -boson decay ($\chi^2 < 1$)	
Analysis regions	Categorization based on p_T and $ \eta $ of the two jets from hadronic W -boson decay	

Two groups of analysis regions are used, the *diagonal* regions, where both jets from W -boson decay have $|\eta| < 0.8$ and fall into the same p_T bin, and the *off-diagonal* regions, where both jets from W -boson decay have $|\eta| < 0.8$ but belong to different p_T bins. Figure 4.5 illustrates the definition of the analysis regions, along with the number of selected events in Run 2 data without the *in situ* correction in each region. The diagonal-region where both jets have p_T between 20 and 35 GeV is excluded from

the measurement, as there are virtually no jets from the W -boson decay due to the kinematic restriction of the p_T and the jet $|\eta|$ requirements.

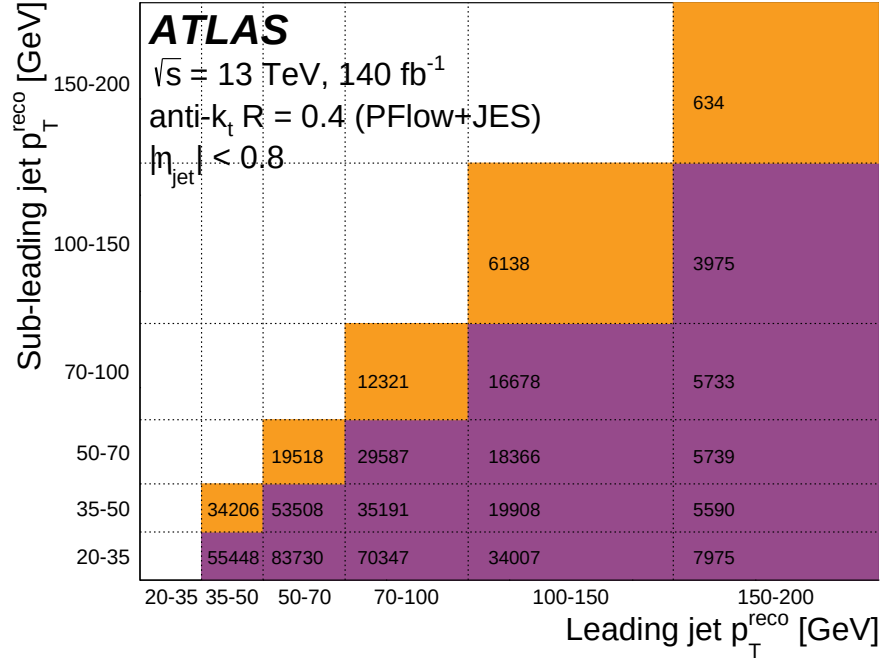


FIGURE 4.5: Definition of the analysis regions based on reconstructed p_T and $|\eta|$ of the two jets identified to originate from the W -boson decay. The two jets from W boson are required to have $|\eta| < 0.8$. The regions shown on the diagonal are the regions where both jets from the W boson belong to the same reconstructed p_T bin, off-diagonal are the regions where the two jets belong to different reconstructed p_T bins. The diagonal region with both jets having reconstructed p_T between 20 and 35 GeV is not used due to kinematic restrictions. The numbers in the bins represent the number of selected events in Run 2 data without the *in situ* correction. [15]

The Run 2 and Run 3 comparisons of W -boson mass distribution between data and prediction for all the analysis regions are provided in Appendix ??, for both versions of data, with and without the *in situ* correction applied. In general, a better agreement between data and prediction is seen for the Run 3 distributions of the W -boson mass, suggesting that JES is better modelled in the Run 3 simulation, possibly due to the improvements in the GEANT4 simulation between Run 2 and Run 3 mentioned in [16].

4.3 W -boson reconstruction efficiency

A jet is considered *matchable* if there is a parton from W -boson decay within $\Delta R < 0.4$ of the reconstructed jet, that can be uniquely assigned. Only unique matches are

considered, i.e. if one parton falls within $\Delta R < 0.4$ of multiple reconstructed jets, these reconstructed jets are labeled as *not matchable*. If both jets from the W -boson decay can be matched to the partons, the event is considered matchable. The efficiency of the W -boson reconstruction, also referred to as the *matching efficiency*, is defined as the fraction of correctly matched events relative to either all events passing a given event selection or all *matchable* events.

Tables 4.2-4.7 show the matching efficiency in the nominal POWHEG+PYTHIA $t\bar{t}$ sample. The tables show the efficiency in the different diagonal analysis regions relative to the efficiency without the $\chi^2 < 1$ requirement, assuming the nominal JES and JER, for both the Run 2 and Run 3 measurements. The MC statistical uncertainties, arising from the limited number of the simulated events, are not displayed as they are negligible.

The $\chi^2 < 1$ requirement significantly increases the fraction of events in which both jets from the hadronic W -boson decay were identified in a *matchable* event. This fraction increases with jet p_T , reaching $\approx 100\%$ in the $100 - 150$ GeV and $150 - 200$ GeV diagonal regions. However, simulation studies reveal the presence of *not matchable* events, where the unique assignment between both partons and reconstructed jets is fundamentally impossible. The fraction of such events relative to the selected events decreases with higher jet p_T , dropping to $\approx 1\%$ in the $150 - 200$ GeV diagonal region. The matching efficiencies are very consistent between Run 2 and Run 3. The matching procedure from Section 4.1 is repeated for each JES and JER assumption because the change in jet p_T can result in different pair of jets being identified as originating from W -boson decay.

TABLE 4.2: Matching efficiency in the 20 - 35 GeV diagonal analysis region for events with $\chi^2 < 1$ and for events with no additional χ^2 requirement, in the Run 2 and Run 3 simulated $t\bar{t}$ events. Fraction of matchable events relative to all events passing the selection is shown as well as the fraction of events with correctly matched jet(s) relative to matchable events or relative to all events passing the selection.

20 - 35 GeV				
20 - 35 GeV	Run 2		Run 3	
Category	$\chi^2 < 1$	No χ^2 cut	$\chi^2 < 1$	No χ^2 cut
Both jets matchable relative to all	49%	27%	49%	28%
Jet 1 matched and Jet 2 failed relative to matchable	21%	31%	21%	31%
Jet 1 failed and Jet 2 matched relative to matchable	9%	14%	10%	15%
Both jets matched relative to matchable	56%	40%	54%	38%
Jet 1 matched and Jet 2 failed relative to all	10%	8%	10%	9%
Jet 1 failed and Jet 2 matched relative to all	5%	4%	5%	4%
Both jets matched relative to all	28%	11%	27%	10%

TABLE 4.3: Matching efficiency in the 35 - 50 GeV diagonal analysis region for events with $\chi^2 < 1$ and for events with no additional χ^2 requirement, in the Run 2 and Run 3 simulated $t\bar{t}$ events. Fraction of matchable events relative to all events passing the selection is shown as well as the fraction of events with correctly matched jet(s) relative to matchable events or relative to all events passing the selection.

35 - 50 GeV				
	Run 2		Run 3	
Category	$\chi^2 < 1$	No χ^2 cut	$\chi^2 < 1$	No χ^2 cut
Both jets matchable relative to all	74%	66%	74%	66%
Jet 1 matched and Jet 2 failed relative to matchable	13%	15%	13%	15%
Jet 1 failed and Jet 2 matched relative to matchable	6%	7%	7%	8%
Both jets matched relative to matchable	72%	69%	72%	69%
Jet 1 matched and Jet 2 failed relative to all	10%	10%	10%	10%
Jet 1 failed and Jet 2 matched relative to all	5%	5%	5%	5%
Both jets matched relative to all	54%	46%	53%	45%

TABLE 4.4: Matching efficiency in the 50 - 70 GeV diagonal analysis region for events with $\chi^2 < 1$ and for events with no additional χ^2 requirement, in the Run 2 and Run 3 simulated $t\bar{t}$ events. Fraction of matchable events relative to all events passing the selection is shown as well as the fraction of events with correctly matched jet(s) relative to matchable events or relative to all events passing the selection.

50 - 70 GeV				
Category	Run 2		Run 3	
	$\chi^2 < 1$	No χ^2 cut	$\chi^2 < 1$	No χ^2 cut
Both jets matchable relative to all	84%	72%	84%	71%
Jet 1 matched and Jet 2 failed relative to matchable	7%	11%	8%	11%
Jet 1 failed and Jet 2 matched relative to matchable	4%	6%	4%	6%
Both jets matched relative to matchable	84%	75%	83%	75%
Jet 1 matched and Jet 2 failed relative to all	6%	8%	6%	8%
Jet 1 failed and Jet 2 matched relative to all	3%	4%	3%	4%
Both jets matched relative to all	70%	54%	70%	53%

TABLE 4.5: Matching efficiency in the 70 - 100 GeV diagonal analysis region for events with $\chi^2 < 1$ and for events with no additional χ^2 requirement, in the Run 2 and Run 3 simulated $t\bar{t}$ events. Fraction of matchable events relative to all events passing the selection is shown as well as the fraction of events with correctly matched jet(s) relative to matchable events or relative to all events passing the selection.

70 - 100 GeV				
Category	Run 2		Run 3	
	$\chi^2 < 1$	No χ^2 cut	$\chi^2 < 1$	No χ^2 cut
Both jets matchable relative to all	91%	76%	91%	76%
Jet 1 matched and Jet 2 failed relative to matchable	3%	7%	3%	8%
Jet 1 failed and Jet 2 matched relative to matchable	2%	4%	2%	5%
Both jets matched relative to matchable	92%	82%	92%	81%
Jet 1 matched and Jet 2 failed relative to all	3%	6%	3%	6%
Jet 1 failed and Jet 2 matched relative to all	2%	3%	2%	4%
Both jets matched relative to all	84%	62%	84%	62%

TABLE 4.6: Matching efficiency in the 100 - 150 GeV diagonal analysis region for events with $\chi^2 < 1$ and for events with no additional χ^2 requirement, in the Run 2 and Run 3 simulated $t\bar{t}$ events. Fraction of matchable events relative to all events passing the selection is shown as well as the fraction of events with correctly matched jet(s) relative to matchable events or relative to all events passing the selection.

100 - 150 GeV				
Category	Run 2		Run 3	
	$\chi^2 < 1$	No χ^2 cut	$\chi^2 < 1$	No χ^2 cut
Both jets matchable relative to all	96%	80%	96%	79%
Jet 1 matched and Jet 2 failed relative to matchable	1%	5%	1%	5%
Jet 1 failed and Jet 2 matched relative to matchable	1%	3%	1%	4%
Both jets matched relative to matchable	96%	86%	96%	84%
Jet 1 matched and Jet 2 failed relative to all	1%	4%	1%	4%
Jet 1 failed and Jet 2 matched relative to all	1%	3%	1%	3%
Both jets matched relative to all	92%	68%	92%	66%

TABLE 4.7: Matching efficiency in the 150 - 200 GeV diagonal analysis region for events with $\chi^2 < 1$ and for events with no additional χ^2 requirement, in the Run 2 and Run 3 simulated $t\bar{t}$ events. Fraction of matchable events relative to all events passing the selection is shown as well as the fraction of events with correctly matched jet(s) relative to matchable events or relative to all events passing the selection.

150 - 200 GeV				
Category	Run 2		Run 3	
	$\chi^2 < 1$	No χ^2 cut	$\chi^2 < 1$	No χ^2 cut
Both jets matchable relative to all	99%	77%	98%	76%
Jet 1 matched and Jet 2 failed relative to matchable	1%	4%	1%	5%
Jet 1 failed and Jet 2 matched relative to matchable	0%	3%	1%	4%
Both jets matched relative to matchable	98%	84%	98%	84%
Jet 1 matched and Jet 2 failed relative to all	1%	3%	1%	3%
Jet 1 failed and Jet 2 matched relative to all	0%	3%	1%	3%
Both jets matched relative to all	97%	65%	96%	64%

4.4 Re-evaluating the event selection

Changes in the reconstructed jet p_T can also affect whether an event passes or fails the event selection criteria. When constructing the W -mass distributions for different assumptions of the JES or JER, it is necessary to evaluate whether each event satisfies the event selection criteria under the corresponding JES or JER assumption. Therefore, a folded version of each event is produced for every JES (s) or JER (r) assumption. In each variation, all jets satisfying the requirements described in Section 2.1 are folded using the same JES or JER assumption, and the event selection criteria are re-evaluated using the resulting *folded* event.

The matching algorithm described in Chapter 4.1 is run for every folded version of event, generating a set of jet indices, identifying the jets from W boson as well as the b -jet, for every s and every r correction assumption.

For the off-diagonal regions, two different JES values (s_1 for the leading jet from the W -boson decay and s_2 for the sub-leading jet) or JER values (r_1 and r_2 , respectively) are associated with each event. In this case, the event selection is re-evaluated using the event folded with $s = s_2$ or $r = r_2$, corresponding to the JES or JER assumption for the sub-leading jet from W -boson decay.

The motivation for this choice is that the jet and b -jet multiplicities can change after folding when the p_T of a jet is close to the 20 GeV or 30 GeV selection thresholds, i.e. the multiplicities are affected by the lower p_T jet. This is an approximation, however it is safe to make this approximation since the JES and JER corrections, s and r , respectively, are not expected to be very different for different p_T bins.

Similarly, E_T^{miss} is evaluated for the event folded with either s_2 or r_2 , even though E_T^{miss} depends on all jets in the event, which may have different p_T values. While variations in E_T^{miss} can affect the event acceptance, their impact on the reconstructed W -boson mass distribution is expected to be negligible. This is supported by the negligible size of the E_T^{miss} systematic uncertainties observed in the measurement (see Section ??). It is therefore also considered safe to make this approximation.

The impact of the folding on E_T^{miss} is estimated approximately by propagating the difference between the folded and original jet four-momenta using the following equation

$$\vec{p}_T^{\text{miss, folded}} = \vec{p}_T^{\text{miss, original}} - \sum_i (\vec{p}_T^{\text{jet folded},i} - \vec{p}_T^{\text{jet original},i}), \quad (4.2)$$

465 where $\vec{p}_T^{\text{miss}}$ is the missing transverse momentum vector and $\vec{p}_T^{\text{jet}}$ is the transverse-
466 momentum vector of jet i .

Chapter 5

W-boson mass templates

Once an event is confirmed to satisfy the selection criteria for a given JES (s) or JER (r) assumption, the W -boson mass is reconstructed from the two jets identified as originating from the W -boson decay under that assumption.

In the off-diagonal regions, where one scale or resolution (s_1 or r_1) is assumed for the leading jet from W and another one (s_2 or r_2) for the sub-leading jet, the event selection criteria are checked for the s_2 or r_2 assumption, as described in Section 4.4. Another approximation is made, when the set of indices identifying the two jets from W and the b -jet (there is a set of indices for each considered s and r value) for $s = s_1$ or $r = r_1$ does not match the indices for $s = s_2$ or $r = r_2$, in which case the indices corresponding to $s = s_2$ or $r = r_2$ are taken.

It is then checked whether the two jets from W -boson decay are well isolated by imposing a cut on the jet isolation variable $f_{\text{iso}} < 0.3$, which rejects jets with large energy fraction outside of the jet radius $R = 0.4$ ¹.

The f_{iso} variable was introduced as the recommended jet isolation criterion for defining jets used in the calibration. It is defined as follows:

$$f_{\text{iso}} = \frac{p_{\text{T}}^{\text{iso}} - \rho A_{\text{iso}}}{p_{\text{T}}^{\text{Jet}} - \rho A_{\text{Jet}}}, \quad (5.1)$$

where $p_{\text{T}}^{\text{iso}}$ is the p_{T} of all constituents within an isolation annulus (green area in Figure 5.1) with radius $R_{\text{iso}} = 1.25R$ around the jet (grey area in Figure 5.1) with radius $R = 0.4$ and $p_{\text{T}}^{\text{Jet}}$ is the p_{T} of all jet constituents. The pile-up contribution is subtracted from both, $p_{\text{T}}^{\text{iso}}$ and $p_{\text{T}}^{\text{Jet}}$. It is obtained by multiplying the jet active area

¹Energy outside the jet radius $R = 0.4$ can arise from radiation associated with the jet that was not clustered into it, or from additional nearby activity that can affect the reconstructed jet energy. In order to reconstruct the W -boson mass reliably, the two jets from its decay should be a good representation of the corresponding decay products. The f_{iso} requirement therefore rejects configurations which introduce possible biases in the reconstructed jet energies and, consequently, in the reconstructed W -boson mass

488 A_{Jet} or the area of the isolation annulus $A_{\text{iso}} = \pi R_{\text{iso}}^2 - A_{\text{Jet}}$ by the pile-up density ρ .
 489 The recommended requirement is $f_{\text{iso}} \leq 0.3$.

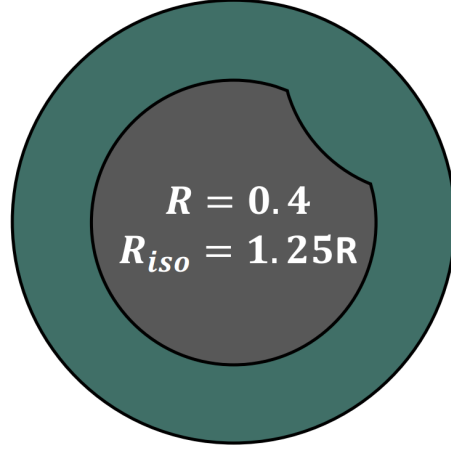


FIGURE 5.1: The definition of isolation annulus (green) with radius R_{iso} around the jet (grey) with radius $R = 0.4$.

490 The invariant mass of W boson m_W for the template distributions is calculated from
 491 folded four-momenta of the two jets identified to originate from W -boson, $\vec{p}_1^{\text{folded}}$ and
 492 $\vec{p}_2^{\text{folded}}$, as:

$$m_W = \sqrt{(E_1^{\text{folded}} + E_2^{\text{folded}})^2 - |\vec{p}_1^{\text{folded}} + \vec{p}_2^{\text{folded}}|^2}. \quad (5.2)$$

493 In diagonal regions, the leading and sub-leading jet from W -boson decay are folded
 494 with the same s or r value, as they belong to the same p_T bin. In off-diagonal region,
 495 the leading jet is folded with s_1 or r_1 and the sub-leading jet with s_2 or r_2 .

496 Before calculating the W -boson mass, the $\chi^2 < 1$ criterion is checked again, using
 497 the folded jets to calculate the χ^2 value defined by Formula 4.1. The b -jet entering
 498 the χ^2 calculation is either folded with the same s or r as one of the light jets from
 499 W -boson decay if it belongs to the same p_T bin or is left unchanged.

500 Finally the events are split into the analysis regions by imposing the p_T and η
 501 requirements corresponding to the analysis regions (defined in Section ??) on the folded
 502 four-momenta of the two jets from W , and the invariant mass of W boson for the
 503 assumed JES or JER is calculated using Equation 5.2.

504 The templates are created from MC simulated $t\bar{t}$ sample as well as all considered
 505 background samples and the contribution from fake leptons is added to each template.

506 In the diagonal regions, 11 templates with different values of s , ranging from 0.95 to
 507 1.05, and 11 templates with different values of r , ranging from 0.60 to 1.4, are produced.

In the off-diagonal regions, 121 templates are produced for the JES variations, corresponding to all combinations of 11 values of s_1 and 11 values of s_2 . Similarly, 121 templates are produced for the JER variations, corresponding to all combinations of 11 values of r_1 and 11 values of r_2 . The ranges of the considered values remain the same as for s and r , that is 0.95–1.05 for the scale variations and 0.6–1.4 for the resolution variations.

Tables 5.1 and 5.2 show the relative fractions of jets identified as originating from the hadronic *W*-boson decay that fail to satisfy the requirements (described in Section 2.1) needed to apply the forward-folding procedure in the Run 2 and Run 3 measurements, respectively.

The fraction of *reco* jets failing the no *truth* lepton within $\Delta R = 0.5$ requirement decreases with growing p_T , same for the unmatched *reco* jet category but here the fraction also increases for the last 150 – 200 GeV region. The fraction of jets failing the *truth* jet isolation criterion is very similar in the first three regions with jet p_T from 20 to 100 GeV, then it grows with growing p_T and it is at 75 % for the last p_T bin 150 – 200 GeV, which is reflected in the increased statistical uncertainty (see Section 6.2) for said region, especially for JER since the resolution is better for higher p_T and thus changing the r value has only a small effect. For JES, even though the separation between different s (JES) assumptions is bad due to low statistical power when only 25 % of jets in the last p_T bin are shifted by the forward-folding formula, it is compensated by the fact that the s variation has a bigger effect for high p_T .

TABLE 5.1: Fractions of jets identified to be from the W boson decay that cannot be forward-folded due to failing some of the required criteria, shown for the different diagonal regions in Run 2 measurement.

jet p_T [GeV]	Inclusive	20 – 35	35 – 50	50 – 70	70 – 100	100 – 150	150 – 200
Category	[%]						
Failed <i>truth</i> lepton in jet	1.4	3.0	1.2	0.5	0.2	0.1	0.0
Failed to match <i>truth</i> jet	2.0	3.4	0.7	0.3	0.2	0.2	1.3
Failed <i>truth</i> jet isolation	4.4	3.7	3.5	3.5	4.4	7.5	74.9
Passed all criteria	92.3	89.9	94.6	95.6	95.2	92.3	23.7

TABLE 5.2: Fractions of jets identified to be from the W boson decay that cannot be forward-folded due to failing some of the required criteria, shown for the different diagonal regions in Run 3 measurement.

jet p_T [GeV]	Inclusive	20 – 35	35 – 50	50 – 70	70 – 100	100 – 150	150 – 200
Category	[%]						
Failed <i>truth</i> lepton in jet	1.4	2.9	1.2	0.5	0.2	0.1	0.0
Failed to match <i>truth</i> jet	1.6	2.8	0.6	0.3	0.2	0.2	1.4
Failed <i>truth</i> jet isolation	4.4	3.9	3.6	3.5	4.6	7.2	74.7
Passed all criteria	92.6	90.5	94.6	95.6	95.1	92.5	23.9

Figures 5.2-5.4 shows the effect of JES and JER variations in one representative diagonal and one representative off-diagonal regions in the Run 2 measurement. The shift of the peak of the reconstructed *W*-mass distribution caused by JES (s , s_1 or s_2 parameter) variation is shown on five template distributions with nominal value of JER ($r = r_1 = r_2 = 1$) and s , s_1 or s_2 ranging from 0.95 to 1.05. The widening of the *W*-mass distribution caused by JER (r , r_1 or r_2 parameter) variation is shown on five template distributions with nominal value of JES ($s = s_1 = s_2 = 1$) and r , r_1 or r_2 ranging from 0.6 to 1.4.

The effect of varying the JES correction factor, s_1 , for the leading jet is more pronounced than that of varying s_2 for the subleading jet, indicating that the JES measurement is more sensitive in the higher- p_T bins. This is because a fixed relative variation of 5% corresponds to a larger absolute change in jet p_T at higher momenta, resulting in a greater separation between the template distributions. The difference in separation between the template distributions for s_1 and s_2 variations is shown in Figure 5.3.

In contrast, the JER measurement is more sensitive in the lower- p_T bins, corresponding to variations of r_2 for the subleading jet. Since the jet p_T resolution is intrinsically poorer at lower p_T , a variation in the resolution produces a larger absolute change in the width of the reconstructed *W*-mass distribution, leading to a better separation between the JER templates. The difference in separation between the template distribution for r_1 and r_2 variations is shown in Figure 5.4.

All the Run 2 *W*-boson mass templates, as well as the Run 3 templates for diagonal regions, are shown in Appendices D and E, respectively.

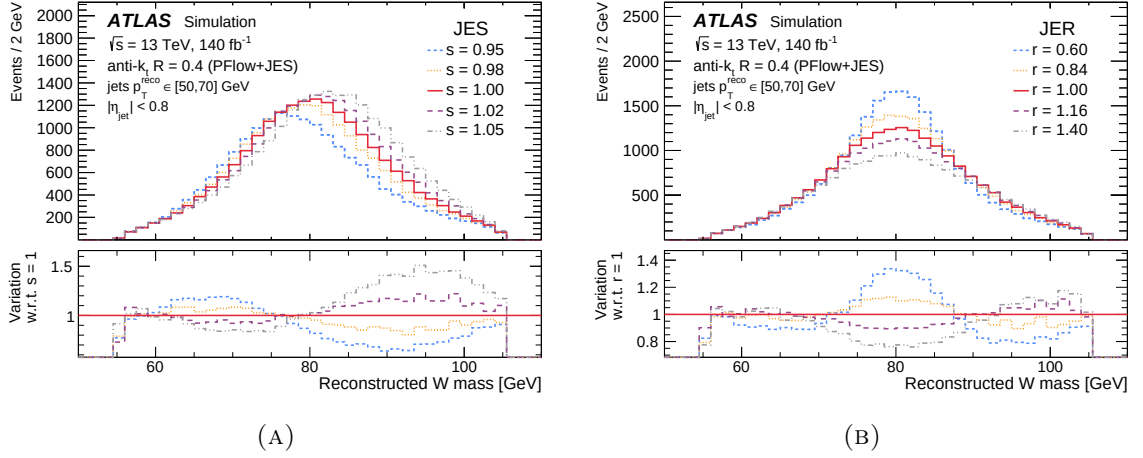


FIGURE 5.2: Reconstructed W -boson mass in the diagonal region with the two jets from the W -boson decay from the 50 – 70 GeV reconstructed p_T bin, in the Run 2 simulation. Templates with different (A) JES and (B) JER assumptions represented by the correction factors s and r from the forward-folding formula are shown. For the JES variations, the JER correction, r , is set to one, and for the JER variations the JES correction, s is set to one. The differences between the total yields for each distribution come from the acceptance effects due to the changes in the reconstructed jet p_T . The bottom panels show the ratios to the template with nominal (A) JES or (B) JER assumption. The solid line in the bottom panels represents the ratio of one. [15]

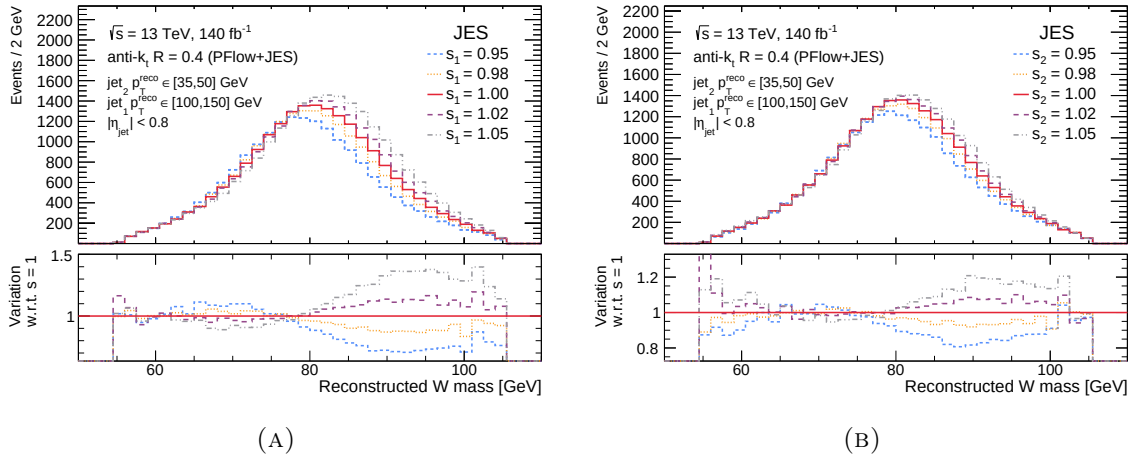


FIGURE 5.3: Reconstructed W -boson mass in a representative off-diagonal region of the Run 2 simulation. The leading jet from the W -boson decay is required to lie in the reconstructed 100–150 GeV p_T bin, while the sub-leading jet lies in the reconstructed 35–50 GeV p_T bin. Templates corresponding to different JES assumptions for (A) the leading jet (s_1 variations, while $s_2 = 1$) and (B) the sub-leading jet (s_2 variations with $s_1 = 1$). The JER correction factor, r , is fixed to unity. [15]

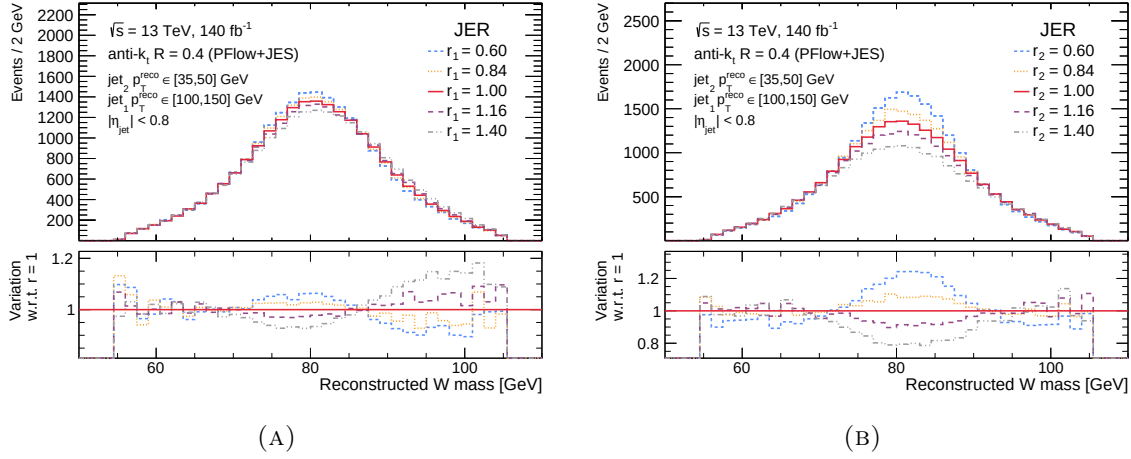


FIGURE 5.4: Reconstructed W -boson mass in a representative off-diagonal region of the Run 2 simulation. The leading jet from the W -boson decay is required to lie in the reconstructed 100–150 GeV p_T bin, while the sub-leading jet lies in the reconstructed 35–50 GeV p_T bin. Templates corresponding to different JER assumptions for (A) the leading jet (r_1 variations, while $r_2 = 1$) and (B) the sub-leading jet (r_2 variations with $r_1 = 1$). The JES correction factor, s , is fixed to unity. [15]

Chapter 6

Extraction of JES and JER

The implementation of the statistical analysis described in this chapter was mostly performed by another member of the analysis team.

In order to extract the optimal JES correction (s) from a fit to data, the mean μ of the W -mass distribution is parametrised as a function of s . Similarly, to extract the optimal JER correction (r) from a fit to data, the standard deviation σ of the W -mass distribution is parametrised as a function of r (Section 6.1).

The motivation to use the μ or σ of the W -mass distribution only, instead of the full distribution, is to simplify the fit setup to allow for two JES (s_1 and s_2) or JER (r_1 and r_2) parameters in the off-diagonal regions, while retaining most of the sensitivity.

Once the parametrisation is obtained, a maximum likelihood fit to μ or σ of W -mass distribution observed in data is performed to extract the JES or JER correction for all regions simultaneously (Section 6.2).

In the JES measurement, the data without JES *in situ* calibration are used, whereas for the JER measurement, the data with *in situ* calibration are used.

6.1 JES and JER parametrisation

For the JES measurement in the diagonal regions, the mean μ of the W -mass template is extracted for every value of s and the obtained values are fitted with a one-dimensional linear function $\mathcal{F}(s)$. In the off-diagonal regions, μ is extracted for every combination of s_1 and s_2 and the obtained values are fitted with two-dimensional quadratic function $\mathcal{F}(s_1, s_2)$.

For the JER measurement in the diagonal regions, the standard deviation σ of the W -mass template is extracted for every value of r and the obtained values are fitted with one-dimensional quadratic function $\mathcal{F}(r)$. In the off-diagonal regions, σ is

extracted for every combination of r_1 and r_2 and the obtained values are fitted with two-dimensional quadratic function $\mathcal{F}(r_1, r_2)$.

The parametrisation was chosen to be as simple as possible while still providing a good description of the data, with a linear form being sufficient for $\mathcal{F}(s)$ and a quadratic form providing a better description for $\mathcal{F}(r)$.

The diagonal and off-diagonal parametrisation functions, $\mathcal{F}(s)$ and $\mathcal{F}(s_1, s_2)$, share a set of six parameters s_i , one for each reconstructed jet p_T bin. Similarly, $\mathcal{F}(r)$ and $\mathcal{F}(r_1, r_2)$ share a set of six parameters r_i .

Example one-dimensional parametrisations, used for the diagonal regions, are shown in Figure 6.1 and two-dimensional parametrisations, used in off-diagonal regions, are shown in Figure 6.2. The Run 2 parametrisations for the remaining analysis regions are shown in Appendix F. The fact that there is linear parametrisation for JES in the diagonal regions shows how strongly is the JES correlated with W mass.

In the JER measurement, the diagonal region with jet p_T between 150 and 200 GeV is excluded from the fit due to a poor statistical power in the separation of the different JER assumed values.

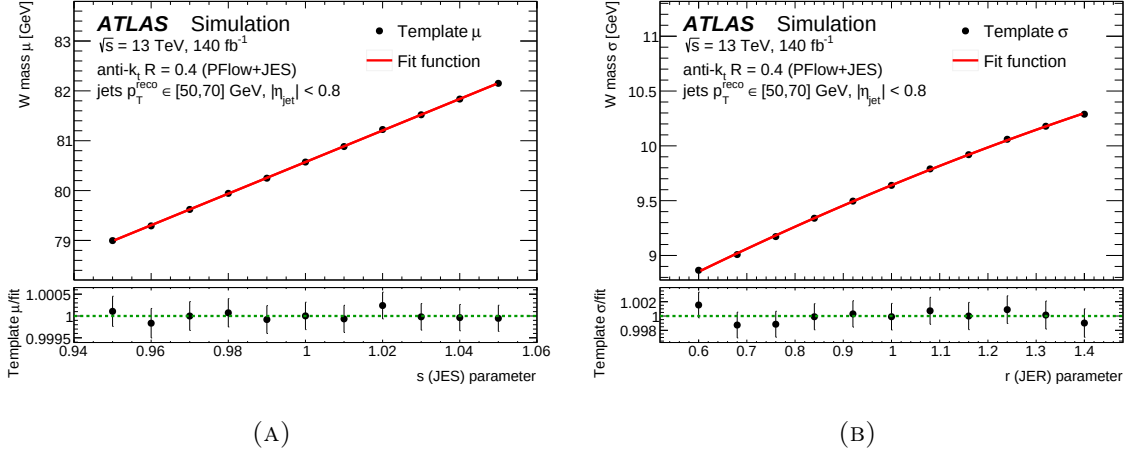


FIGURE 6.1: The mean (A) and standard deviation (B) of the W -mass distribution in a representative diagonal region with reconstructed jet p_T between 50 – 70 GeV as a function of (A) JES (s) or (B) JER (r) parameters for the Run 2 measurement. The fitted function $\mathcal{F}$ is shown by solid red line. The bottom panel shows the ratio of the (A) mean or (B) standard deviation value to the fitted function $\mathcal{F}$. [15]

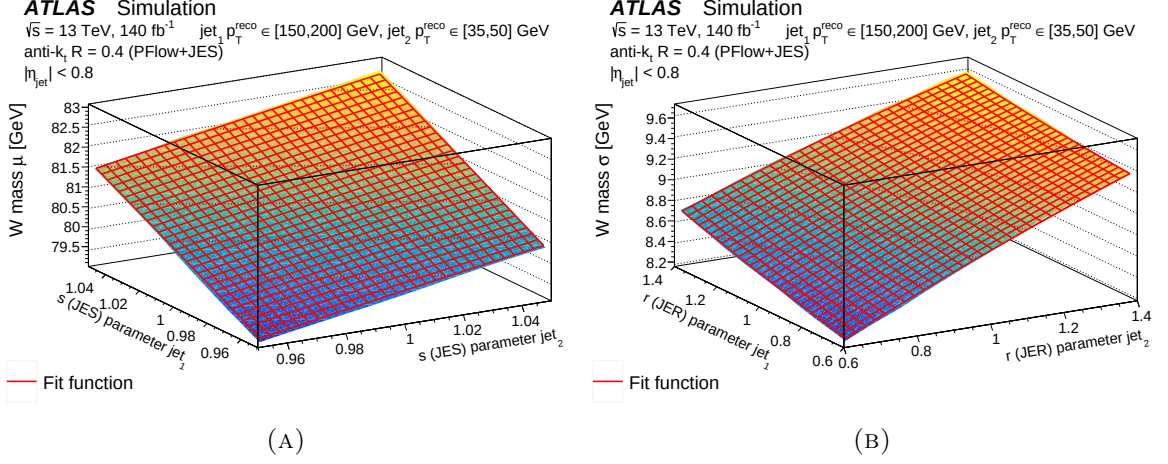


FIGURE 6.2: The mean (A) and standard deviation (B) of the W -mass distribution in a representative off-diagonal region with the leading jet (jet₁) with reconstructed p_T between 150 – 200 GeV and the sub-leading jet (jet₂) with reconstructed p_T between 35 – 50 GeV, as a function of the (A) JES parameters, s_1 and s_2 , and (B) JER parameters, r_1 and r_2 , for the Run 2 measurement. The fitted two-dimensional quadratic function $\mathcal{F}$ is shown in red. [15]

6.2 Fit setup

Once the parametrisation is available, a maximum-likelihood fit¹ to μ or σ of the W -mass distribution reconstructed from recorded data is performed to extract the values of the JES and JER corrections, s_i and r_i , for each p_T bin i simultaneously.

The likelihood is defined as

$$\mathcal{L}(\vec{\mu}^{\text{data}}|\vec{s}) \equiv \prod_{a \in \text{reg}} G[\mu_a | \mathcal{F}_a(\vec{s}), \sigma_{\mu_a}], \quad (6.1)$$

where $\vec{\mu}^{\text{data}}$ is the vector of the μ_a^{data} or σ_a^{data} values measured in data, $\vec{s}$ is the vector of the scale or resolution correction factors s_i or r_i to be estimated, G represents a Gaussian function, $\mathcal{F}_a$ is the parametrisation formula from Section 6.1 for region a that replaces the μ_a or σ_a values in G , and σ_{μ_a} is the data statistical uncertainty on the μ_a or σ_a values in analysis region a for the JES or JER fit, respectively. Gaussian function G is used under assumption that the distributions of μ_a and σ_a are Gaussian. For the JES measurement, the function takes the form:

$$G \propto \exp \left[-\frac{(\mu_a^{\text{data}} - \mathcal{F}_a(s_i))^2}{\sigma_{\mu_a}^2} \right]. \quad (6.2)$$

¹Technically, this is done by minimising the negative logarithm of the likelihood performed by the MINUIT2 library [17] implemented in RooFit [18].

Similarly, for the JER measurement, it takes the form:

$$G \propto \exp \left[\frac{\sigma_a^{\text{data}} - \mathcal{F}_a(r_i)}{\sigma_{\sigma_a}} \right]. \quad (6.3)$$

The fit is performed simultaneously across all analysis regions, indexed by a . Since the regions are statistically independent, the total likelihood function is constructed as a product of individual region likelihoods. Each analysis region corresponds to a particular combination of two p_T bins of the jets from W -boson decay. The fit is performed separately for the JES and JER measurement. In the JES fit, six JES correction factors, s_i ($i = 1, \dots, 6$), are extracted, with one correction factor associated with each jet p_T bin. Similarly, the JER fit extracts six JER correction factors, r_i ($i = 1, \dots, 6$), one for each jet p_T bin. The statistical uncertainty is obtained as the difference between the correction factors for which the likelihood is maximised and the ones for which the likelihood is half of the maximum value (which corresponds to one σ variation).

Chapter 7

Results

In this chapter, the *in situ* jet energy scale (JES) and jet energy resolution (JER) correction factors extracted from the hadronic W -boson mass reconstruction, s_i and r_i , respectively, are presented for each considered p_T bin i . Results for Run 2 and Run 3 are shown, together with their total uncertainties.

In ATLAS, the JES calibration is applied to data while the JER calibration is applied to MC. However, in the presented measurement, the correction factors for the MC simulation were derived for both, JES and JER. Thus, to correct the JES in data, the inverse of the fitted values, alongside their uncertainties, needs to be applied to data.

7.1 Pre-fit and post-fit W -mass distribution parameters

The likelihood fit (Section 6.2) to data is performed separately for JES and JER measurements and for Run 2 and Run 3, using the likelihood model described by Equation 6.1.

For the JES measurements, the fitted parameters are the JES correction factors, s_i , which determine the predicted mean, μ , of the W -mass distribution in each analysis region through the parametrisation $\mathcal{F}$ described in Section 6.1. Analogous procedure is performed in the JER measurement to obtain the JER correction factors, r_i , using the parametrisation of the W -mass distribution width, σ . Figures 7.1 and 7.2 show the distributions entering these fits (*pre-fit* distributions), together with their corresponding *post-fit* values. Figure 7.1 compares the mean, μ , of the original (*pre-fit*) W mass distributions in each analysis region with the corresponding (*post-fit*) value, obtained by substituting the fitted s_i into the parametrisation $\mathcal{F}$. The comparison is presented

separately for the Run 2 and Run 3 JES measurements. Similarly, Figure 7.2 compares the *pre-fit* and *post-fit* values of the W -mass distribution width, σ , obtained in the presented Run 2 and Run 3 JER measurements. It can be seen that the *post-fit* prediction agreement with the observed data improves significantly (especially for the JES fit in Run 2). All remaining discrepancies between the *post-fit* prediction and data are covered by the considered uncertainties.

The uncertainty bands in Figures 7.1 and 7.2 are estimated from *pseudo-experiments*. This approach is adopted because, although the uncertainties on the individual JES and JER correction factors, s_i and r_i , are Gaussian, the propagated uncertainties on μ or σ are non-Gaussian when the parametrisation is non-linear.

For each *pseudo-experiment*, JES or JER correction factors are sampled from Gaussian distributions whose means are given by the fitted correction factors, s_i or r_i , and whose standard deviations corresponds to their total uncertainties. The sampled correction factors are then propagated through the parametrisation described in Section 6.1 to obtain the corresponding values of the W -mass distribution mean, μ , or standard deviation, σ . After performing 1000 *pseudo-experiments*, distribution of the resulting μ and σ values are obtained. The uncertainty bands are defined by the standard deviation of these distributions.

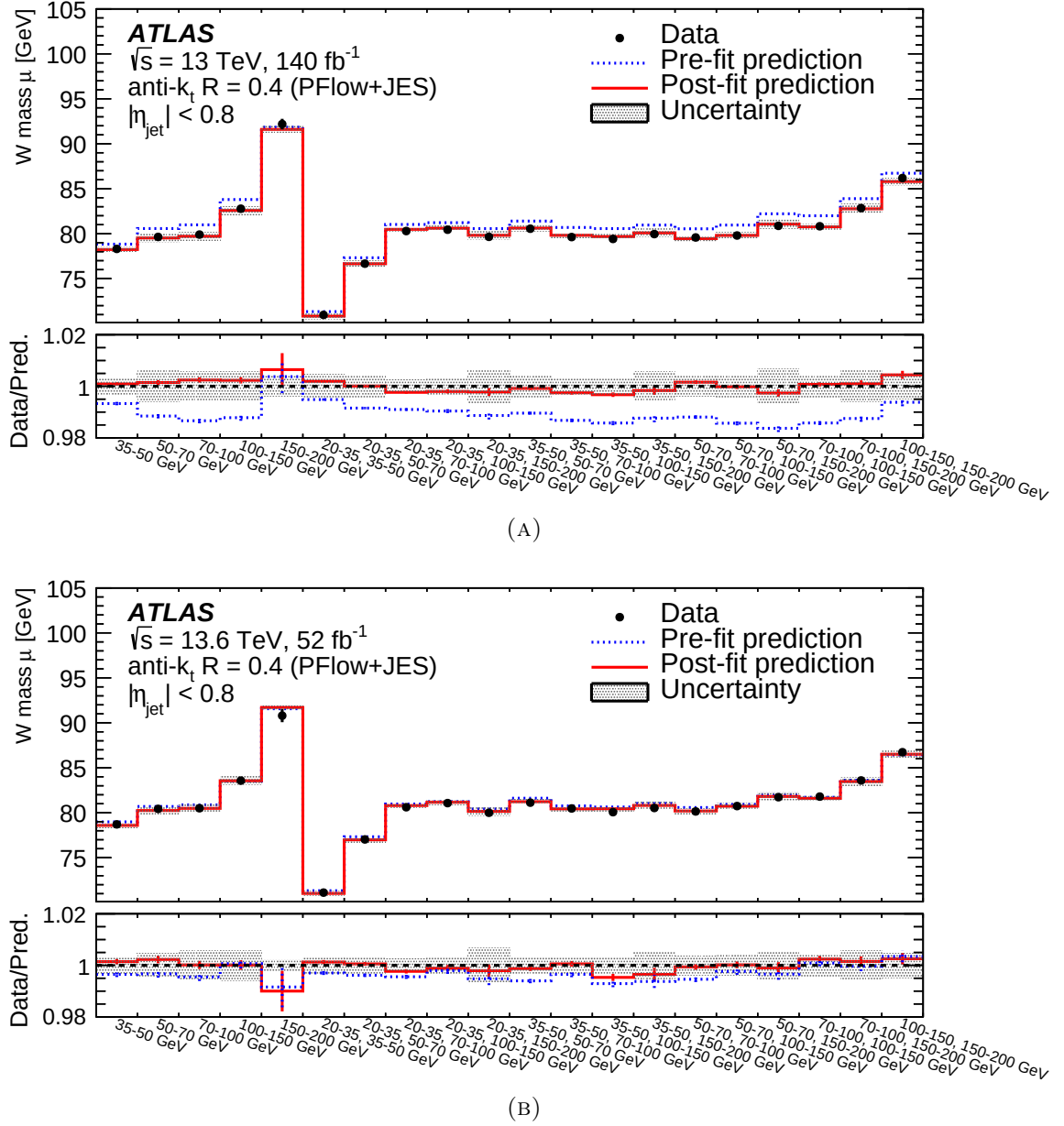


FIGURE 7.1: Mean of the W -mass distribution shown for the analysis regions defined in Section ?? in the (A) Run 2 and (B) Run 3 JES measurement. The values obtained in data are compared to those obtained in simulation without the JES corrections (*pre-fit*) and those after applying the fitted JES corrections, s_i , (*post-fit*). The first five bins show the diagonal regions, while the remaining bins show the off-diagonal regions. The bottom panel shows the ratio of the data without the *in situ* JES calibration (PFlow+JES) to the *pre-fit* (dotted line) and *post-fit* (solid line) predictions, with the dashed line representing the ratio of one.

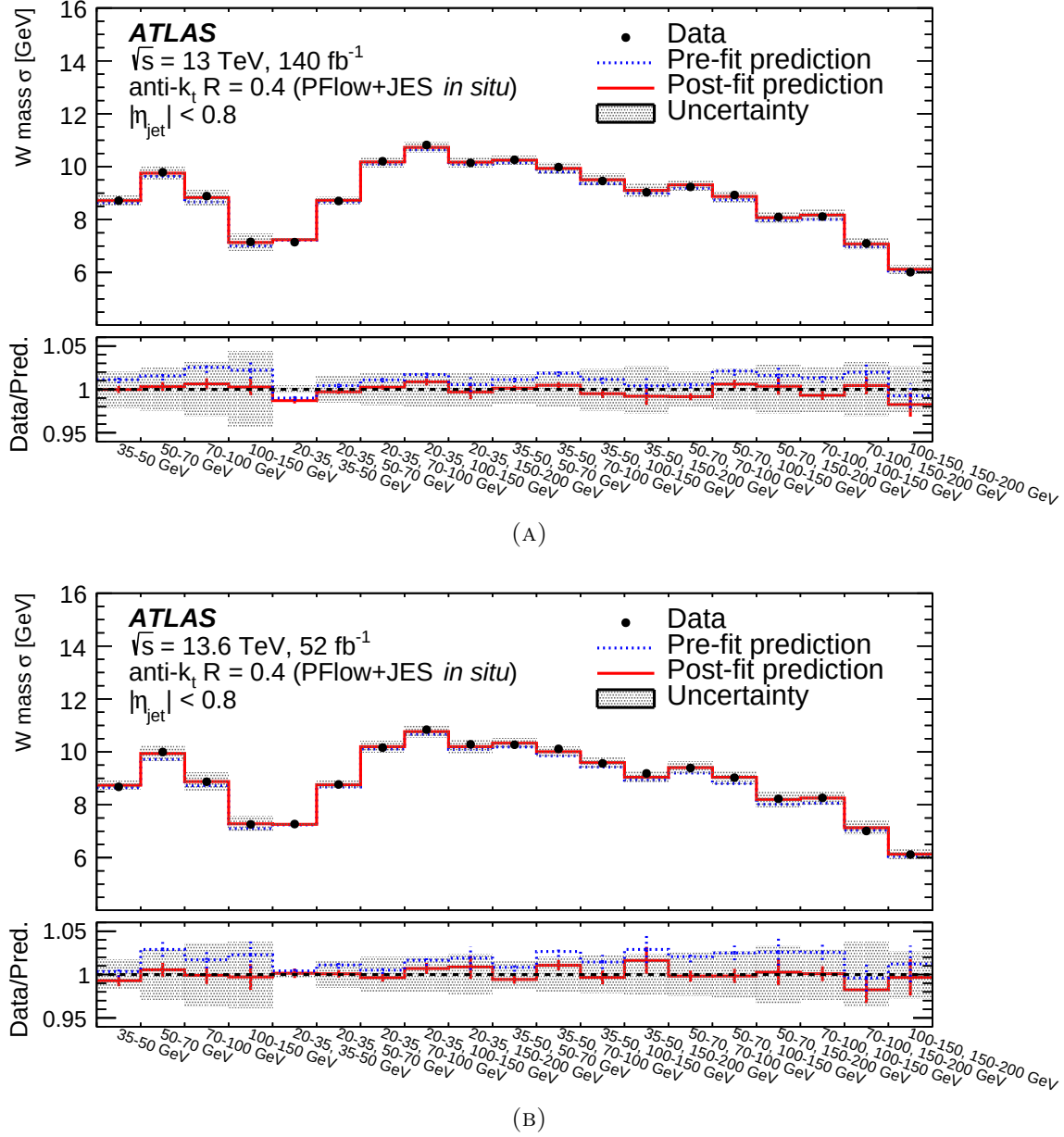


FIGURE 7.2: Standard deviation of the W -mass distribution shown for the analysis regions defined in Section ?? in the (A) Run 2 and (B) Run 3 JER measurement. The values obtained in data are compared to those obtained in simulation without the JER corrections (*pre-fit*) and those after applying the fitted JER corrections, r_i , (*post-fit*). The first five bins show the diagonal regions, while the remaining bins show the off-diagonal regions. The bottom panel shows the ratio of the data without the *in situ* JES calibration (PFlow+JES) to the *pre-fit* (dotted line) and *post-fit* (solid line) predictions, with the dashed line representing the ratio of one.

7.2 JES and JER correction factors

The fitted central values of the JES correction factors, s_i , with their uncertainties, are shown in Figure 7.3 for Run 2 and Run 3. Although the overall uncertainties are similar between the Run 2 and Run 3 measurements, the central values for the measured JES corrections are closer to unity in the Run 3 measurement, suggesting an improved detector simulation (GEANT4) in Run 3. Additionally, the JES corrections are larger for low p_T bins compared to high p_T bins, which could be attributed to the QCD effects that are more difficult to model for low p_T jets. The obtained JER correction factors are shown in Figure 7.3, also with the uncertainties and for both, Run 2 and Run 3. The JES and JER corrections are derived in discrete p_T bins, however a continuous correction function will be later constructed by interpolation across these bins, in order to apply the derived corrections. The discrete correction factors must be therefore assigned to a representative p_T value. Since the jet p_T spectrum falls steeply, as shown in Figure 7.5, the jet population within any p_T bin is skewed towards the lower edge and the average correction derived for a bin corresponds to the mean p_T in the bin rather than to its geometric centre. Bin-centre corrections are therefore applied to the horizontal values in Figures 7.3 and 7.4, to shift the derived corrections from bin centre to the mean of the jet p_T distribution of the jets from the W -boson decay in each p_T bin in data. Figure 7.5 shows the p_T distribution of jets from W -boson decays in data, compared with the corresponding distribution obtained from the MC simulation, for all considered p_T bins.

The central values of the obtained correction factors and the statistical, systematic and total uncertainties in Run 2 and Run 3 are summarised in Tables 7.1 and 7.2, for the JES and JER estimate, respectively. For the JES, the 20 – 35 GeV p_T bin reaches about 5(7)% uncertainty, while the uncertainty drops to about 1.6(1.3)% for the 35 – 50 GeV bin and reaches about 0.9(0.9)% for the 100 – 150 GeV bin in the case of excluding (including) pre-recommendation JES and JER components. For the JER, the uncertainty ranges between 14 to 22% in the whole considered p_T range.

Figure 7.6 shows the statistical correlations between the correction factors associated with the individual reconstructed jet p_T bins. These correlations arise from the use of the off-diagonal analysis regions, which simultaneously constrain the correction factors associated with the p_T bins of the leading and sub-leading jets. The correlations are obtained from the covariance matrix of the fitted parameters, which is determined from the inverse of the Hessian matrix at the likelihood minimum. The correlation coefficients

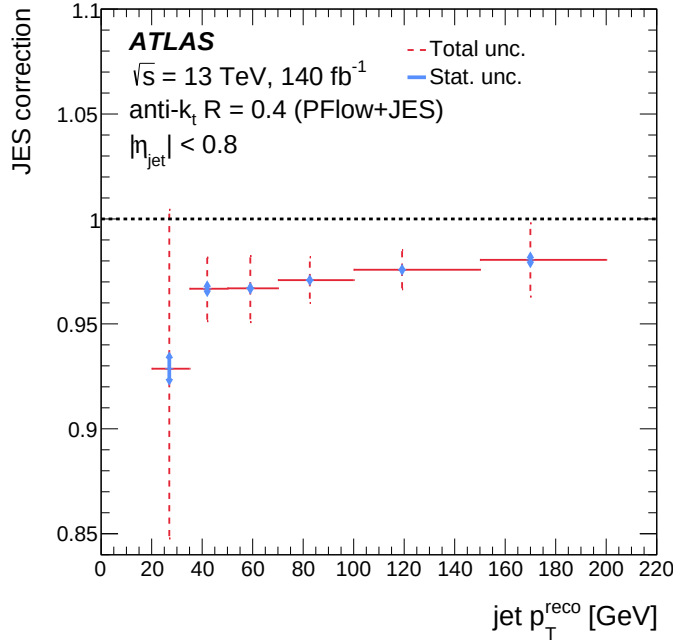
TABLE 7.1: The JES correction factors, s_i , obtained from the fit to data for each reconstructed jet p_T bin i , are shown for the Run 2 and Run 3 measurement, together with their uncertainties. The total uncertainty is calculated by adding the statistical and systematic uncertainties in quadrature and includes the pre-recommendation JER uncertainties.

Region i [GeV]	Central		Statistical		Systematic		Total	
	Run 2	Run 3	Run 2	Run 3	Run 2	Run 3	Run 2	Run 3
$20 < p_T < 35$	0.929	0.950	0.0064	0.0119	0.0780	0.0730	0.0783	0.0739
$35 < p_T < 50$	0.967	0.979	0.0023	0.0037	0.0150	0.0150	0.0151	0.0154
$50 < p_T < 70$	0.967	0.985	0.0013	0.0020	0.0157	0.0120	0.0157	0.0121
$70 < p_T < 100$	0.971	0.991	0.0011	0.0017	0.0108	0.0105	0.0109	0.0107
$100 < p_T < 150$	0.976	1.001	0.0012	0.0018	0.0092	0.0097	0.0093	0.0098
$150 < p_T < 200$	0.981	1.002	0.0023	0.0036	0.0173	0.0162	0.0174	0.0165

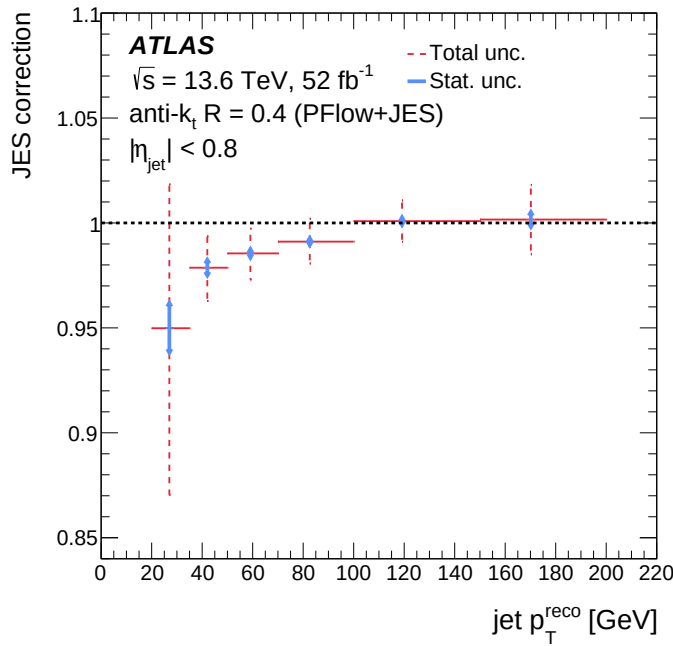
TABLE 7.2: The JER correction factors, r_i , obtained from the fit to data for each reconstructed jet p_T bin i , are shown for the Run 2 and Run 3 measurement, together with their uncertainties. The total uncertainty is calculated by adding the statistical and systematic uncertainties in quadrature and includes the pre-recommendation JES uncertainties.

Region i [GeV]	Central		Statistical		Systematic		Total	
	Run 2	Run 3	Run 2	Run 3	Run 2	Run 3	Run 2	Run 3
$20 < p_T < 35$	1.085	1.101	0.0214	0.0373	0.2029	0.2323	0.2041	0.2352
$35 < p_T < 50$	1.082	1.088	0.0178	0.0299	0.1784	0.1616	0.1793	0.1644
$50 < p_T < 70$	1.065	1.129	0.0181	0.0296	0.1396	0.1748	0.1408	0.1773
$70 < p_T < 100$	1.081	1.085	0.0204	0.0333	0.1405	0.1807	0.1420	0.1838
$100 < p_T < 150$	1.078	1.103	0.0275	0.0450	0.1839	0.1641	0.1859	0.1701
$150 < p_T < 200$	0.997	0.995	0.0765	0.1220	0.2116	0.2528	0.2250	0.2808

are obtained by dividing the corresponding covariance terms by the uncertainties of the two parameters. These correlations need to be taken into account when combining the correction factors s_i and r_i obtained from the hadronic W -boson decay measurement with the corrections obtained using the standard *in situ* techniques (see Section ??). It is worth noting that in the JES measurement, there are relatively large anti-correlations between s_1 and s_2 or s_3 . This is because the diagonal 20 – 35 GeV bin is not included in the measurement and consequently, s_1 is only estimated from the off-diagonal regions, where the effects of a JES variation in the 20 – 35 GeV bin are difficult to distinguish from the effect of JES variation in the 35 – 50 GeV or 50 – 70 GeV bin, due to the effect of JES variation on low p_T jets being smaller compared to high p_T jets (see Appendix ??).



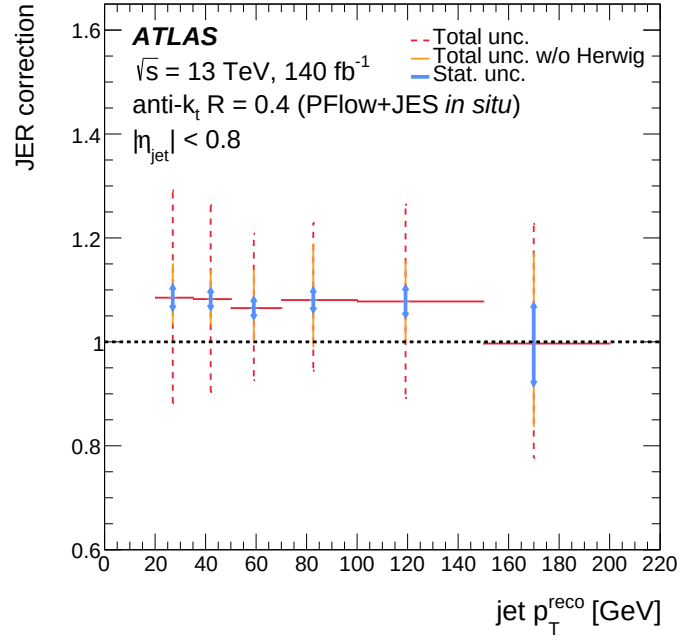
(A)



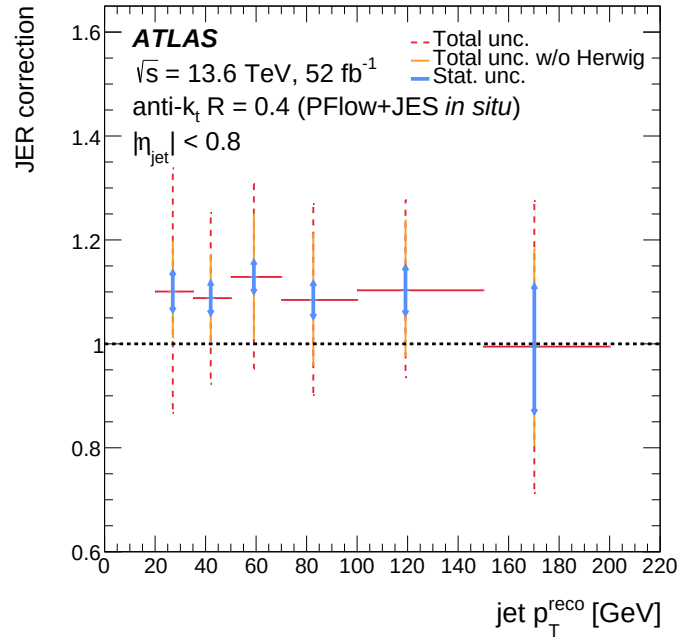
(B)

FIGURE 7.3: The JES correction factors s_i and the corresponding uncertainties shown for the different reconstructed jet p_T bins, in the Run 2 (A) and Run 3 (B) measurement. The statistical uncertainty is shown with a thick solid bar with arrows on both ends, while the total uncertainty is shown with a thin dashed line. The horizontal values are taken from the mean of the reconstructed jet p_T distribution of the jets from W -boson decay in each p_T bin.

The dashed horizontal line represents JES correction $s_i = 1$, i.e. no correction needed.



(A)



(B)

FIGURE 7.4: The JER correction factors r_i and the corresponding uncertainties shown for the different reconstructed jet p_T bins, in the Run 2 (A) and Run 3 (B) measurement. The statistical uncertainty is shown with a thick solid bar with arrows on both ends, while the total uncertainty is shown with a thin dashed line. The thin solid line represents the total uncertainty without the PYTHIA versus HERWIG component. The horizontal values are taken from the mean of the reconstructed jet p_T distribution of the jets from W -boson decay in each p_T bin. The dashed horizontal line represents JER correction $r_i = 1$, i.e. no correction needed.

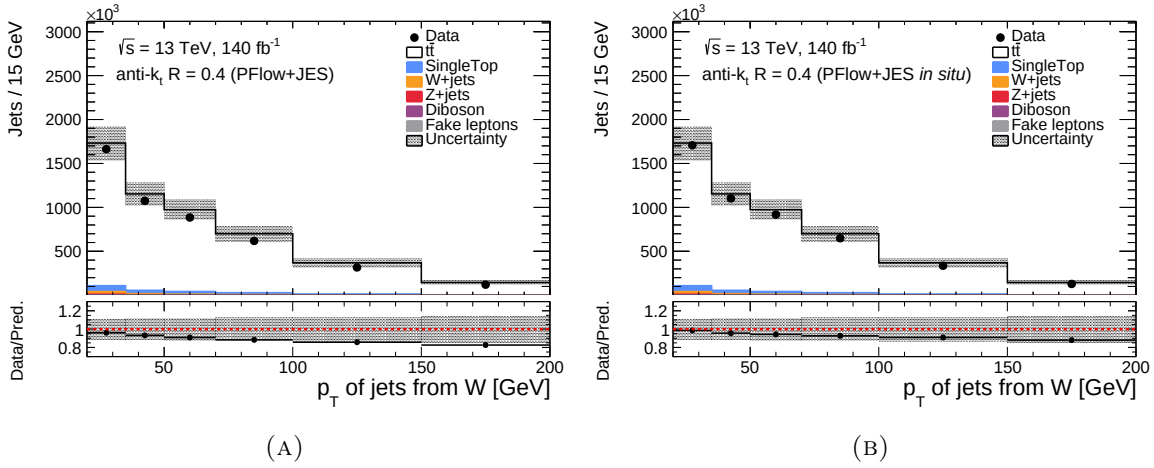


FIGURE 7.5: The p_T distribution of jets coming from the W -boson decay in data without (A) and with (B) *in situ* JES calibration compared with the corresponding distributions in simulation, across all considered reconstructed jet p_T bins in the Run 2 measurement.

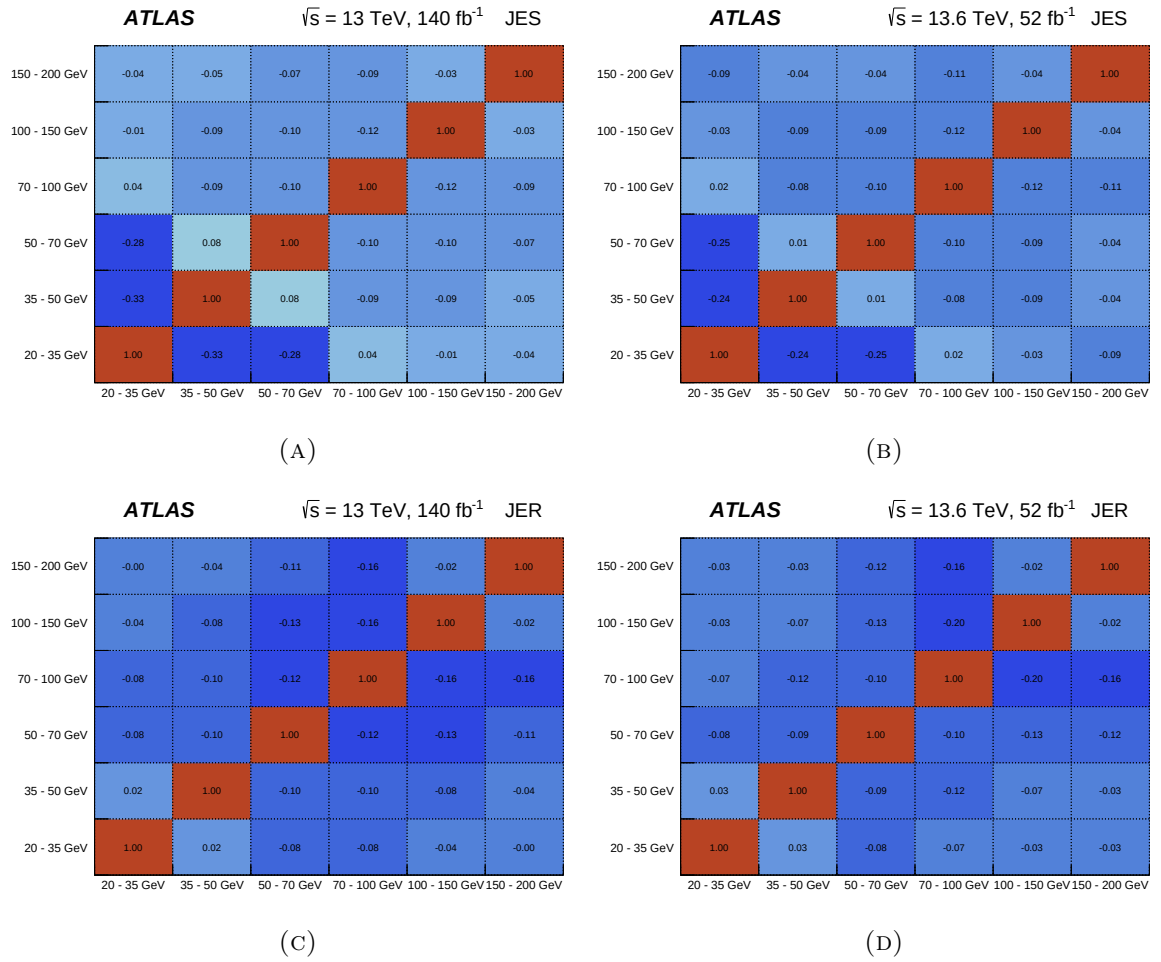


FIGURE 7.6: The statistical correlations between different reconstructed jet p_T bins estimated from the fit in the Run 2 (left) and Run 3 (right) JES (top) and JER (bottom) measurements.

7.3 Fit validation

Figures 7.7 and 7.8 compare the *pre-fit* and *post-fit* predictions of the reconstructed W -boson mass distribution with the corresponding distribution in data for a representative diagonal analysis region in the JES and JER measurements, respectively. The *post-fit* predictions are obtained by propagating the fitted JES or JER correction factor, s or r , through the parametrisation of the predicted yield in each bin as a function of the corresponding correction factor. This parametrisation is derived specifically to obtain the *post-fit* W -boson mass distribution, as the W -boson mass distribution itself is not the distribution that is fitted in the measurement. The associated uncertainty bands are derived from the predicted yield when using the fitted correction factor varied by its total uncertainty, i.e. $s_i \pm \sigma_{s_i}$ or $r_i \pm \sigma_{r_i}$. An improved agreement between the *post-fit* prediction and the data is observed in Run 2. The improvement is seen across the full W -mass distribution, even though only the mean of the distribution was fitted. In contrast, no such improvement is observed in Run 3, where the corrections are substantially smaller compared to the Run 2 corrections. No improvement is observed in the JER distributions either, which is consistent with the corresponding JER corrections being close to unity.

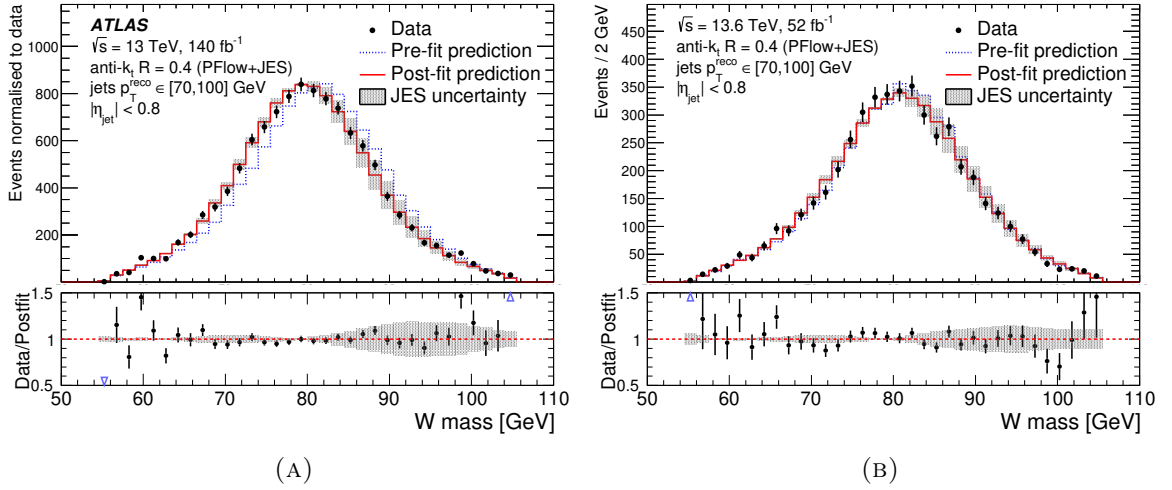


FIGURE 7.7: The *pre fit* and *post fit* predictions for the W -boson mass distribution in the representative diagonal region with reconstructed jet p_T from 70 – 100 GeV in the (A) Run 2 and (B) Run 3 JES measurements. The uncertainty band is estimated by propagating the total uncertainties, σ_{s_i} , on the fitted JES correction factors, s_i , to the yield predictions for each bin parametrised as a function of JES correction factors. The predictions are normalised to the same yield as data. The bottom panels show the ratio of the data without the *in situ* JES calibration (PFlow + JES) to the *post-fit* prediction, with the dashed horizontal line representing the ratio of one. The arrows indicate a value outside of the displayed range [15].

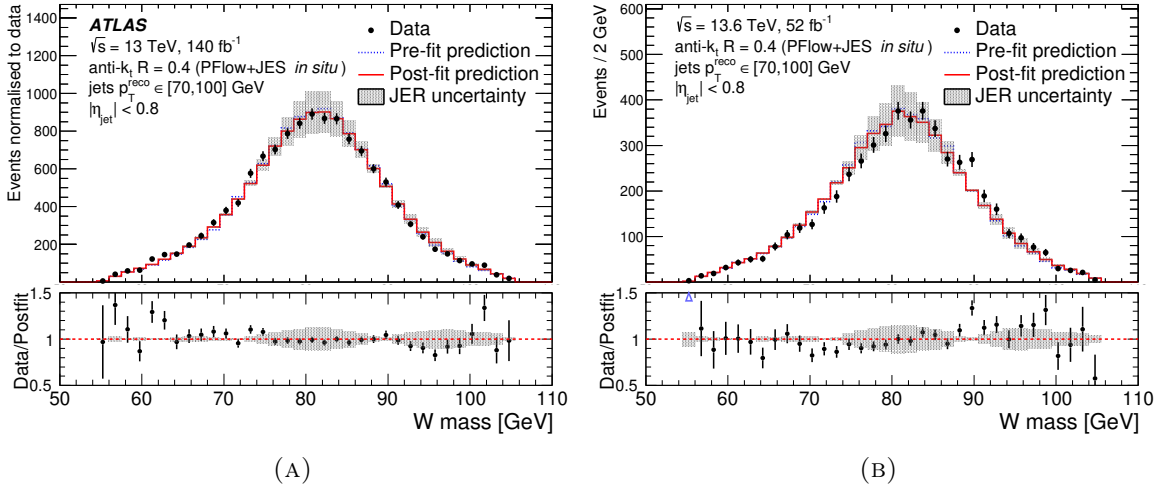


FIGURE 7.8: The *pre fit* and *post fit* predictions for the W -boson mass distribution in the representative diagonal region with reconstructed jet p_T from 70 – 100 GeV in the (A) Run 2 and (B) Run 3 JER measurement. The uncertainty band is estimated by propagating the total uncertainties, σ_{r_i} , on the fitted JER correction factors, r_i , to the yield predictions for each bin parametrised as a function of JER correction factors. The predictions are normalised to the same yield as data. The bottom panels show the ratio of the data with the *in situ* JES calibration (PFlow + JES *in situ*) to the *post-fit* prediction, with the dashed horizontal line representing the ratio of one. The arrows indicate a value outside of the displayed range [15].

Appendix A

Run 2 response distributions

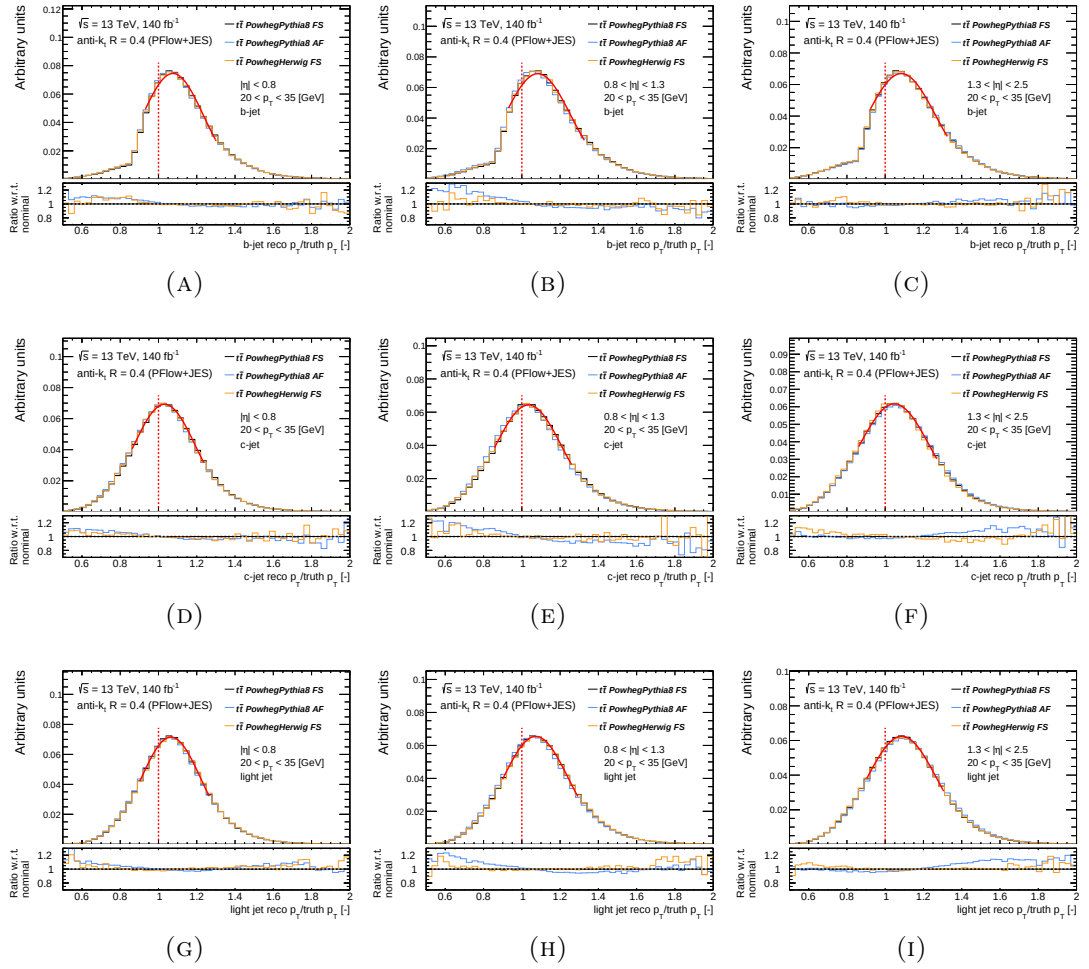


FIGURE A.1: The Run 2 simulated jet response distribution for jets from different p_T bins and $|\eta|$ bins. The distributions are shown for different $t\bar{t}$ samples. The red line represents the Gaussian fit to the response distribution of the nominal $t\bar{t}$ sample. The bottom panel shows the ratios of jet response distributions in the alternative samples over the jet response in the nominal $t\bar{t}$ sample.

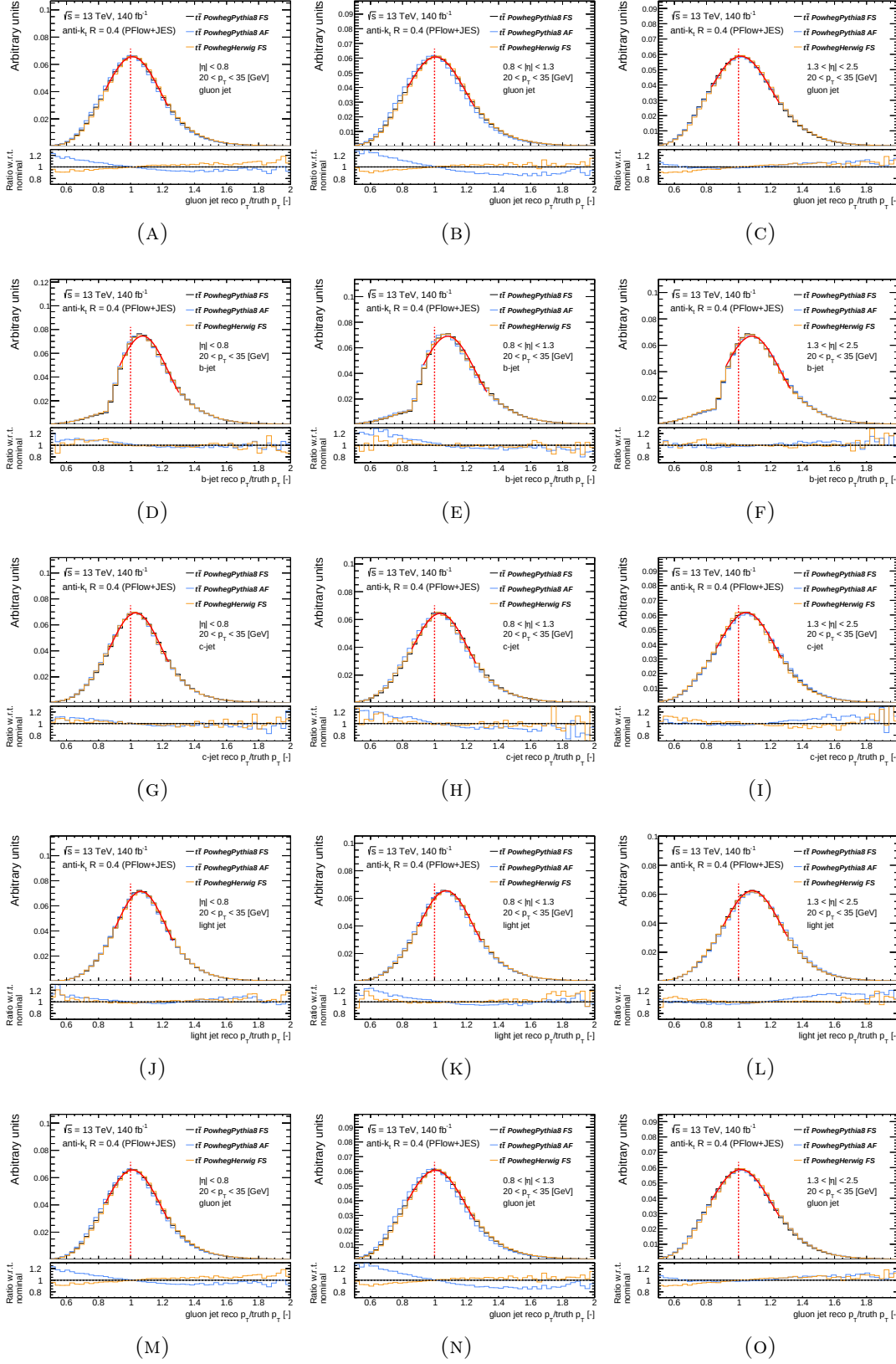


FIGURE A.2: The Run 2 simulated jet response distribution for jets from different p_T bins and $|\eta|$ bins. The distributions are shown for different $t\bar{t}$ samples. The red line represents the Gaussian fit to the response distribution of the nominal $t\bar{t}$ sample. The bottom panel shows the ratios of jet response distributions in the alternative samples over the jet response in the nominal $t\bar{t}$ sample.

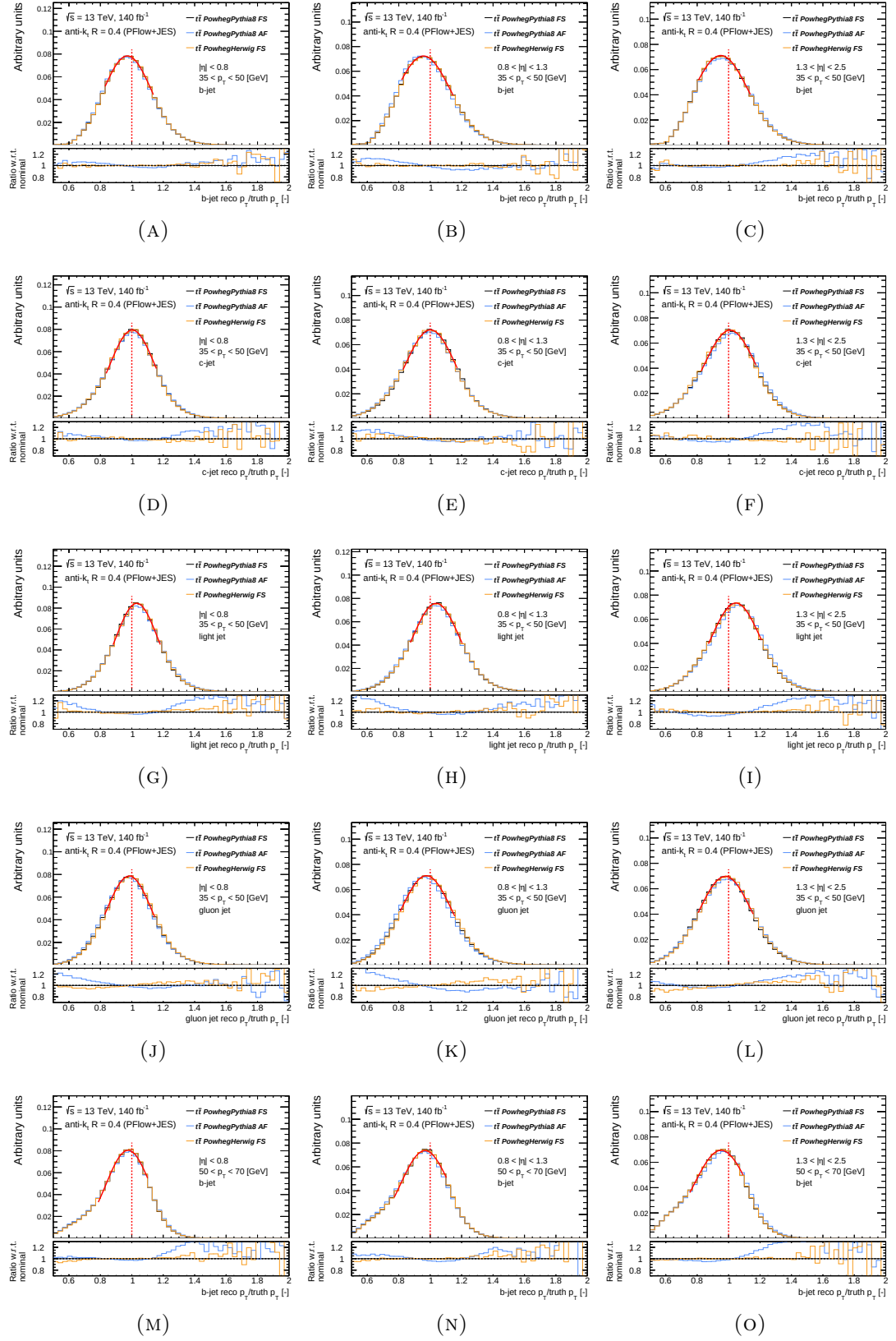


FIGURE A.3: The Run 2 simulated jet response distribution for jets from different p_T and $|\eta|$ bins. The distributions are shown for different $t\bar{t}$ samples. The red line represents the Gaussian fit to the response distribution of the nominal $t\bar{t}$ sample. The bottom panel shows the ratios of jet response distributions in the alternative samples over the jet response in the nominal $t\bar{t}$ sample.

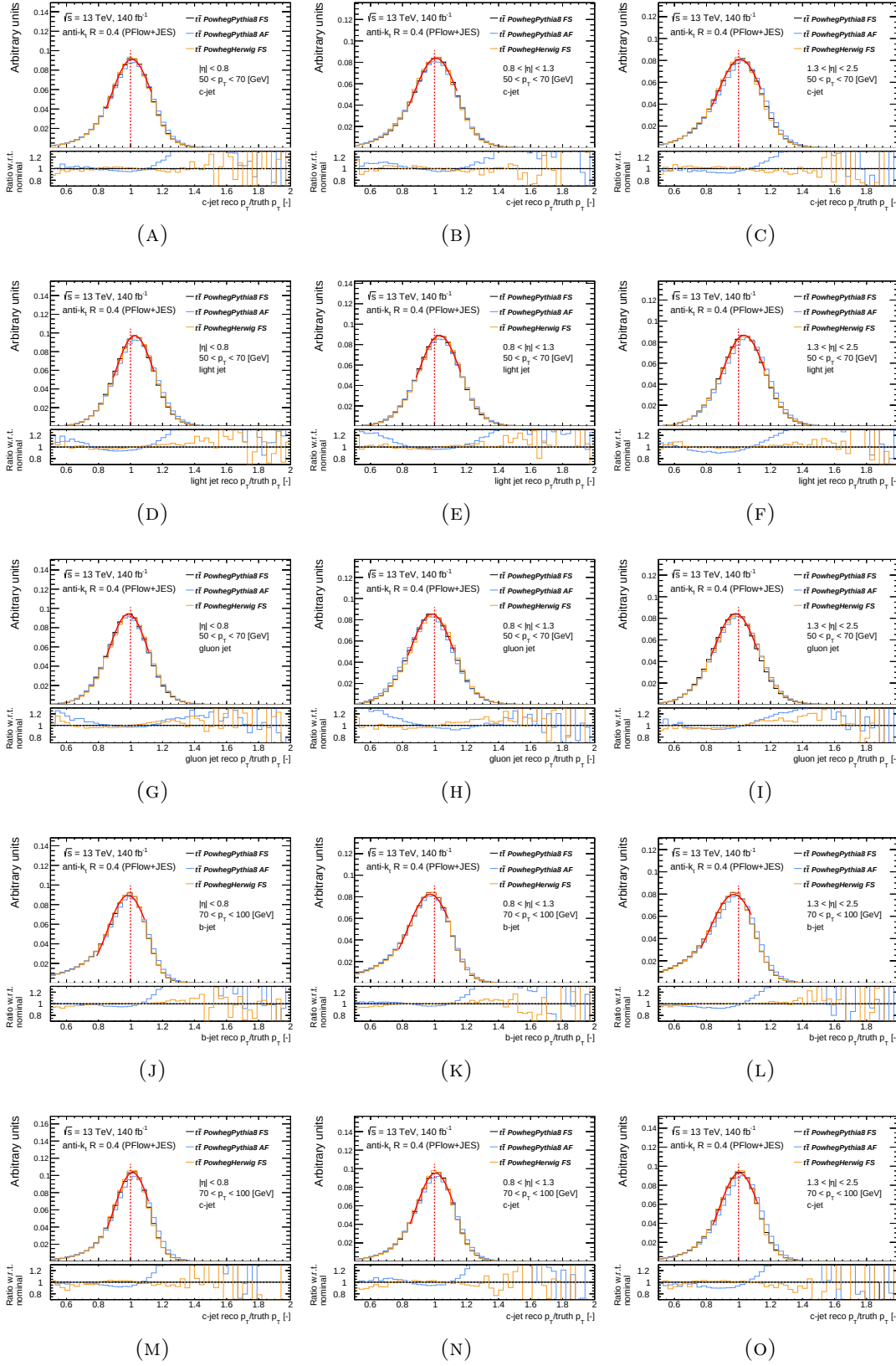


FIGURE A.4: The Run 2 simulated jet response distribution for jets from different p_T bins and $|\eta|$ bins. The distributions are shown for different $t\bar{t}$ samples. The red line represents the Gaussian fit to the response distribution of the nominal $t\bar{t}$ sample. The bottom panel shows the ratios of jet response distributions in the alternative samples over the jet response in the nominal $t\bar{t}$ sample.

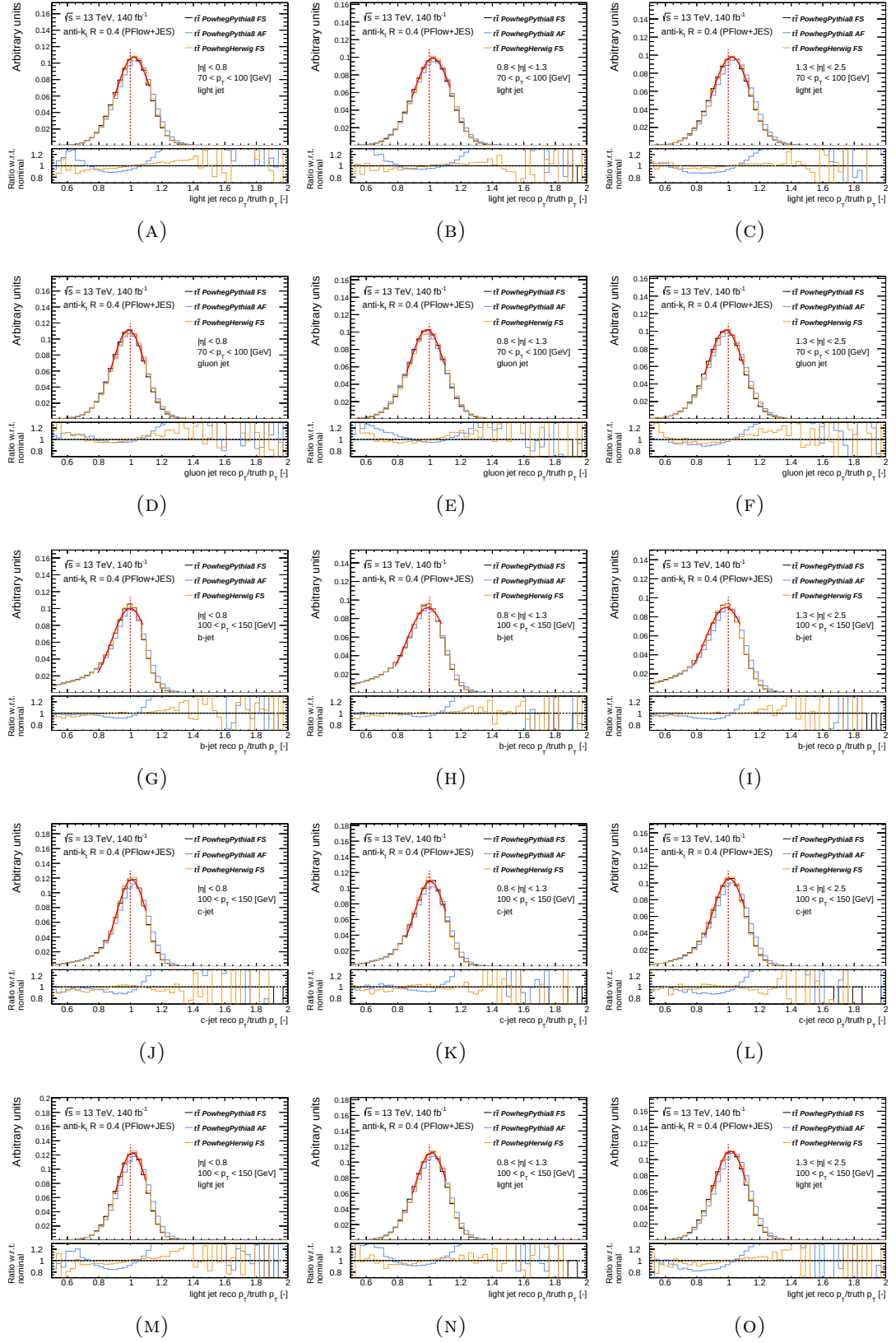


FIGURE A.5: The Run 2 simulated jet response distribution for jets from different p_T bins and $|\eta|$ bins. The distributions are shown for different $t\bar{t}$ samples. The red line represents the Gaussian fit to the response distribution of the nominal $t\bar{t}$ sample. The bottom panel shows the ratios of jet response distributions in the alternative samples over the jet response in the nominal $t\bar{t}$ sample.

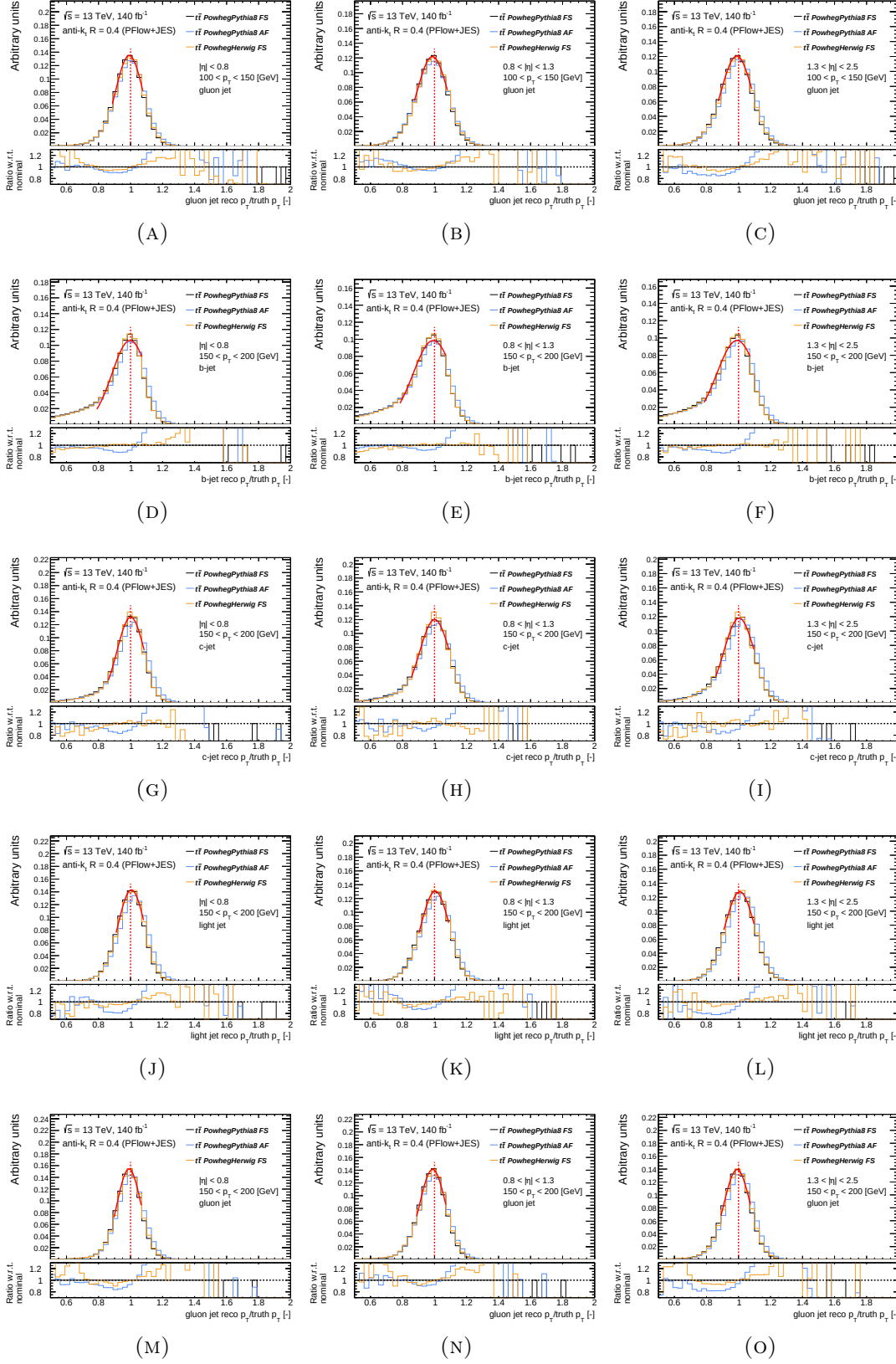


FIGURE A.6: The Run 2 simulated jet response distribution for jets from different p_T bins and $|\eta|$ bins. The distributions are shown for different $t\bar{t}$ samples. The red line represents the Gaussian fit to the response distribution of the nominal $t\bar{t}$ sample. The bottom panel shows the ratios of jet response distributions in the alternative samples over the jet response in the nominal $t\bar{t}$ sample.

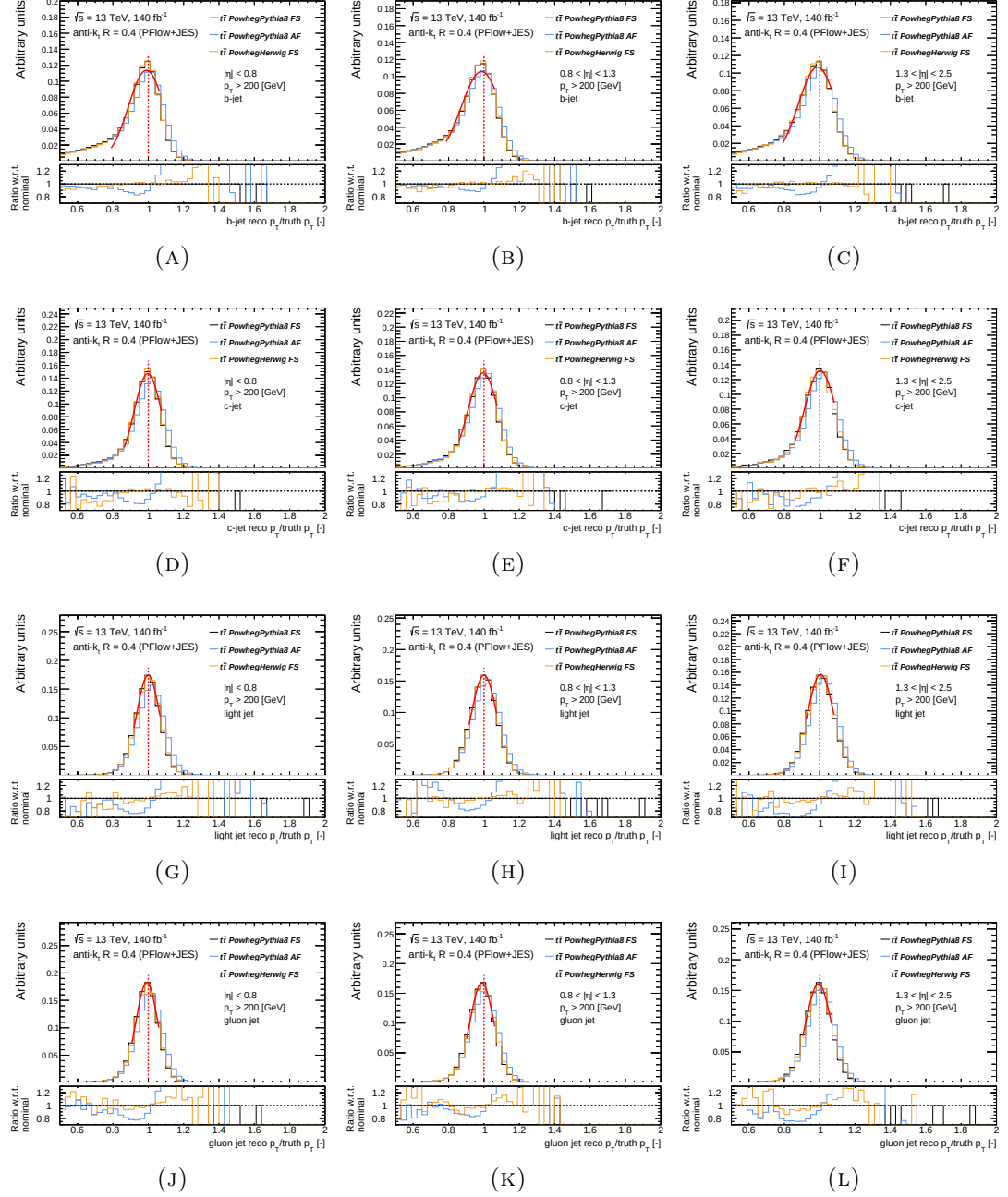


FIGURE A.7: The Run 2 simulated jet response distribution for jets from different p_T bins and $|\eta|$ bins. The distributions are shown for different $t\bar{t}$ samples. The red line represents the Gaussian fit to the response distribution of the nominal $t\bar{t}$ sample. The bottom panel shows the ratios of jet response distributions in the alternative samples over the jet response in the nominal $t\bar{t}$ sample.

Appendix B

Run 2 Data/MC control plots

B.1 Data/MC control plots without *in situ*

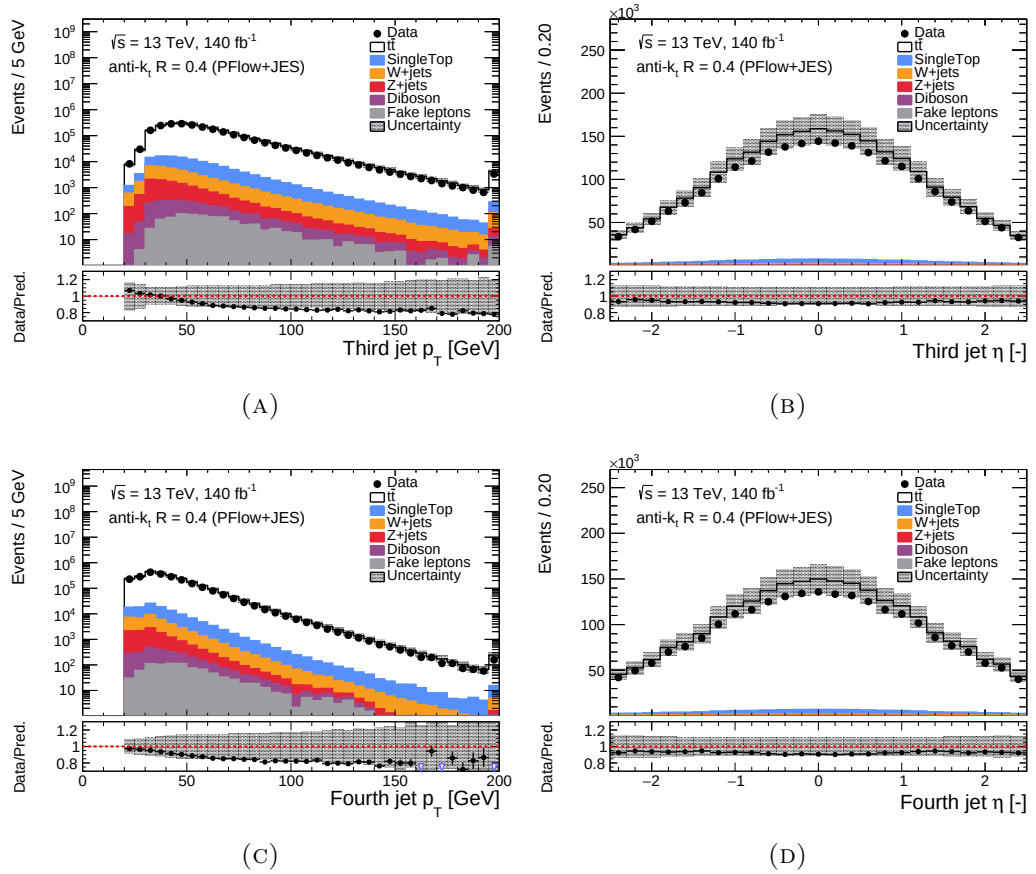


FIGURE B.1: The Run 2 data (without JES *in situ* correction) to prediction comparisons in the Inclusive region with no p_T or η cut on the jets from W boson. The uncertainty represents all considered systematic uncertainties as described in Section ??.

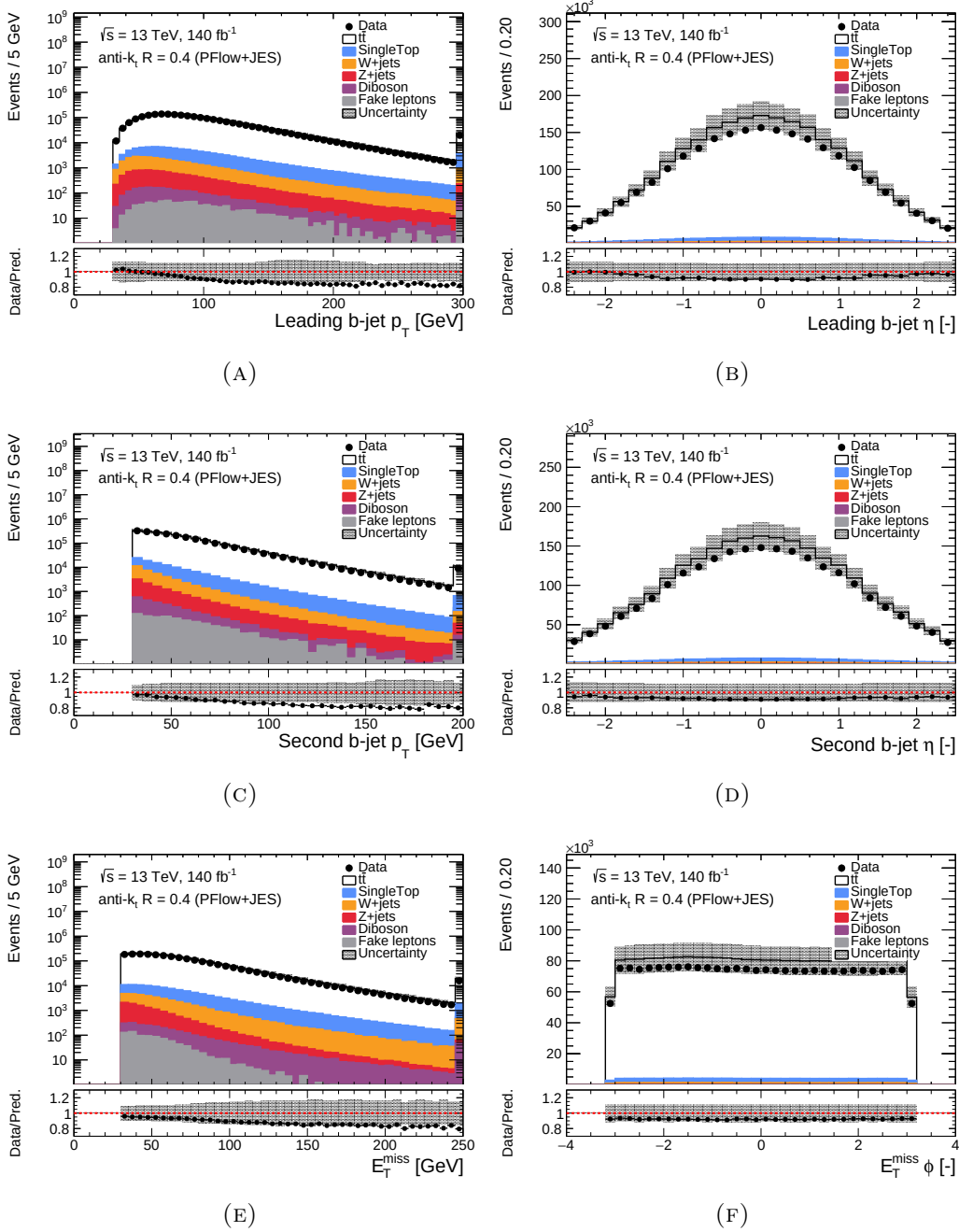


FIGURE B.2: The Run 2 data (without JES *in situ* correction) to prediction comparisons in the Inclusive region with no p_T or η cut on the jets from W boson. The uncertainty represents all considered systematic uncertainties as described in Section ??.

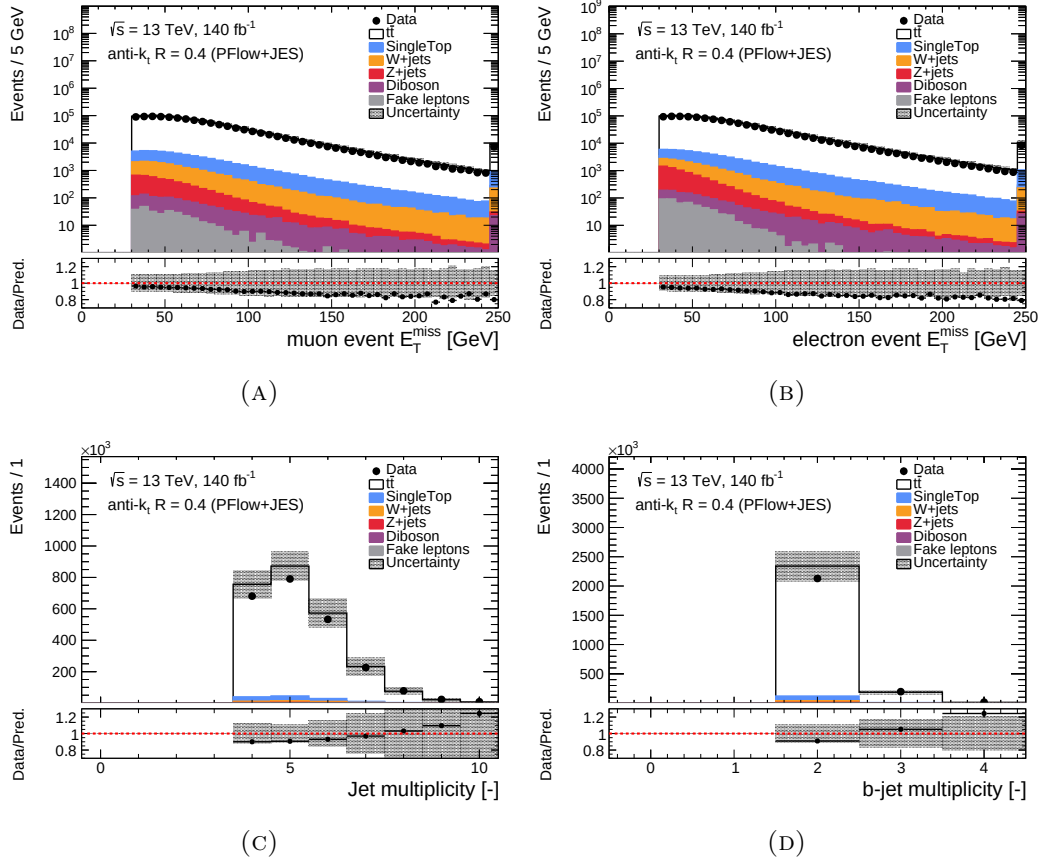


FIGURE B.3: The Run 2 data (without JES *in situ* correction) to prediction comparisons in the Inclusive region with no p_T or η cut on the jets from W boson. The uncertainty represents all considered systematic uncertainties as described in Section ??.

Appendix C

Run 3 Data/MC control plots

C.1 Data/MC control plots without *in situ*

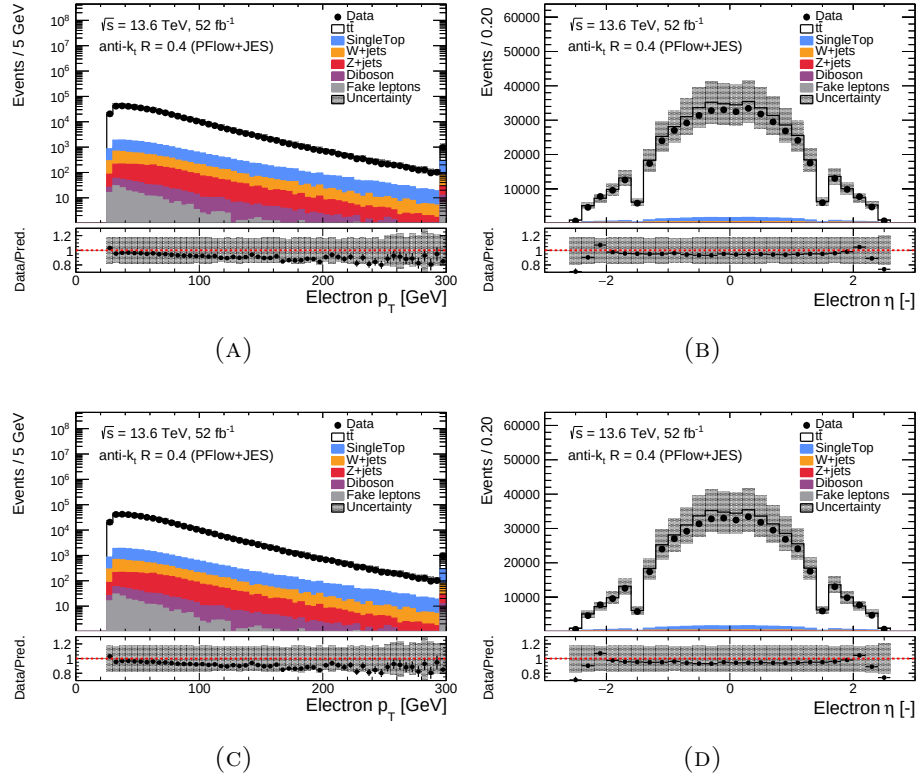


FIGURE C.1: The Run 3 data (without JES *in situ* correction) to prediction comparisons in the Inclusive region with no p_T or η cut on the jets from W boson. The uncertainty represents all considered systematic uncertainties as described in Section ??.

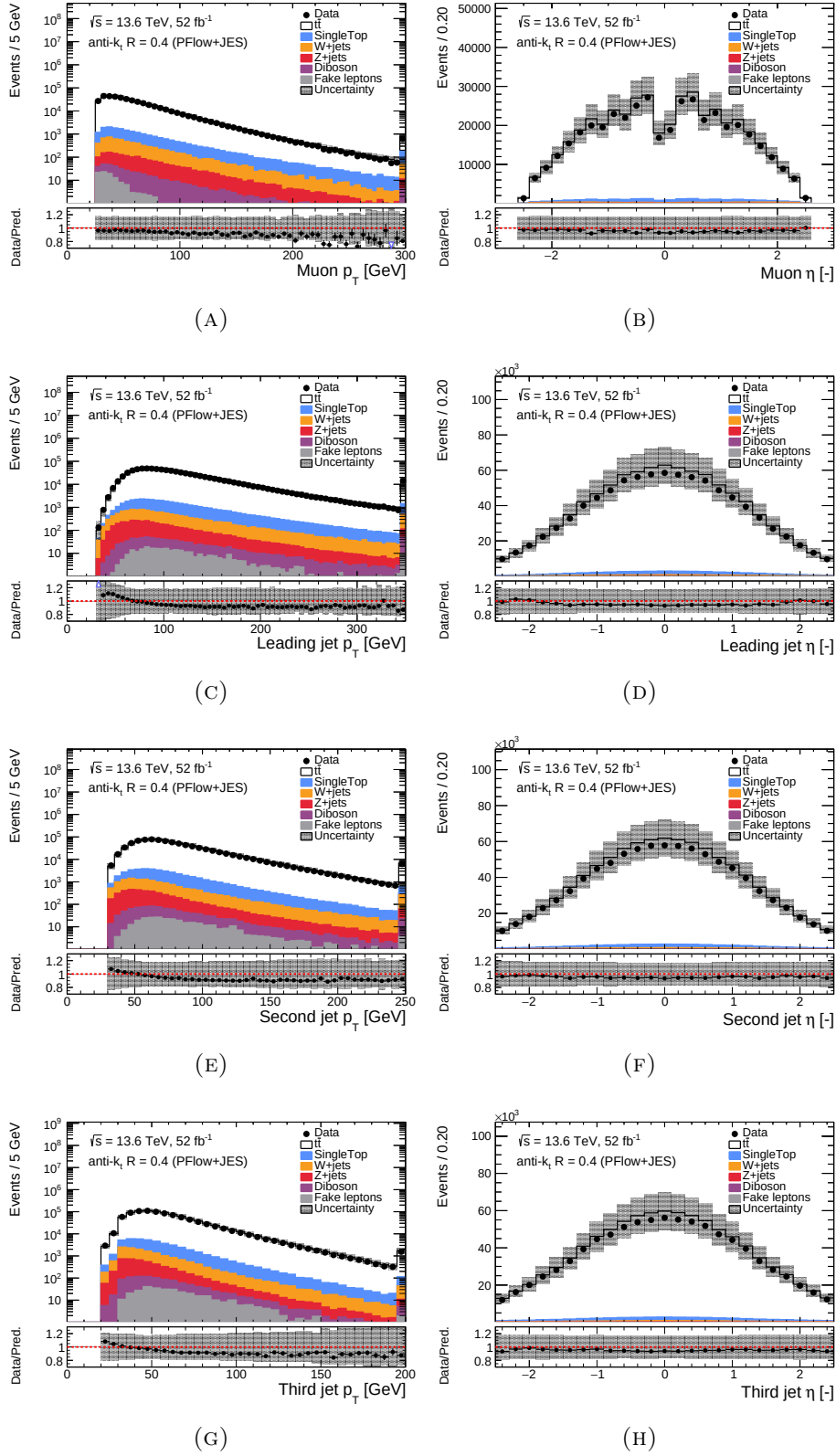


FIGURE C.2: The Run 3 data (without JES *in situ* correction) to prediction comparisons in the Inclusive region with no p_T or η cut on the jets from W boson. The uncertainty represents all considered systematic uncertainties as described in Section ??.

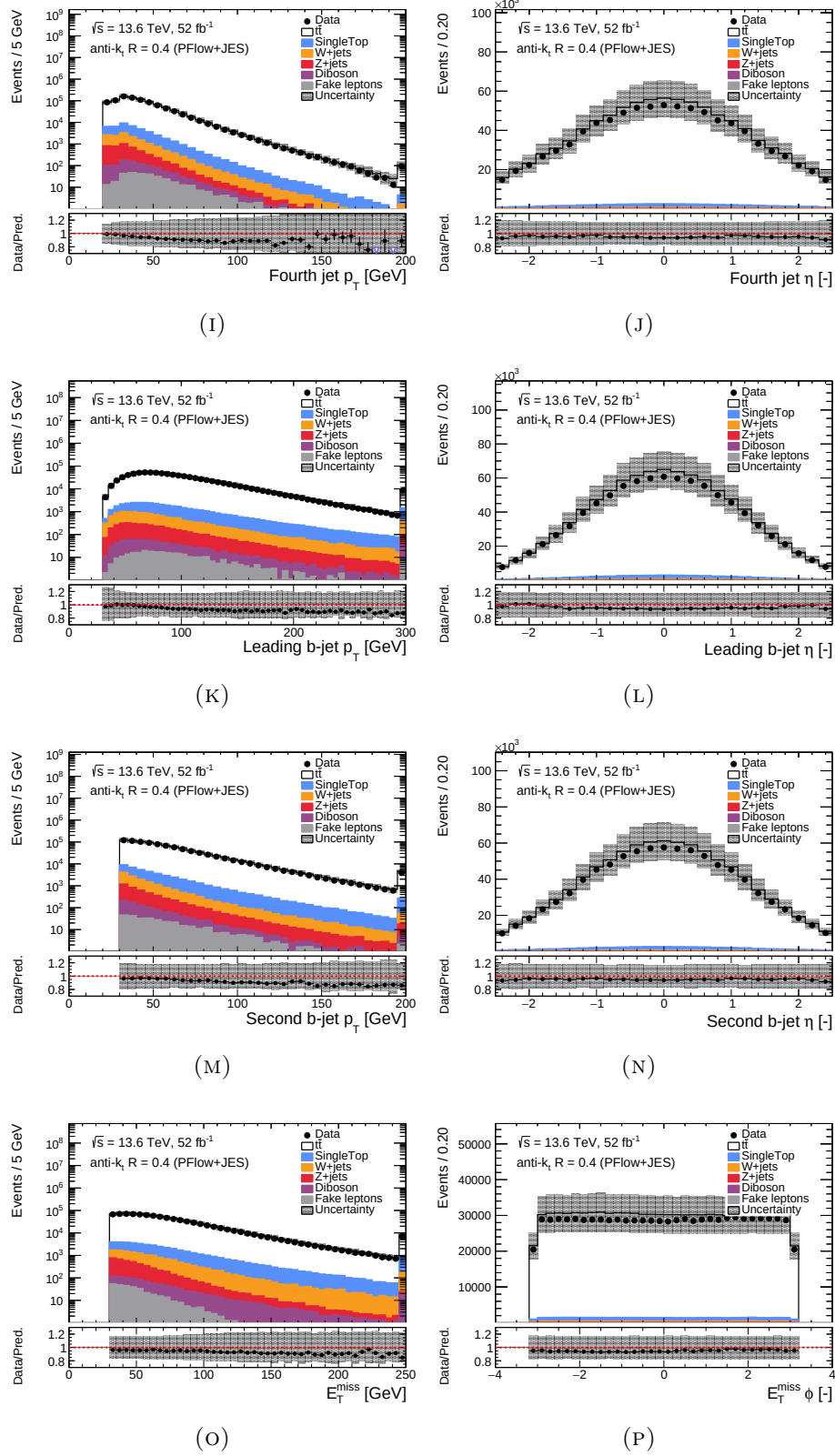


FIGURE C.2: The Run 3 data (without JES *in situ* correction) to prediction comparisons in the Inclusive region with no p_T or η cut on the jets from W boson. The uncertainty represents all considered systematic uncertainties as described in Section ??.

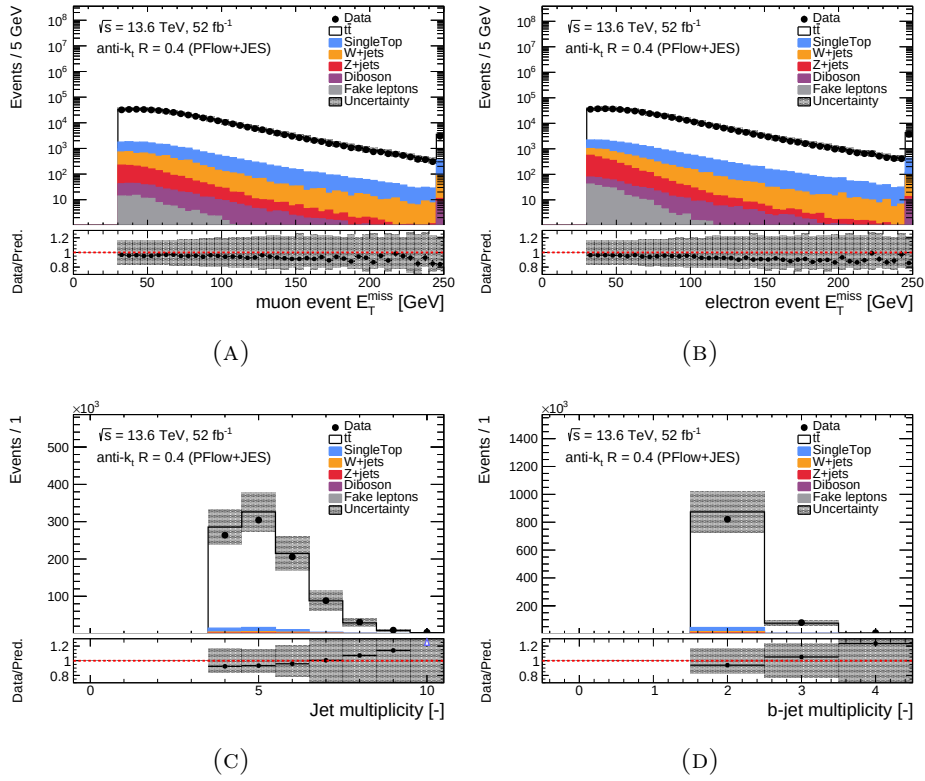


FIGURE C.3: The Run 3 data (without JES *in situ* correction) to prediction comparisons in the Inclusive region with no p_T or η cut on the jets from W boson. The uncertainty represents all considered systematic uncertainties as described in Section ??.

Appendix D

Run 2 W -mass templates

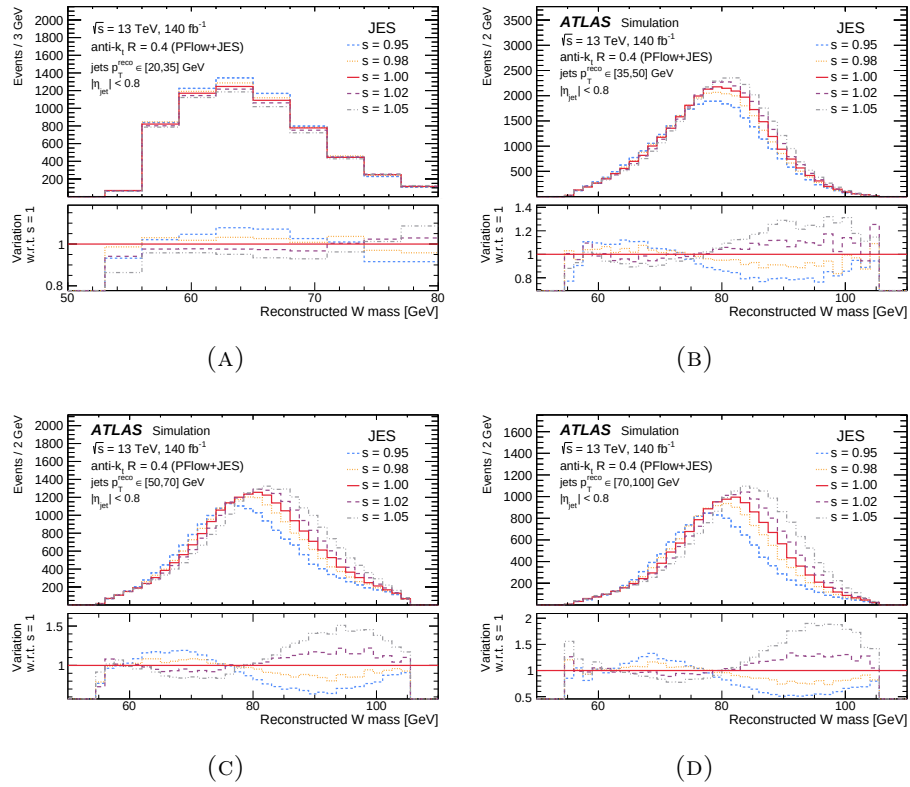


FIGURE D.1: Template W -boson mass distributions for various corrections s or r , to JES or JER, respectively. Run 2 templates in the different diagonal regions are shown. For JES variations, the JER correction, r , is set to one, and vice versa.

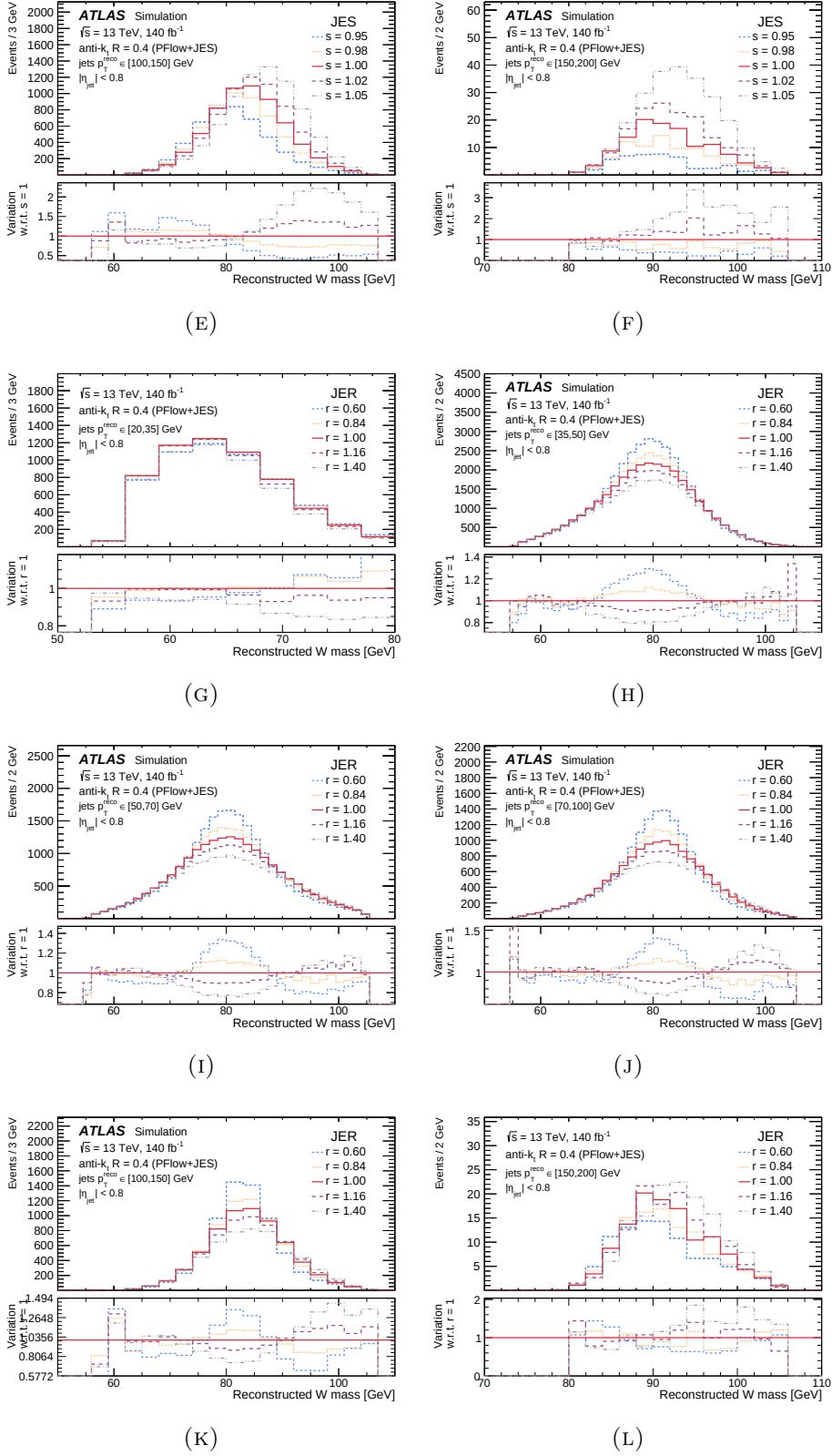


FIGURE D.1: Template W -boson mass distributions for various corrections s or r , to JES or JER, respectively. Run 2 templates in the different diagonal regions are shown. For JES variations, the JER correction, r , is set to one, and vice versa.

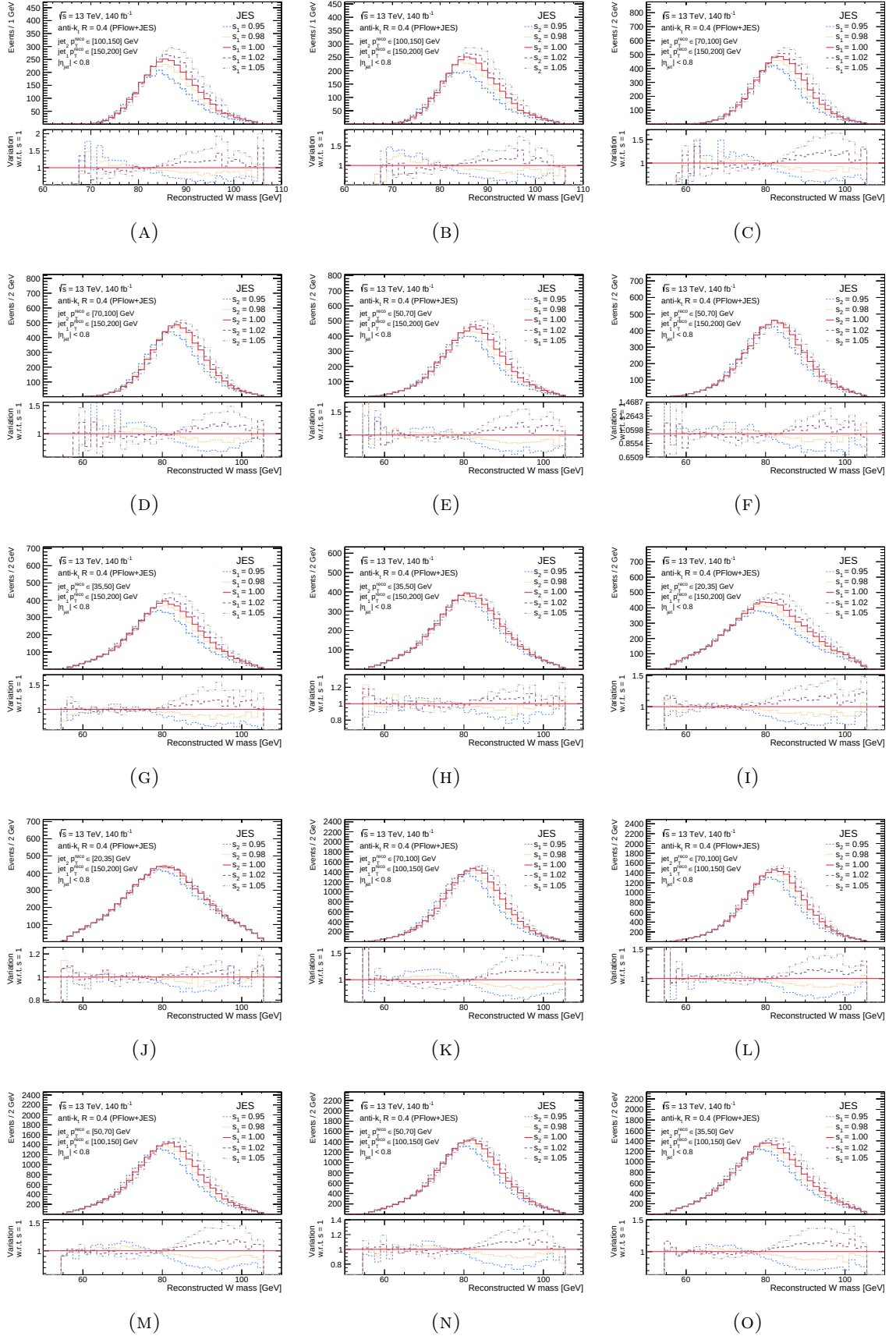


FIGURE D.2: Template W -boson mass distributions for various corrections s_1 or s_2 to the JES of the leading or sub-leading jet from W decay, respectively. Run 2 templates in the different off-diagonal regions are shown. The JER correction factor, r , is fixed to unity.

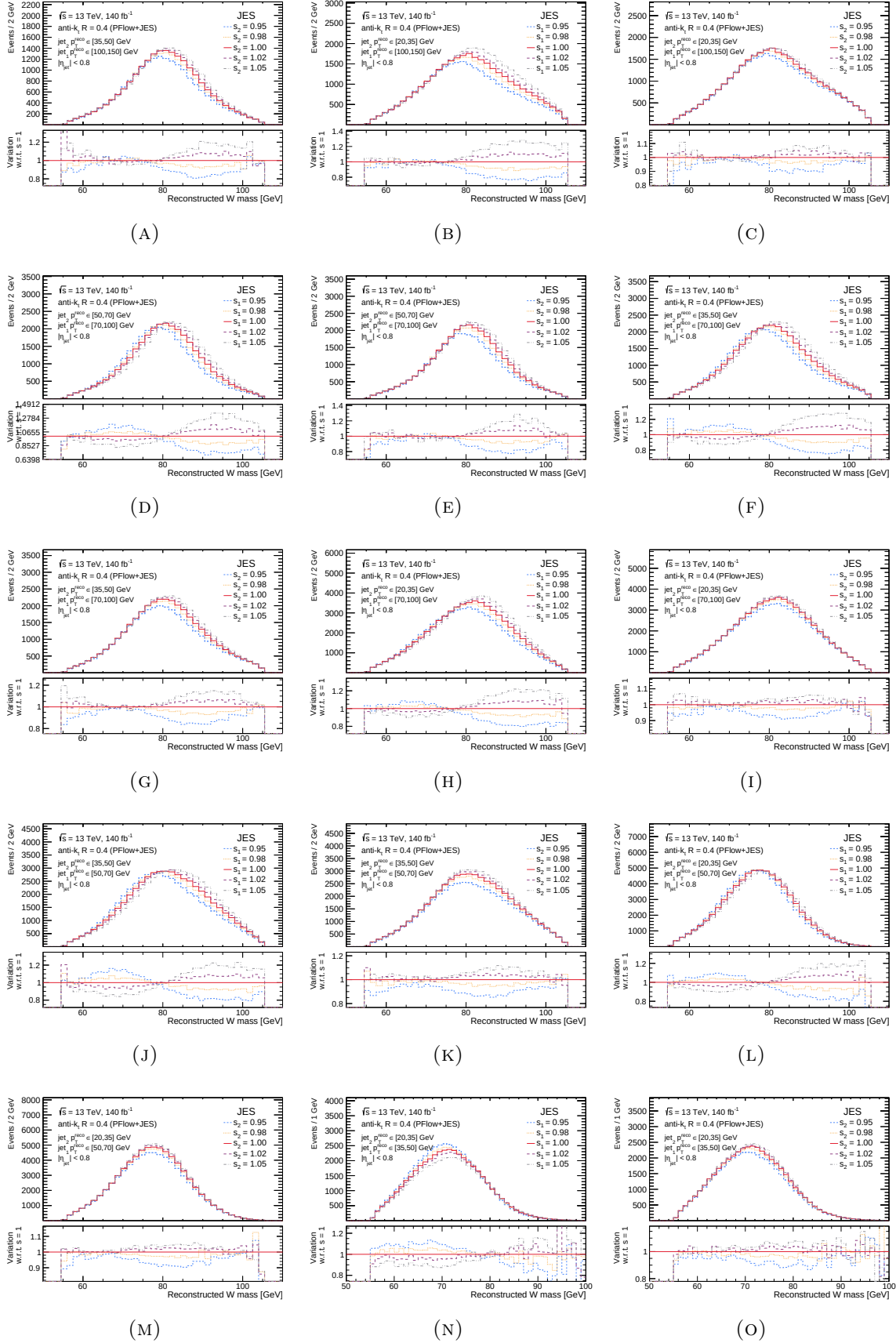


FIGURE D.3: Template W -boson mass distributions for various corrections s_1 or s_2 to the JES of the leading or sub-leading jet from W decay, respectively. Run 2 templates in the different off-diagonal regions are shown. The JER correction factor, r , is fixed to unity.

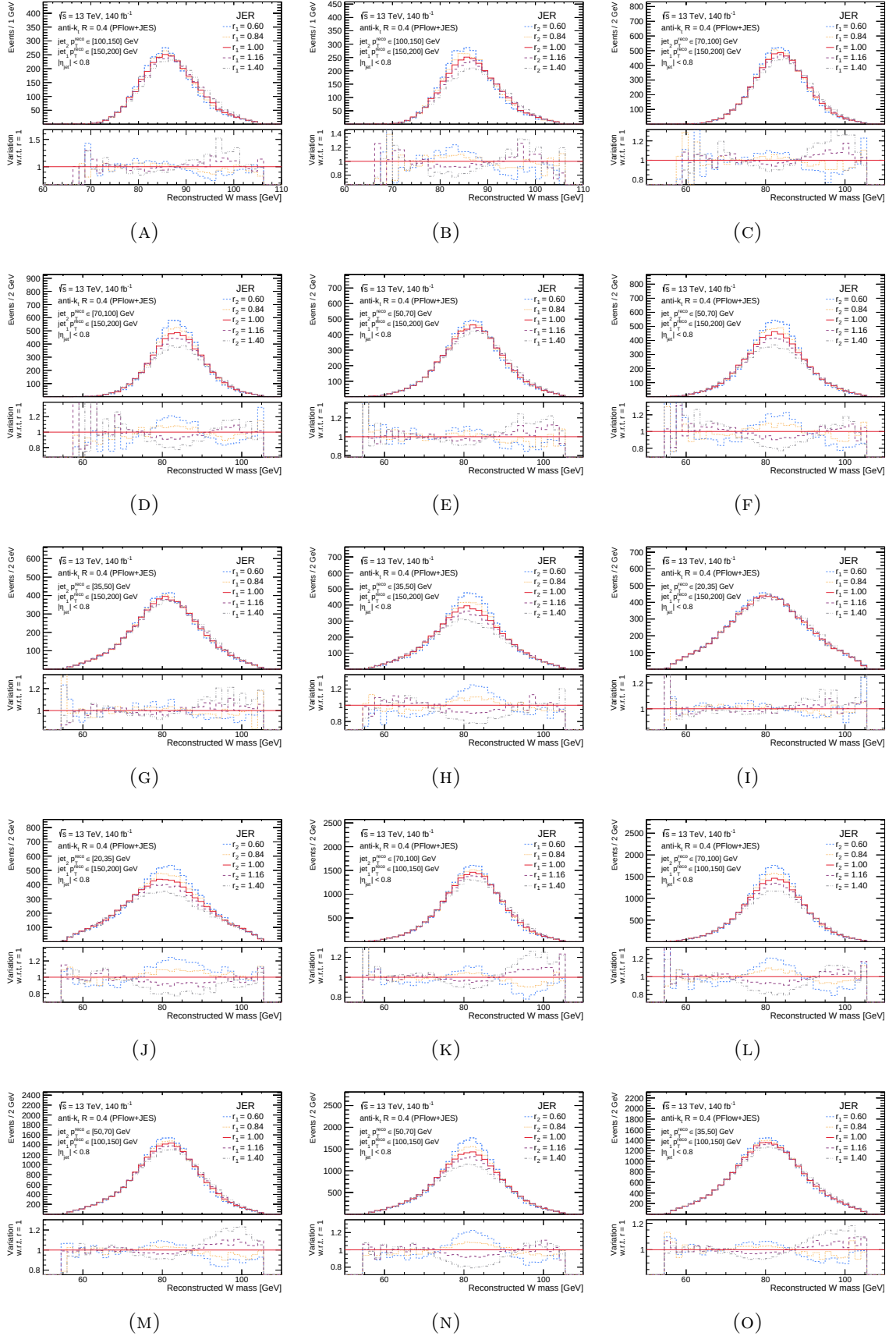


FIGURE D.4: Template W -boson mass distributions for various corrections r_1 or r_2 to the JER of the leading or sub-leading jet from W decay, respectively. Run 2 templates in the different off-diagonal regions are shown. The JES correction factor, s , is fixed to unity.

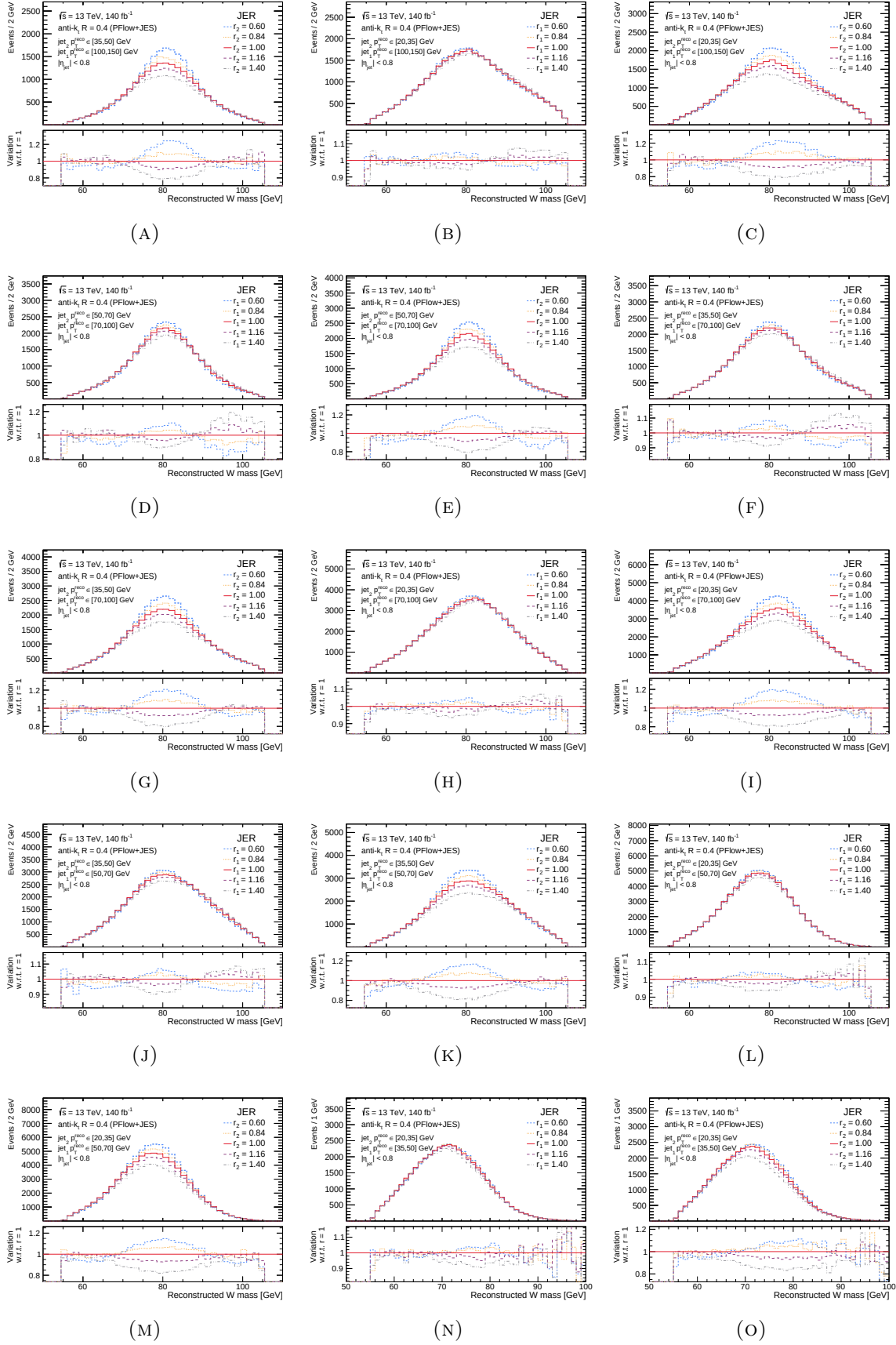


FIGURE D.5: Template W -boson mass distributions for various corrections r_1 or r_2 to the JER of the leading or sub-leading jet from W decay, respectively. Run 2 templates in the different off-diagonal regions are shown. The JES correction factor, s , is fixed to unity.

Appendix E

Run 3 W -mass templates

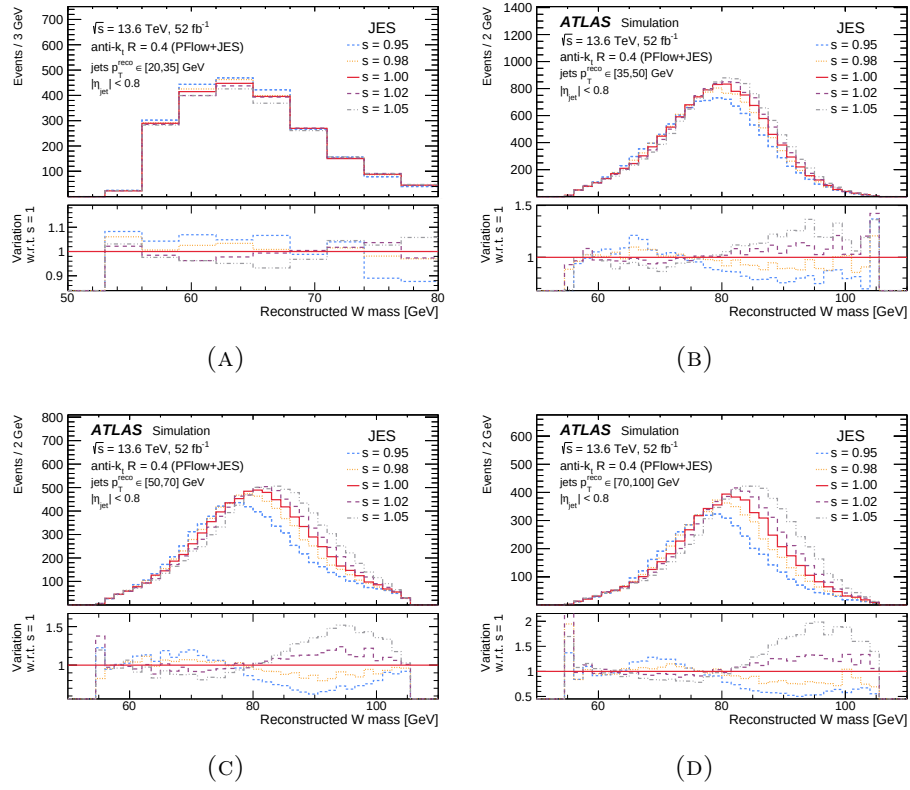


FIGURE E.1: Template W -boson mass distributions for various corrections s or r , to JES or JER, respectively. Run 3 templates in the different diagonal regions are shown. For JES variations, the JER correction, r , is set to one, and vice versa.

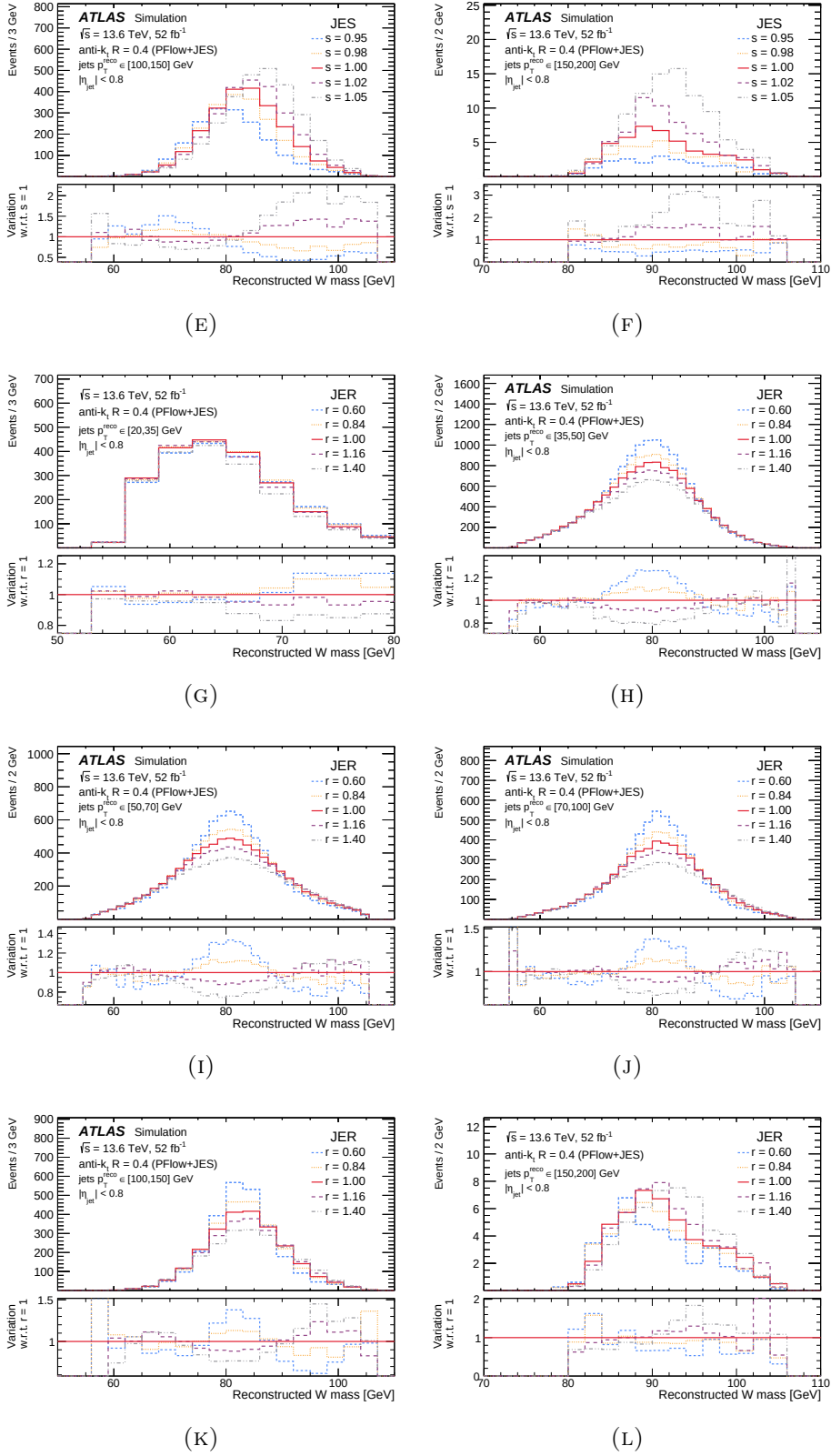


FIGURE E.1: Template W -boson mass distributions for various corrections s or r , to JES or JER, respectively. Run 3 templates in the different diagonal regions are shown. For JES variations, the JER correction, r , is set to one, and vice versa.

Appendix F

JES and JER parametrisation

F.1 JES parametrisation

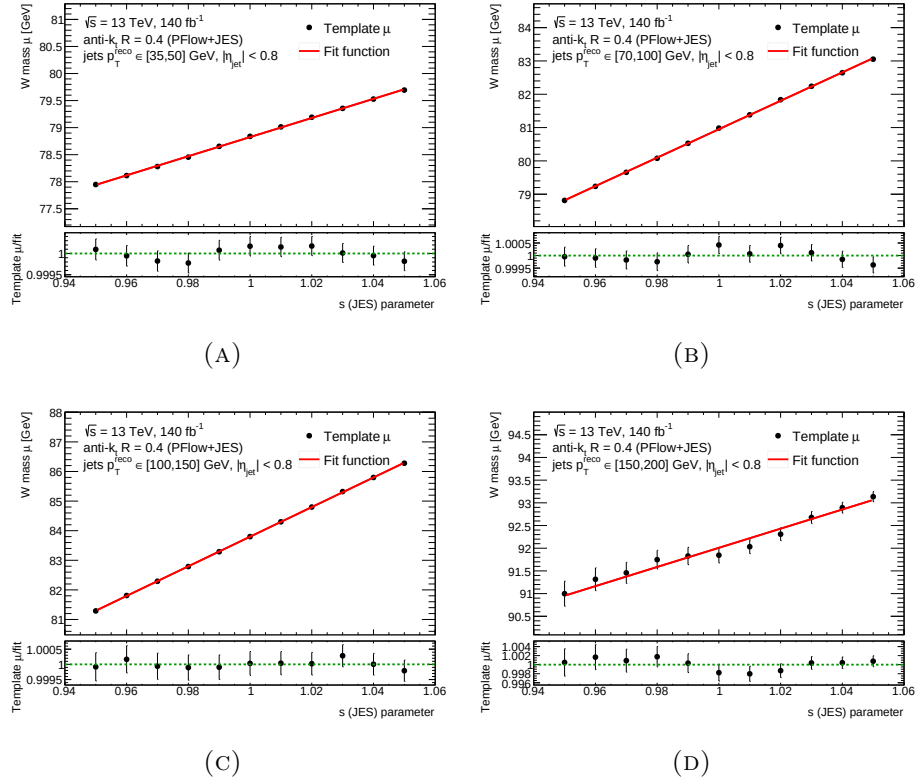


FIGURE F.1: The mean of the W -mass distribution as a function of JES (s), in the different diagonal regions, for the Run 2 measurement. The fitted linear function $\mathcal{F}$ is shown by solid red line. The bottom panel shows the ratio of the mean to the fitted function $\mathcal{F}$.

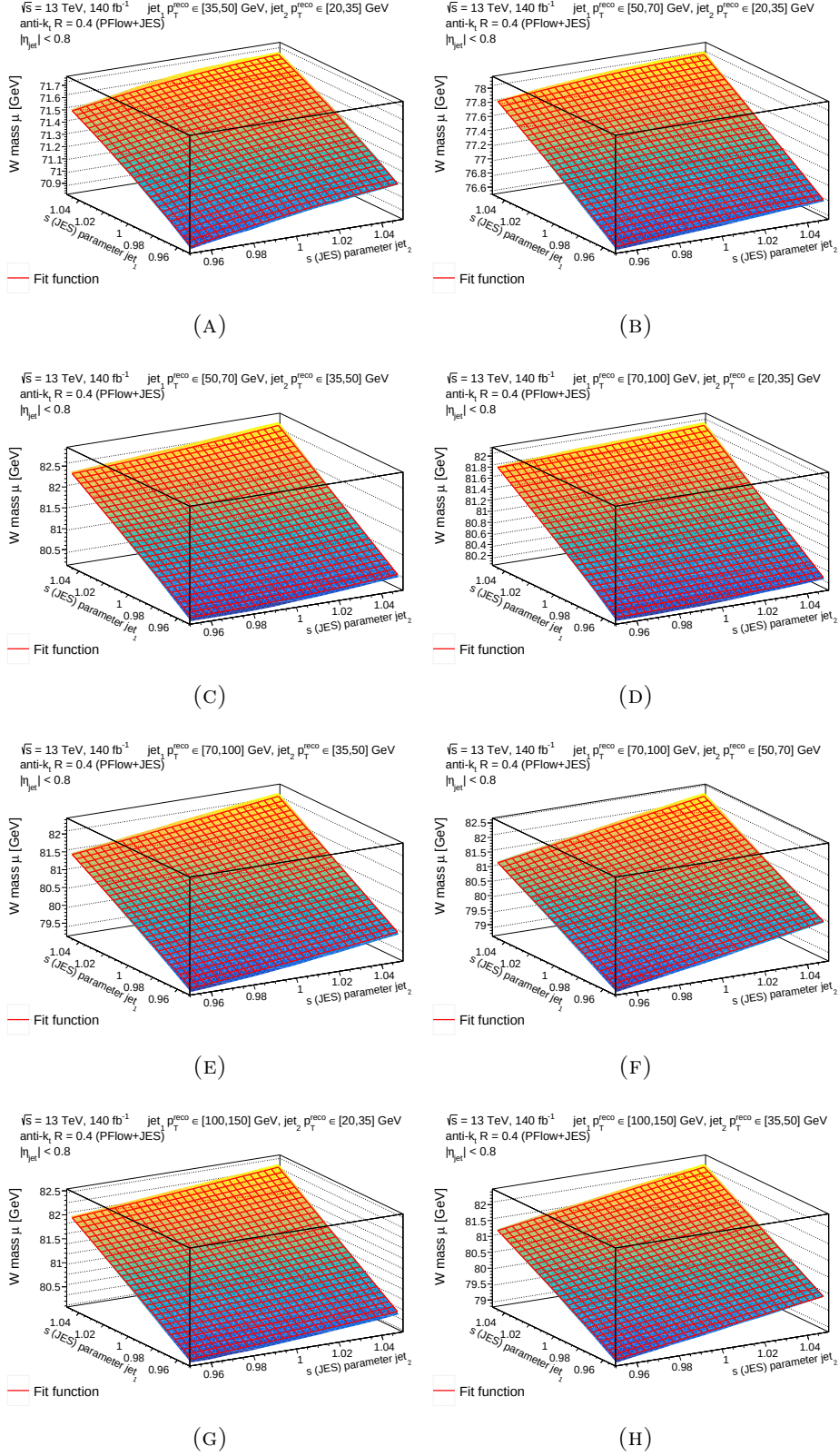


FIGURE F.2: The mean of the W -mass distribution as a function of the JES parameters, s_1 and s_2 , in the different off-diagonal regions with the leading jet denoted as jet_1 and the sub-leading jet denoted as jet_2 , for the Run 2 measurement. The fitted two-dimensional quadratic function $\mathcal{F}$ is shown in red.

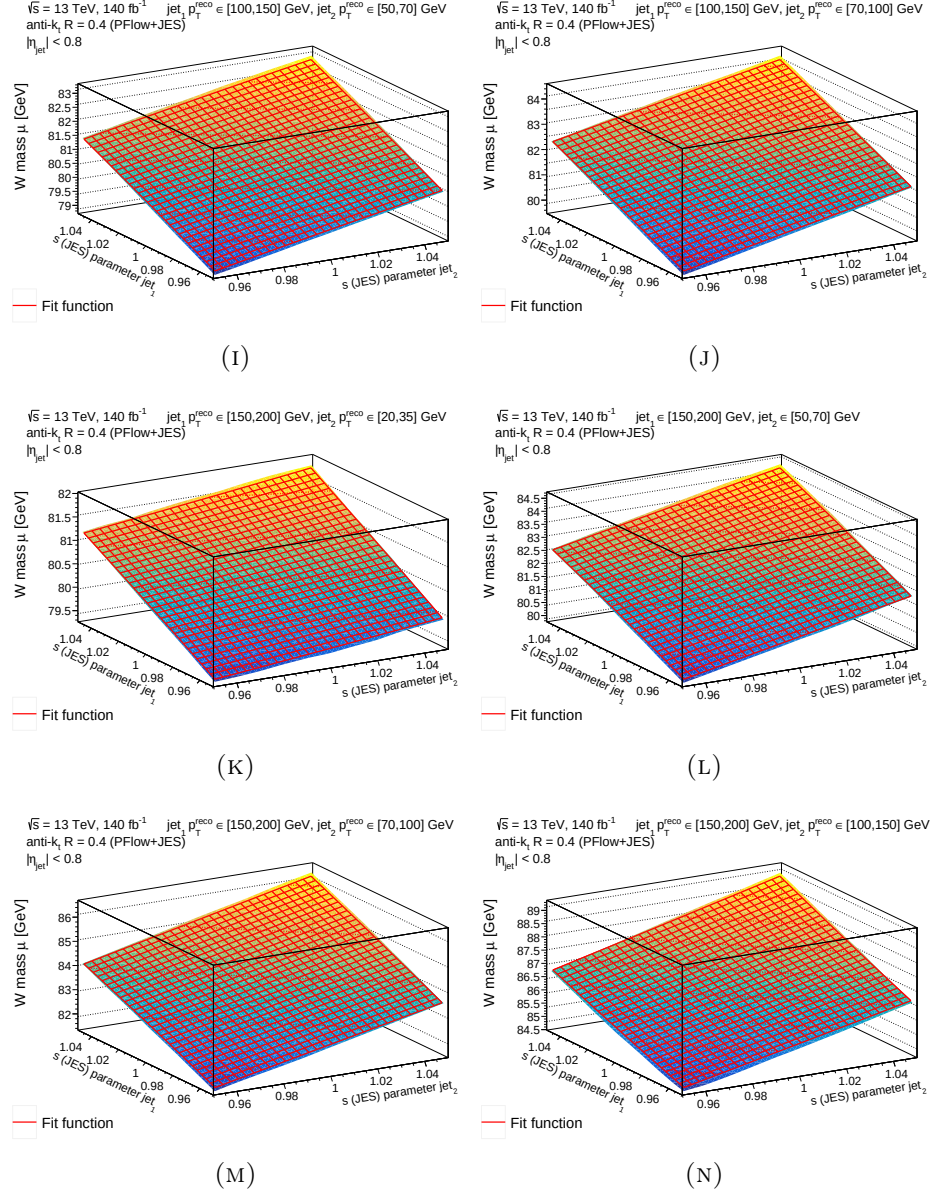


FIGURE F.2: The mean of the W -mass distribution as a function of the JES parameters, s_1 and s_2 , in the different off-diagonal regions with the leading jet denoted as jet_1 and the sub-leading jet denoted as jet_2 , for the Run 2 measurement. The fitted two-dimensional quadratic function $\mathcal{F}$ is shown in red.

737 F.2 JER parametrisation

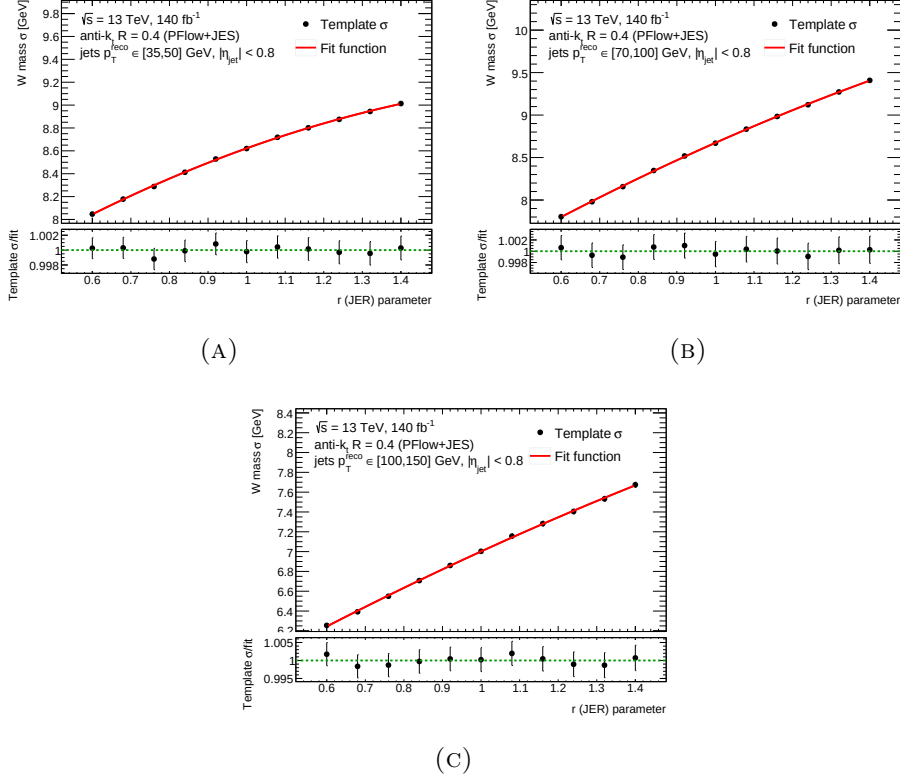


FIGURE F.3: The standard deviation of the W -mass distribution as a function of JER (r), in the different diagonal regions, for the Run 2 measurement. The fitted quadratic function $\mathcal{F}$ is shown by solid red line. The bottom panel shows the ratio of the standard deviation to the fitted function $\mathcal{F}$.

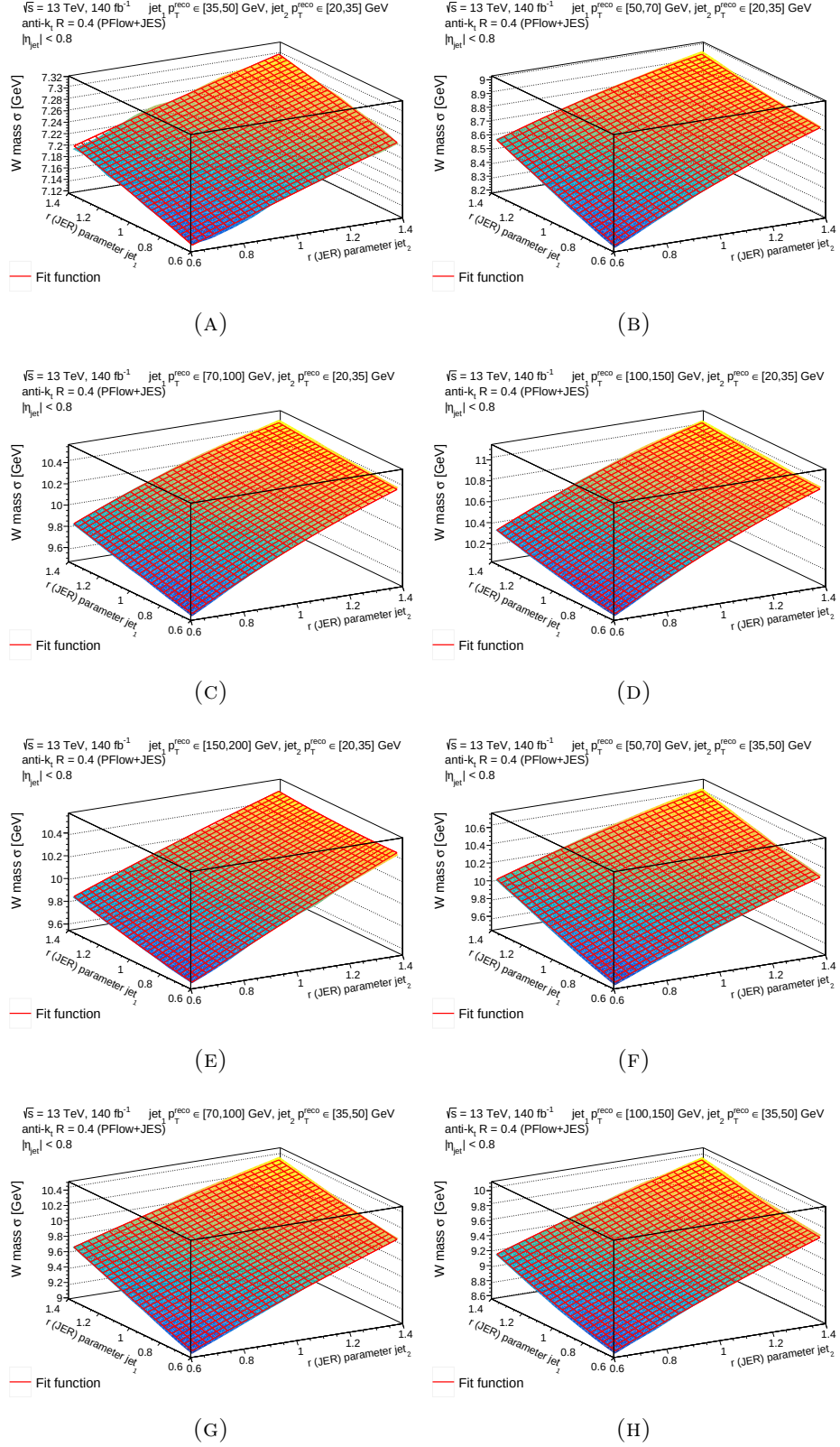


FIGURE F.4: The standard deviation of the W -mass distribution as a function of the JER parameters, r_1 and r_2 , in the different off-diagonal regions with the leading jet denoted as jet₁ and the sub-leading jet denoted as jet₂, for the Run 2 measurement. The fitted two-dimensional quadratic function $\mathcal{F}$ is shown in red.

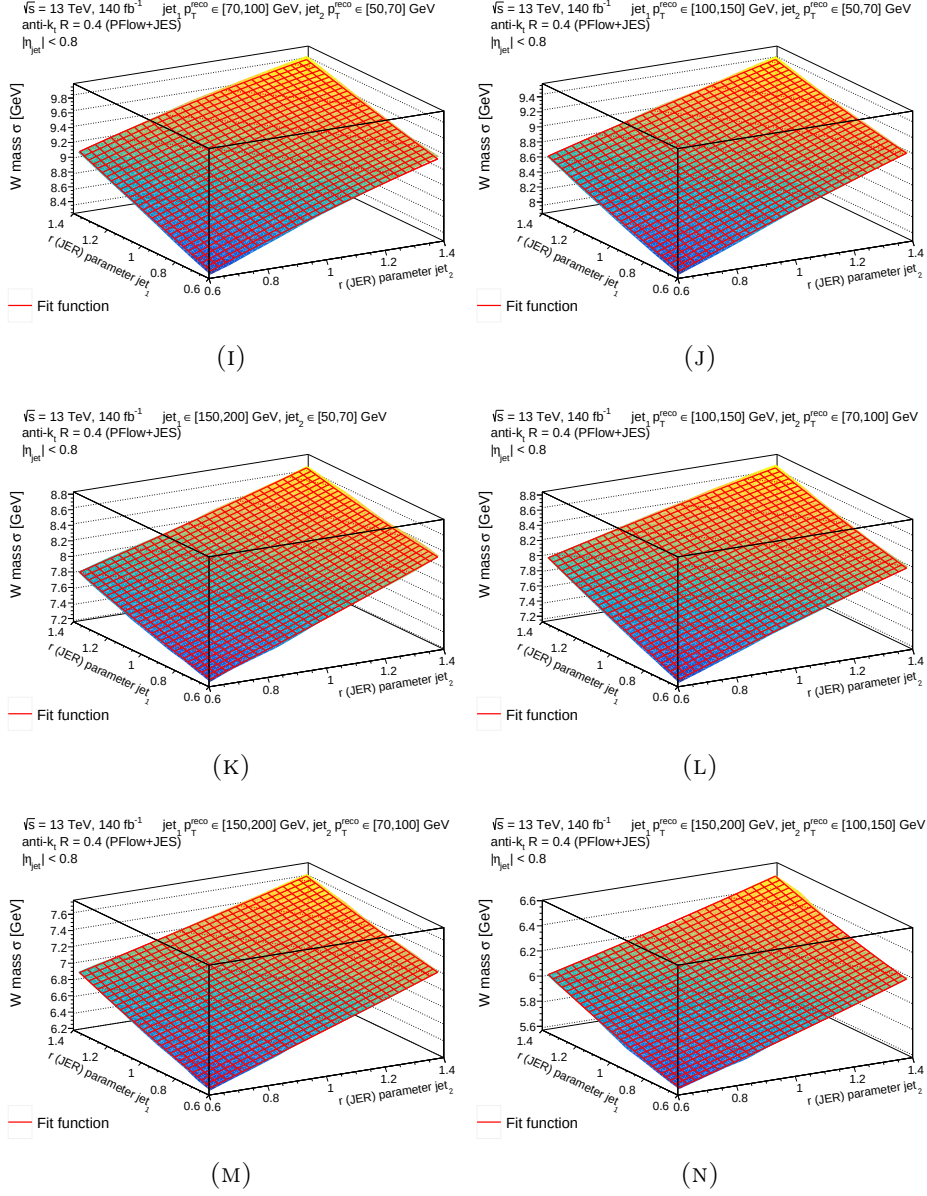


FIGURE F.4: The standard deviation of the W -mass distribution as a function of the JER parameters, r_1 and r_2 , in the different off-diagonal regions with the leading jet denoted as jet_1 and the sub-leading jet denoted as jet_2 , for the Run 2 measurement. The fitted two-dimensional quadratic function $\mathcal{F}$ is shown in red.

Bibliography

- [1] ATLAS Collaboration, *ATLAS calorimeter performance Technical Design Report*, (1996) (cit. on p. 1).
- [2] *Jet energy measurement and its systematic uncertainty in proton–proton collisions at $\sqrt{s} = 7$ TeV with the ATLAS detector*, *The European Physical Journal C* **75** (2015), ISSN: 1434-6052, URL: <http://dx.doi.org/10.1140/epjc/s10052-014-3190-y> (cit. on p. 1).
- [3] ATLAS Collaboration, *Measurement of the top quark mass with the ATLAS detector using $t\bar{t}$ events with a high transverse momentum top quark*, *Phys. Lett. B* **867** (2025) 139608, arXiv: 2502.18216 [hep-ex] (cit. on p. 1).
- [4] ATLAS Collaboration, *Measurement of the top quark mass in the $t\bar{t} \rightarrow$ dilepton channel from $\sqrt{s} = 8$ TeV ATLAS data*, *Phys. Lett. B* **761** (2016) 350, arXiv: 1606.02179 [hep-ex] (cit. on p. 1).
- [5] ATLAS Collaboration, *Measurement of the top quark mass in the $t\bar{t} \rightarrow$ lepton+jets and $t\bar{t} \rightarrow$ dilepton channels using $\sqrt{s} = 7$ TeV ATLAS data*, *Eur. Phys. J. C* **75** (2015) 330, arXiv: 1503.05427 [hep-ex] (cit. on p. 1).
- [6] CMS Collaboration, *Measurement of the top quark mass with lepton+jets final states using pp collisions at $\sqrt{s} = 13$ TeV*, *The European Physical Journal C* **78** (2018), ISSN: 1434-6052, URL: <http://dx.doi.org/10.1140/epjc/s10052-018-6332-9> (cit. on p. 1).
- [7] CDF Collaboration, *Measurement of the top-quark mass in the lepton+jets channel using a matrix element technique with the CDF II detector*, *Physical Review D* **84** (2011), ISSN: 1550-2368, URL: <http://dx.doi.org/10.1103/PhysRevD.84.071105> (cit. on p. 1).
- [8] ATLAS Collaboration, *Measurement of the $t\bar{t}$ production cross-section in the lepton+jets channel at $\sqrt{s} = 13$ TeV with the ATLAS experiment*, *Phys. Lett. B* **810** (2020) 135797, arXiv: 2006.13076 [hep-ex] (cit. on p. 1).

- [9] B. Nachman, *Investigating the Quantum Properties of Jets and the Search for a Supersymmetric Top Quark Partner with the ATLAS Detector*, Thesis (Ph.D.)–Stanford University, 2016, arXiv: [1609.03242 \[hep-ex\]](#), URL: <https://arxiv.org/abs/1609.03242> (cit. on p. 5).
- [10] ATLAS Collaboration, *Measurement of the ATLAS Detector Jet Mass Response using Forward Folding with 80 fb^{-1} of $\sqrt{s} = 13\text{ TeV}$ pp data*, ATLAS-CONF-2020-022, 2020, URL: <https://cds.cern.ch/record/2724442> (cit. on p. 5).
- [11] ATLAS Collaboration, *Measurement of large radius jet mass reconstruction performance at $\sqrt{s} = 8\text{ TeV}$ using the ATLAS detector*, ATLAS-CONF-2016-008, 2016, URL: <https://cds.cern.ch/record/2139642> (cit. on p. 5).
- [12] ATLAS Collaboration, *Measurements of differential cross-sections in top-quark pair events with a high transverse momentum top quark and limits on beyond the Standard Model contributions to top-quark pair production with the ATLAS detector at $\sqrt{s} = 13\text{ TeV}$* , *Journal of High Energy Physics* **2022** (2022), ISSN: 1029-8479, URL: [http://dx.doi.org/10.1007/JHEP06\(2022\)063](http://dx.doi.org/10.1007/JHEP06(2022)063) (cit. on p. 16).
- [13] A. Shmakov et al., *SPANet: Generalized permutationless set assignment for particle physics using symmetry preserving attention*, *SciPost Physics* **12** (2022), ISSN: 2542-4653, URL: <http://dx.doi.org/10.21468/SciPostPhys.12.5.178> (cit. on p. 20).
- [14] J. Erdmann et al., *A likelihood-based reconstruction algorithm for top-quark pairs and the KL Fitter framework*, *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* **748** (2014) 18, ISSN: 0168-9002, URL: <http://dx.doi.org/10.1016/j.nima.2014.02.029> (cit. on p. 20).
- [15] ATLAS Collaboration, *Calibration of the jet energy scale and resolution of small-radius jets using semileptonic $t\bar{t}$ events with the ATLAS detector*, 2025, arXiv: [2512.17482 \[hep-ex\]](#), URL: <https://arxiv.org/abs/2512.17482> (cit. on pp. 23, 37, 38, 40, 41, 55, 56).

- [16] ATLAS Collaboration,
Software and computing for Run 3 of the ATLAS experiment at the LHC,
Eur. Phys. J. C **85** (2025) 234, arXiv: 2404.06335 [hep-ex] (cit. on p. 23).
- [17] F. James and M. Winkler, *MINUIT User's Guide*, (2004) (cit. on p. 41).
- [18] W. Verkerke and D. Kirkby, *The RooFit toolkit for data modeling*,
eConf **C0303241** (2003) MOLT007, arXiv: physics/0306116 (cit. on p. 41).
- [19] ATLAS Collaboration, *Electron and photon performance measurements with the ATLAS detector using the 2015–2017 LHC proton–proton collision data*,
JINST **14** (2019) P12006, arXiv: 1908.00005 [hep-ex].
- [20] ATLAS Collaboration, *Muon reconstruction and identification efficiency in ATLAS using the full Run 2 pp collision data set at $\sqrt{s} = 13$ TeV*,
Eur. Phys. J. C **81** (2021) 578, arXiv: 2012.00578 [hep-ex].
- [21] ATLAS Collaboration,
Performance of electron and photon triggers in ATLAS during LHC Run 2,
Eur. Phys. J. C **80** (2020) 47, arXiv: 1909.00761 [hep-ex].
- [22] ATLAS Collaboration, *Performance of the ATLAS muon triggers in Run 2*,
JINST **15** (2020) P09015, arXiv: 2004.13447 [physics.ins-det].
- [23] ATLAS Collaboration, *Muon reconstruction performance of the ATLAS detector in proton–proton collision data at $\sqrt{s} = 13$ TeV*, *Eur. Phys. J. C* **76** (2016) 292,
arXiv: 1603.05598 [hep-ex].
- [24] ATLAS Collaboration, *Jet energy scale and resolution measured in proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*,
The European Physical Journal C **81** (2021) 73, arXiv: 2007.02645 [hep-ex].
- [25] ATLAS Collaboration, *ATLAS b-jet identification performance and efficiency measurement with $t\bar{t}$ events in pp collisions at $\sqrt{s} = 13$ TeV*,
Eur. Phys. J. C **79** (2019) 970, arXiv: 1907.05120 [hep-ex].
- [26] ATLAS Collaboration, *Luminosity determination in pp collisions at $\sqrt{s} = 13$ TeV using the ATLAS detector at the LHC*,
Eur. Phys. J. C **83** (2023) 982, arXiv: 2212.09379 [hep-ex].
- [27] ATLAS Collaboration, *Summary of ATLAS Pythia 8 tunes*,
ATL-PHYS-PUB-2012-003, 2012,
URL: <https://cds.cern.ch/record/1474107>.

- [28] ATLAS Collaboration, *Studies on the improvement of the matching uncertainty definition in top-quark processes simulated with POWHEG+PYTHIA8*, ATL-PHYS-PUB-2023-029, 2013,
URL: <https://cds.cern.ch/record/2872787>.
- [29] ATLAS Collaboration, *Comparison of Monte Carlo generator predictions to ATLAS measurements of top pair production at 7 TeV*, ATL-PHYS-PUB-2015-002, 2015,
URL: <https://cds.cern.ch/record/1981319>.
- [30] ATLAS Collaboration, *Comparison of Monte Carlo generator predictions from Powheg and Sherpa to ATLAS measurements of top pair production at 7 TeV*, ATL-PHYS-PUB-2015-011, 2015,
URL: <https://cds.cern.ch/record/2020602>.
- [31] M. Czakon et al.,
Top-pair production at the LHC through NNLO QCD and NLO EW,
JHEP **10** (2017) 186, arXiv: 1705.04105 [hep-ph].
- [32] A. Collaboration, *Measurements of observables sensitive to colour reconnection in $t\bar{t}$ events with the ATLAS detector at $\sqrt{s} = 13$ TeV*,
The European Physical Journal C **83** (2023), ISSN: 1434-6052,
URL: <http://dx.doi.org/10.1140/epjc/s10052-023-11479-x>.
- [33] ATLAS Collaboration,
Measurement of the top-quark mass in $t\bar{t} \rightarrow$ dilepton events with the ATLAS experiment using the template method in 13 TeV pp collision data,
ATLAS-CONF-2022-058, 2022, URL: <https://cds.cern.ch/record/2826701>.
- [34] ATLAS Collaboration, *Measurement of the top-quark mass using a leptonic invariant mass in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*,
JHEP **06** (2023) 019, arXiv: 2209.00583 [hep-ex].
- [35] J. Butterworth et al., *PDF4LHC recommendations for LHC Run II*,
J. Phys. G **43** (2016) 023001, arXiv: 1510.03865 [hep-ph].
- [36] R. D. Ball et al.,
The PDF4LHC21 combination of global PDF fits for the LHC Run III,
J. Phys. G **49** (2022) 080501, arXiv: 2203.05506 [hep-ph].
- [37] B. Andersson, G. Gustafson, and B. Söderberg,
A General Model for Jet Fragmentation, **Z. Phys. C** **20** (1983) 317.

-
- 857 [38] M. G. Bowler, e^+e^- Production of Heavy Quarks in the String Model,
858 *Z. Phys. C* **11** (1981) 169.
- 859 [39] ATLAS Collaboration, *Performance of top-quark and W-boson tagging with*
860 *ATLAS in Run 2 of the LHC*, *Eur. Phys. J. C* **79** (2019) 375,
861 arXiv: 1808.07858 [hep-ex].
- 862 [40] ATLAS Collaboration,
863 *ATLAS measurements of the properties of jets for boosted particle searches*,
864 *Phys. Rev. D* **86** (2012) 072006, arXiv: 1206.5369 [hep-ex].